An investigation as to how AGE, DWELLING, health condition, living arrangement, method of travel to work and communal establishment affected covid-19 deaths in England

DS7006 2526 CW2-Quantitative Data Analysis

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Abstract:

This individual research project conducts a quantitative analysis of COVID-19 mortality trends across English local authorities, aiming to uncover the socio-economic and demographic factors influencing immunity and death rates during the pandemic.

Data was sourced from www.nomisweb.co.uk and processed using exploratory data analysis techniques, including normality tests and visualization methods such as box plots, histograms, and Q-Q plots. Further normality tests were run with Shapiro Wilk tests and Kolmogorov-Smirnov tests.

After which, the researcher ran Correlation and Partial correlation tests to check the variables that had strong positive correlation with covid total deaths and strong negative correlation with covid total deaths and the factors that could have been responsible.

Six key variables were examined: Age, Dwelling, Health Condition, Living arrangements, Method of Travel to Work, and Communal Establishment.

The literature review highlights the disproportionate impact of COVID-19 on older adults, care home residents, individuals with preexisting medical conditions, and those living in deprived or densely populated areas. The analysis reveals that age groups 15-25 experienced higher mortality rates, attributed to lifestyle factors such as increased social activity, smoking, and drinking. In contrast, mortality rates declined among those aged 35-50, likely due to greater responsibility, family life, and improved health awareness. Further reductions were observed in the 60-75+ age group, associated with enhanced medical care and restricted movement.

Transport mode emerged as a significant factor, with private vehicle users experiencing lower mortality rates compared to public transport users, due to reduced social contact. Living arrangements also played a crucial role: Individuals living as couples had better survival outcomes than those living alone, benefiting from mutual support and care. Additionally, good dwelling conditions and hygiene were linked to lower mortality rates.

The study concludes that targeted interventions- such as promoting vaccination, regular health check-ups, minimizing unnecessary social contact, and improving living and transport conditions can mitigate future pandemic risks.

The findings underscore the importance of socio-economic and demographic variables in shaping COVID-19 outcomes and provide actionable recommendations for public health policy.

Section 1: Introduction.

This Individual Data Analysis project aims to study COVID 19 data trends and extract insights on the factors that could have been responsible to making different categories of people immune or less immune to the virus, and it also aims to study what was responsible for the death toll difference across various categories of people in the UK.

Coronavirus disease 2019 (COVID-19), is an illness caused by a virus. The virus is called severe acute respiratory syndrome coronavirus 2 or, more commonly, SARS-CoV-2. It started spreading at the end of 2019 and became a pandemic disease in 2020.

The virus that causes COVID-19 spreads between people in close contact. Most commonly, it spreads through the air in tiny droplets of fluid. Many people with COVID-19 have mild illness or no symptoms. But for older adults and people with certain medical conditions, COVID-19 can lead to the need for care in the hospital or death.

Staying up to date on your COVID-19 vaccine helps prevent serious illness, the need for hospital care due to COVID-19 and death from COVID-19. Other things that may help prevent the spread of this coronavirus include good indoor airflow, physical distancing, wearing a mask in the right setting and good hygiene.

Medicine can limit the seriousness of the viral infection. Most people recover without long-term effects, but some people have symptoms that continue for months.

Typical COVID-19 symptoms often show up 2 to 14 days after contact with the virus.

Symptoms can include:

- Dry cough.
- Shortness of breath.
- Loss of taste or smell.
- Extreme tiredness, called fatigue.
- Digestive symptoms such as upset stomach, vomiting or loose stools, also called diarrhoea.
- Pain, such as headaches and body or muscle aches.
- Fever or chills.
- Cold like symptoms such as congestion, runny nose or sore throat.

People may have only a few symptoms or none. People who have no symptoms but test positive for COVID-19 are called asymptomatic. For example, many children who test positive don't have symptoms of COVID-19 illness. People who go on to have symptoms are considered presymptomatic. People in both groups can still spread the COVID-19 virus to others.

Some people may have symptoms that get worse about 5 to 7 days after symptoms start.

Most people with COVID-19 have mild to moderate symptoms. But COVID-19 can cause serious medical complications and lead to death. Older adults and people who already have medical conditions are at greater risk of serious illness.

COVID-19 may be a mild, moderate, severe or critical illness.

- In broad terms, mild COVID-19 doesn't affect the ability of the lungs to get oxygen to the body.
- In moderate COVID-19 illness, the lungs work properly but there are signs that the infection is deep in the lungs.
- Severe COVID-19 means that the lungs don't work correctly, and the person needs oxygen and other medical help in the hospital.
- Critical COVID-19 illness means the lungs and breathing system, called the respiratory system, have failed and there is damage throughout the body.

Rarely, people who catch this coronavirus can develop a group of symptoms linked to inflamed organs or tissues. This illness is called multisystem inflammatory syndrome. When children have this illness, it is called multisystem inflammatory syndrome in children (MIS-C). In adults, the name is MIS-A.
(Mayo Clinic (2024). *Coronavirus Disease 2019 (COVID-19)*. [online] Mayo Clinic. Available at: https://www.mayoclinic.org/diseases-conditions/coronavirus/symptoms-causes/syc-20479963.)

There have been so many concerns and questions as to why England experienced high mortality rate during covid, but there are plenty of clues that, when pieced together, help build a picture of why the UK has reached this devastating number.

Some will point a finger at the government - its decision to lock-down later than much of western Europe, the stuttering start to its test-and-trace network and the lack of protection afforded to care home residents.

Others will spotlight deeper rooted problems with British society - its poor state of public health, with high levels of obesity, for example.

Others, still, will note that some of the UK's great strengths - its position as a vibrant

hub for international air travel, its ethnically diverse and densely packed urban populations - exposed its vulnerability to a virus that spreads effortlessly between people. (UK Covid deaths: Why the 100,000 toll is so bad. (2021). *BBC News*. [online] 26 Jan. Available at: https://www.bbc.co.uk/news/health-55757790.)
Examining the socio-economic and demographic factors surrounding the covid death statistics will help the researcher to bring out hidden insights and proffer solutions and recommendations to ensure such disaster is mitigated in future.

Data was collected from www.nomisweb.co.uk and an exploratory data analytical approach was deployed to carry out several normality tests for the different variables to have an in-depth understanding as to what was responsible for the patterns and outliers.

The data collected from www.nomisweb.co.uk was cleaned and processed to a satisfactory degree and converted to a .csv file (u3059227_DS7006_CW2_data.csv) which was imported to R and a series of normality hypotheses tests(box plots, q-q plots and histograms) were ran for the independent variables.

Then we run the Shapiro Wilk tests and Kolmogorov-Smirnov test to test for normality as well.

We then run the Correlation analysis and Partial correlation to check the variables with strong positive correlation with covid total deaths and strong negative correlation and examine the reasons.

Then we start the dimensionality reduction process with Kaiser-Meyer-Olkin Statistics (KMO and MSA)

If MSA > 0.6, we proceed to factor analysis.

We then run the Principal Component Analysis.

My themes which I downloaded from Nomis, joined and normalized are Age, Dwelling, Health Condition, Living arrangements, Method of travel to work and Communal Establishment.

This project aims to identify which socio economic variables had the strongest effect on covid-19 deaths across English local authorities.

Section 2: Literature Review

The researcher chose 6 variables which are Age, Dwelling, Health Condition, living arrangements, Method of travel to work and Communal Establishment: to study the trends in covid deaths.

Age is an important factor because the body's immunity is not uniform across all age levels of an average human being. And as such, the researcher needs to understand the age groups that were most vulnerable and least vulnerable to the virus.

Since infectious agents have their own unique age profile for death, we use a 9-year time series and several different methods to adjust single-year-of-age deaths in England and Wales during 2019 (the pre-COVID-19 base year) to a pathogen-neutral single-year-of-age baseline. This adjusted base year is then used to confirm the widely reported higher deaths in males for most ages above 43 in both 2020 and 2021. During 2020 (+COVID-19 but no vaccination), both male and female population -adjusted deaths significantly increased above age 35. A significant reduction in all-cause mortality among both males and females aged 75+ could be demonstrated in 2021 during the widespread COVID-19 vaccination period; however, deaths below age 75 progressively increased. This finding arises from a mix of vaccination coverage and year-of-age profiles of deaths for the different SARS-CoV-2 variants. In addition, specific effects of age around puberty were demonstrated, where females had higher deaths than males. There is evidence that year-of-birth cohorts may also be involved, indicating that immune priming to specific pathogen outbreaks in the past may have led to lower deaths for some birth cohorts. To specifically identify the age profile for the COVID-19 variants from 2020 to 2023, we employ the proportion of total deaths at each age that are potentially due to or 'with' COVID-19. The original Wuhan strain and the Alpha variant show somewhat limited divergence in the age profile, with the Alpha variant shifting to a moderately higher proportion of deaths below age 84. The Delta variant specifically targeted individuals below age 65. The Omicron variants showed a significantly lower proportion of overall mortality, with a markedly higher relative proportion of deaths above age 65, steeply increasing with age to a maximum around 100 years of age.

(Jones, R.P. and Ponomarenko, A. (2023). COVID-19-Related Age Profiles for SARS-CoV-2 Variants in England and Wales and States of the USA (2020 to 2022): Impact on All-Cause Mortality. *Infectious Disease Reports*, 15(5), pp.600–634. doi:https://doi.org/10.3390/idr15050058.)

The researcher also noticed variations in covid deaths across different dwelling types, and this could be influenced by the hygiene differences between decent accommodations and shanties.

Underlying mortality of care home residents was higher than community controls (adjusted rate ratio 5.59, 95% confidence interval 5.23–5.99, $P < .001$). During April 2020, there was a net increase in mortality of care home residents over that of

controls. The mortality rate of care home residents was 27.2 deaths per 1000 patients per week, compared with 2.31 per 1000 for controls. Excess deaths for care home residents, above that predicted from pre-pandemic years, peaked between April 13 and 19 (men, 27.7, 95% confidence interval 25.1–30.3; women, 17.4, 15.9–18.8 per 1000 per week). Compared with care home residents, long-term conditions and frailty were differentially associated with greater mortality in community-dwelling controls.

Individual-patient data from primary care electronic health records may be used to estimate mortality in care home residents. Mortality is substantially higher than for community-dwelling comparators and showed a disproportionate increase in the first wave of the COVID-19 pandemic. Care home residents require particular protection during periods of high infectious disease transmission.

(Gulliford, M.C., Prevost, A.T., Clegg, A. and Rezel-Potts, E. (2022). Mortality of Care Home Residents and Community-Dwelling Controls During the COVID-19 Pandemic in 2020: Matched Cohort Study. *Journal of the American Medical Directors Association*, [online] 23(6), pp.923-929.e2. doi:https://doi.org/10.1016/j.jamda.2022.04.003.)

The researcher also seeks to understand if and how different pre-existing medical conditions affected covid deaths.
The percentage of SARS-CoV2 deaths among various age groups with and without preexisting conditions was also compared during the Delta variant surge since June 2021 (4 June through 8 October 2021) and prior to the Delta variant surge in 2021 [5]. The relative risk (RR), 95% confidence interval and P value were calculated to compare these outcomes prior to and during the Delta variant surge. RR is calculated using the formula, P value (<0.05) was considered significant for the differentiation between the groups.

Our study demonstrates that preexisting conditions were associated with most deaths among older populations (>60 years of age), with a slight increase in deaths in those without preexisting conditions, especially among younger populations, during the Delta variant surge. Our study included a detailed analysis of a large sample of 94 624 SARS-CoV2 deaths among various age groups from the UK NHS database. A limitation of our study is that it is an observational retrospective study of publicly reported data, and the generalisability of the findings are limited to the UK population. Additionally, we are unable to analyse the mode of COVID-19 deaths and timing of deaths relative to the onset of symptoms and or hospitalisation, as that information is not available.
In conclusion, the majority of deaths (93.5%–94.8%) among those older than 60 years of age during the Delta variant surge were associated with preexisting conditions, with a slight but statistically significant increased proportion of deaths in

those without preexisting conditions (7.7% vs 4.1%) during the Delta variant surge than the prior comparison period before the Delta variant surge, and the majority of this slight increase was seen among those aged 20 to 39 and 40 to 59 years.
(Emani, V.R., Reddy, R., Goswami, K.K., Nakka, A.S., Emani, S.R., Reddy, N.K., Maddula, K.R., Nandanoor, D. and Goswami, S. (2022). Preexisting conditions among the SARS -CoV2 deaths during the SARS-CoV2 B.1.617.2 (Delta variant) outbreak in the United Kingdom. *Journal of Global Antimicrobial Resistance*, 29, pp.97–98. doi:https://doi.org/10.1016/j.jgar.2022.02.014.)

The next variable the researcher investigates is how living arrangements affected covid deaths.
The outbreak of Covid-19 in the UK has seen many families unexpectedly brought back together. The circumstances and stories of individual families have been picked up in the press focussing on the difficulties of people adjusting to their changed living arrangements. Yet, there have been few empirical analyses on how such changes might influence people's health and wellbeing. Therefore, this study explored the changing patterns of perceived stress by living arrangements change during the first wave of the Covid-19 pandemic in 2020.
The data used relates to 12,532 cohort members in the special Covid-19 surveys conducted with the participants of the 1958, 1970, 2000-01 British birth cohorts and Next Steps (born in 1989–90) in May 2020. Probit regression models were used to assess the chance of reporting increased stress amongst those who had experienced changes in living arrangements during the pandemic compared to those who had not.
The results provide strong evidence that those individuals whose living arrangements changed during the Covid-19 pandemic have a higher likelihood of reported increased stress than those whose living arrangements remained unchanged. This was most clearly seen for older cohorts. Increased interpersonal conflict plays a role in mediating the association, especially for the younger cohort. The findings confirm that during the first lockdown, changing living arrangements were negatively associated with individuals' mental wellbeing. As prolonged periods of stress can lead to serious health problems and policymakers need to be mindful that services may need to take these new, albeit for many temporary, forms of living arrangements into account.
(Evandrou, M., Falkingham, J., Qin, M. and Vlachantoni, A. (2021). Changing living arrangements and stress during Covid-19 lockdown: evidence from four birth cohorts in the UK. *SSM - Population Health*, 13, p.100761. doi:https://doi.org/10.1016/j.ssmph.2021.100761.)

The next variable the researcher looks at is how method of travel to work affected covid deaths.
Traveler responses to transport disruptions can be used to understand individual travel choices and the potential barriers restricting the uptake of sustainable travel behaviors. Using the Cardiff Capital Region as the study area, this research employed a cross-sectional survey to determine the immediate and anticipated long-term impacts of the COVID-19 pandemic, as a transport disruption, on work-related travel behaviors. This research identified that COVID-19 had significantly reduced travel frequencies for office-based workers, with reductions identified both during and in the expected frequencies following COVID-19. Similarly, modal changes and intentions to modal shift were identified as an immediate and potential post-lockdown impact. Meanwhile, departure times before and after the pandemic are expected to remain between similar hours. Additionally, no statistical relationships were identified between respondents' socio-demographics and attitudes towards the avoidance of public transport and the exclusive use of a private car in the future. Overall, the research indicated the potential for low-frequency work travel but increased car dependency following the pandemic. This signifies the importance for local policymakers and planners to continue to improve existing active and public transport infrastructure to secure a low-carbon recovery and future.
(Angell, C. and Potoglou, D. (2022). An insight into the impacts of COVID-19 on work-related travel behaviours in the Cardiff Capital Region and following the UK's first national lockdown. *Cities*, [online] 124, p.103602. doi:https://doi.org/10.1016/j.cities.2022.103602.)

The final variable the researcher investigates is how covid deaths varied across different communal establishments.
Recent studies have established the key individual-level risk factors of COVID-19 mortality such as age, gender, ethnicity, and socio-economic status. However, the spread of infectious diseases is a spatial and temporal process implying that COVID-19 mortality and its determinants may vary sub-nationally and over time. We investigate the spatial patterns of age-standardised death rates due to COVID-19 and their correlates across local authority districts in England, Wales, and Scotland across three waves of infection. Using a Spatial Durbin model, we explore within- and between-country variation and account for spatial dependency. Areas with a higher share of ethnic minorities and higher levels of deprivation had higher rates of COVID-19 mortality. However, the share of ethnic minorities and population density in an

area were more important predictors of COVID-19 mortality in earlier waves of the pandemic than in later waves, whereas area-level deprivation has become a more important predictor over time. Second, during the first wave of the pandemic, population density had a significant spillover effect on COVID-19 mortality, indicating that the pandemic spread from big cities to neighbouring areas. Third, after accounting for differences in ethnic composition, deprivation, and population density, initial cross-country differences in COVID-19 mortality almost disappeared. COVID-19 mortality remained higher in Scotland than in England and Wales in the third wave when COVID-19 mortality was relatively low in all three countries. Interpreting these results in the context of higher overall (long-term) non-COVID-19 mortality in Scotland suggests that Scotland may have performed better than expected during the first two waves. Our study highlights that accounting for both spatial and temporal factors is essential for understanding social and demographic risk factors of mortality during pandemics.
(Júlia Mikolai, Dorey, P., Keenan, K. and Hill Kulu (2023). Spatial patterns of COVID-19 and non-COVID-19 mortality across waves of infection in England, Wales, and Scotland. *Social Science & Medicine*, 338, pp.116330–116330. doi:https://doi.org/10.1016/j.socscimed.2023.116330.)

The six variables leave us with pending questions as to if the different variables have key impacts on covid mortality rates and if measures could be put in place to prevent future risks.

Section 3: Research Questions

- Does the proportion of residents aged 65 and above significantly influence COVID-19 mortality?
- Do people living as a couple stand a higher chance of surviving the virus more than people living alone?
- Are public transport users more exposed to the virus than private commuters?
- Are there preexisting medical conditions that could have impacted covid mortality rates?
- Are there lifestyle patterns common with certain age groups that could have impacted covid mortality rates?

Section 3.1: Research aim and independent and dependent variables:

The major aim of this research is to carefully examine all the possible factors that could have been responsible for covid 19 deaths in England and access them based on their degree of significance while possibly looking at what could be the reason behind the deaths and proffer solutions going forward that will prevent a similar occurrence in future.
The Dependent variables in this study is Total_deaths.
The Independent variables are Age group variables, Housing and living arrangements, Health status, Type of residence and Work and commuting characteristics.

Section 4: Methodology

Data was collected from www.nomisweb.co.uk and an exploratory data analytical approach was deployed to carry out several normality tests for the different variables to have an in-depth understanding as to what was responsible for the patterns and outliers.

The data collected from www.nomisweb.co.uk was cleaned and processed to a satisfactory degree and converted to a .csv file (u3059227_DS7006_CW2_data.csv) which was imported to R and a series of normality hypotheses tests(box plots, q-q plots and histograms) were ran for the independent variables.

Then we run the Shapiro Wilk tests and Kolmogorov-Smirnov test to test for normality as well.

We then run the Correlation analysis and Partial correlation to check the variables with strong positive correlation with covid total deaths and strong negative correlation and examine the reasons.

Then we start the dimensionality reduction process with Kaiser-Meyer-Olkin Statistics (KMO and MSA)

If MSA > 0.6, we proceed to factor analysis.

We then run the Principal Component Analysis.

My themes which I downloaded from Nomis, joined and normalized are Age, Dwelling, Health Condition, Living arrangements, Method of travel to work and Communal Establishment.

The data LA_names and LA_code were updated with SQL queries to ensure compliance with real time data.
The data variables downloaded from www.nomisweb.co.uk were joined by LA_code and LA_name with SQL queries and copied as a csv file.

The variables were normalized by dividing the total per 1000. This is essential to create a level playing field for like-to-like comparisons. It is helpful in revealing trends and percentage differences for effective analysis.

Section 4.2:

Techniques, Interpretation and Analysis of plots:

We start with the box plots, histograms and Q-Q plots to test for normality.

Then we run the Shapiro Wilk tests and Kolmogorov-Smirnov test to test for normality as well.

We then run the Correlation analysis and Partial correlation to check the variables with strong positive correlation with covid total deaths and strong negative correlation and examine the reasons.

Then we start the dimensionality reduction process with Kaiser-Meyer-Olkin Statistics (KMO and MSA)

If MSA > 0.6, we proceed to factor analysis.

We then run the Principal Component Analysis.

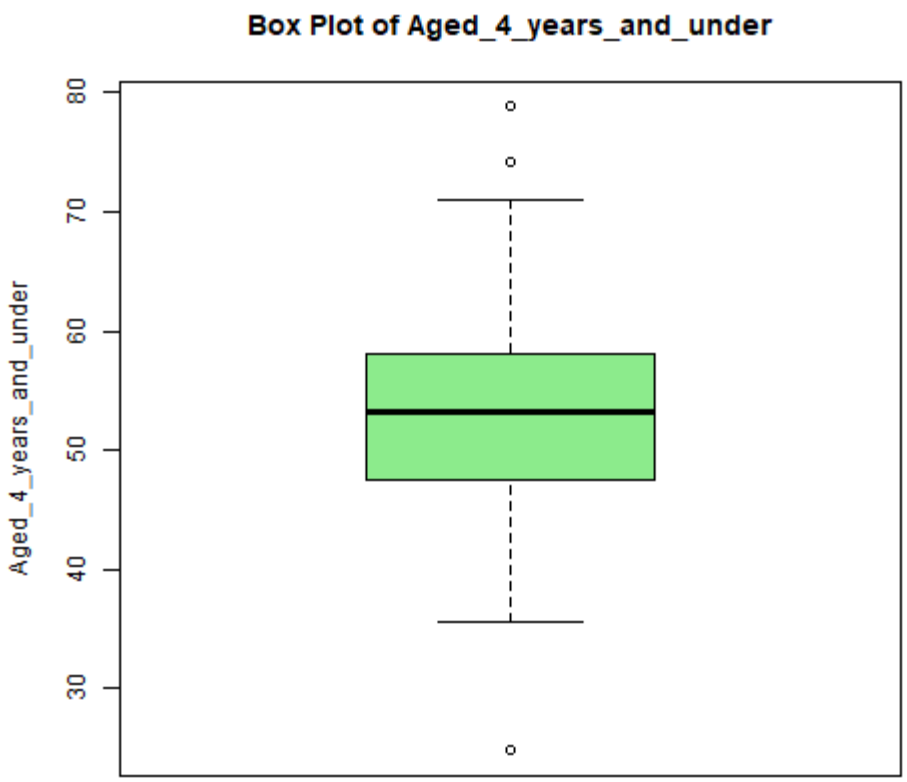

Figure 1box plot of Aged 4 and under

The above box plot of covid deaths for Ages 4 and under shows a normal distribution with a few outliers, which tells us that infants and toddlers of 4 years and under have almost equal mortality rate when exposed to COVID.

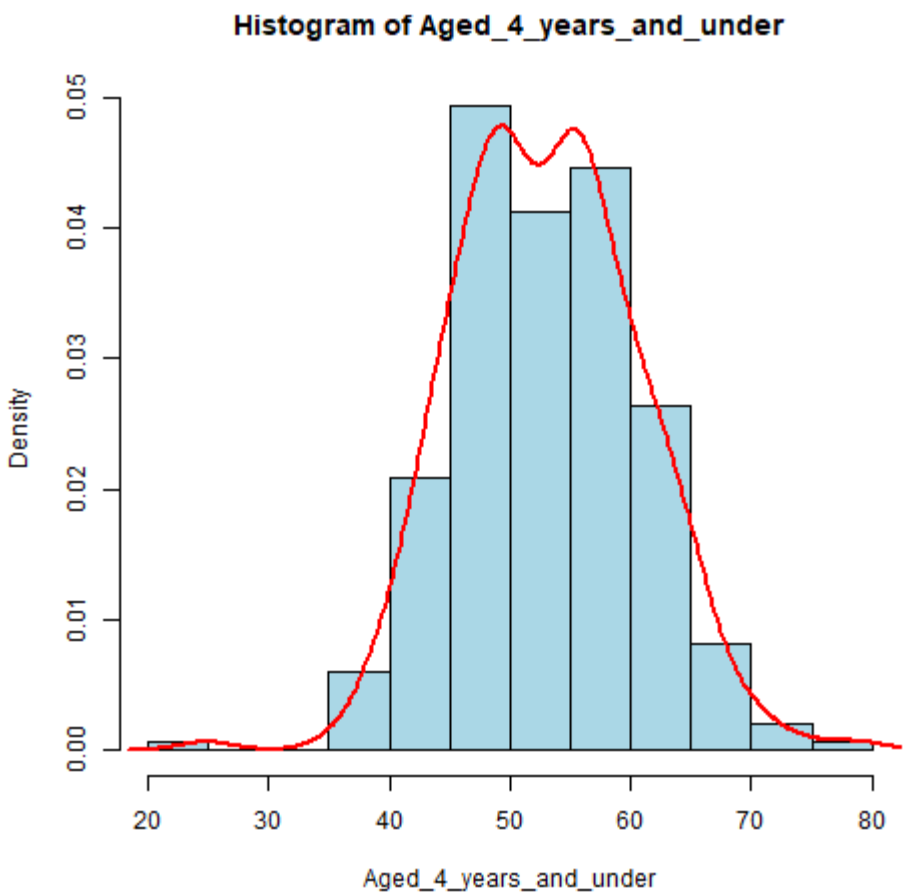

Figure 2 histogram of age 4 and under

The above histogram of covid deaths for Ages 4 and under shows a normal distribution, which tells us that infants and toddlers of 4 years and under have almost equal mortality rate when exposed to COVID.

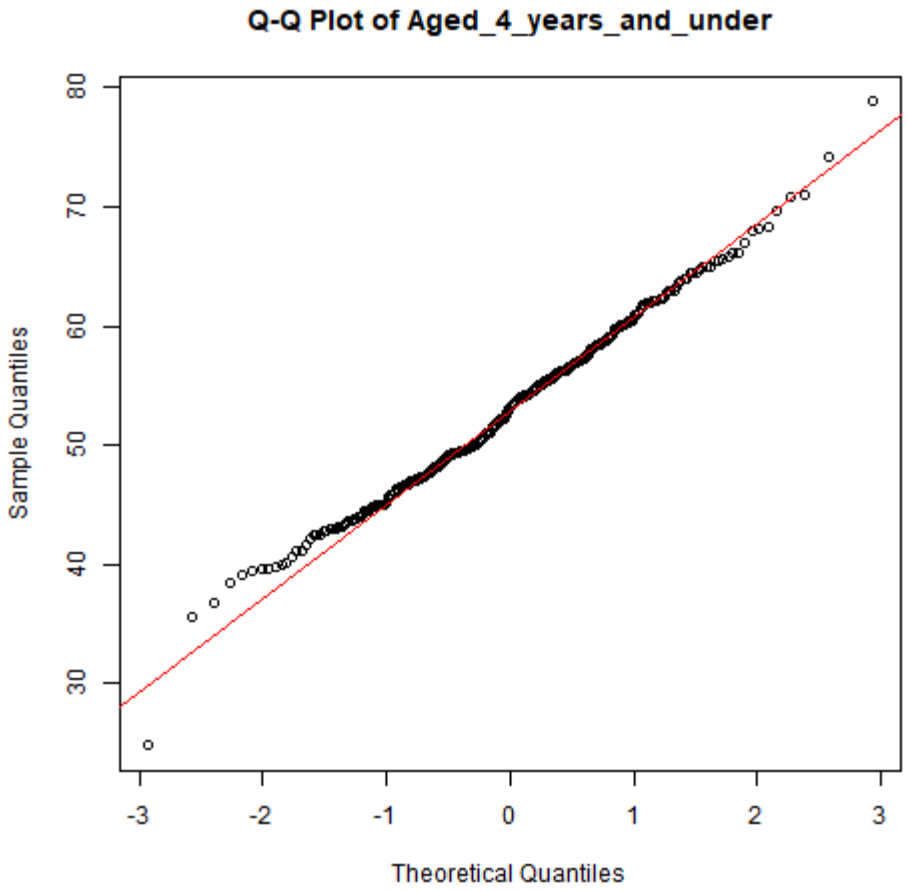

Figure 3 Q-Q plot of age 4 and under

The above Q-Q plot of covid deaths for Ages 4 and under shows a normal distribution, which tells us that infants and toddlers of 4 years and under have almost equal mortality rate when exposed to COVID.

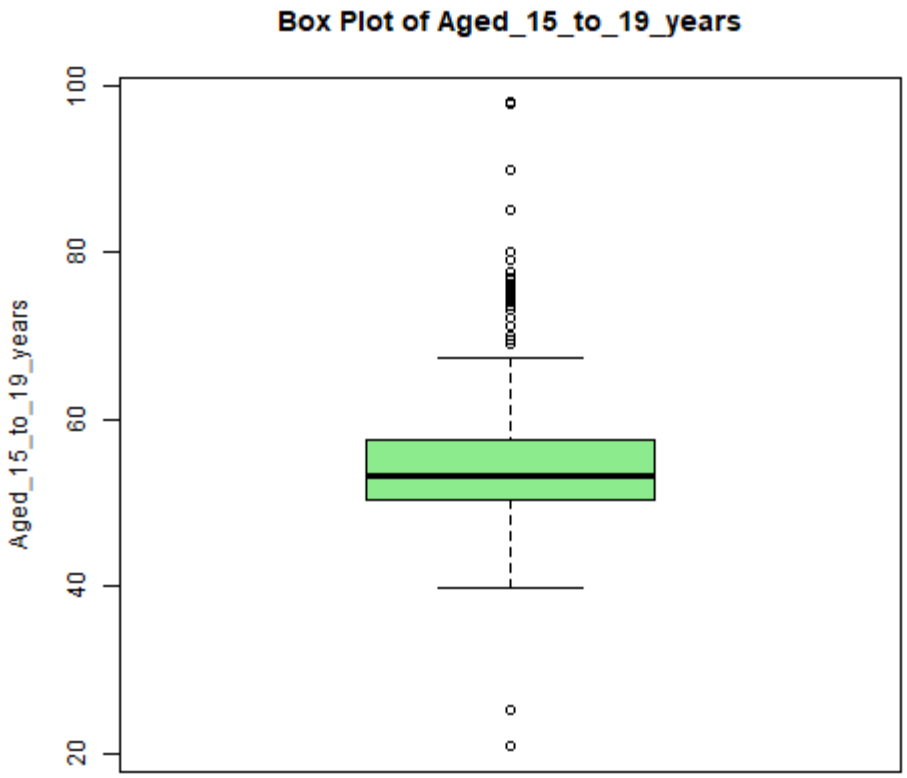

Figure 10 box plot of ages 15-19

The box plot of COVID deaths for ages 15-19 has so many upper outliers indicating increase in death rate which could be due to several factors ranging from underlying health conditions which could be due to exposure to hard drugs and alcohol (which is a common phenomenon among teens of this age group), living arrangements, house parties which increase the risk of contact contamination and available access to health care.

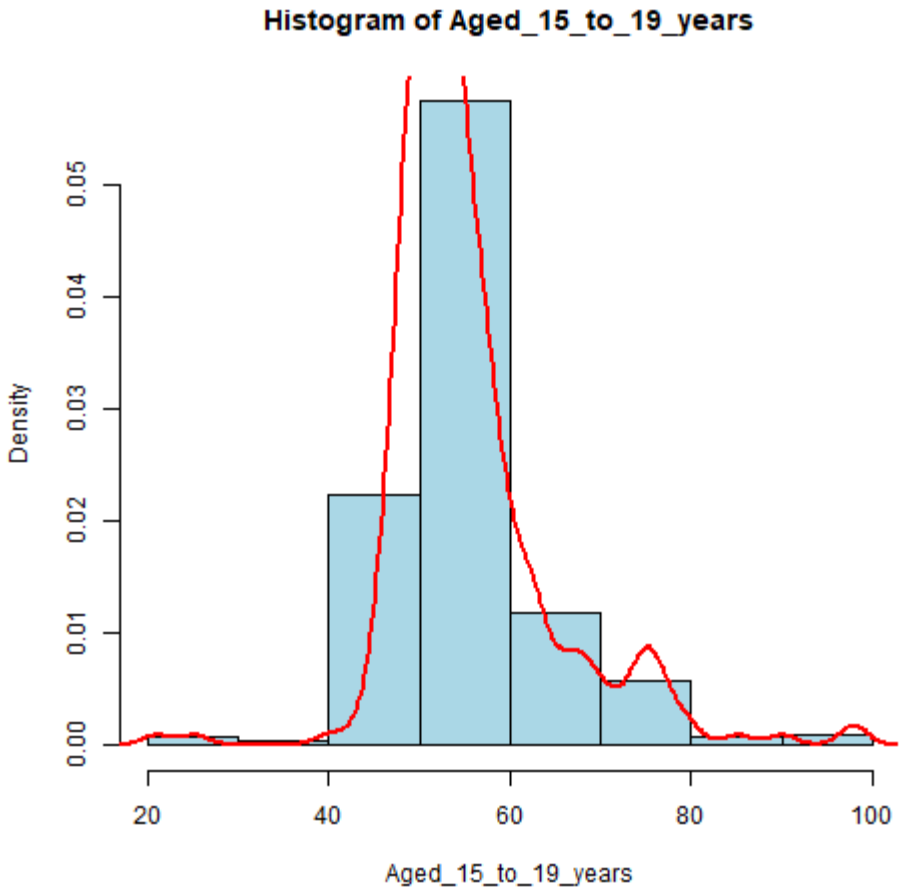

Figure 11 hist plot of ages 15-19

The histogram of COVID deaths for ages 15-19 has so many upper outliers indicating increase in death rate which could be due to several factors ranging from underlying health conditions which could be due to exposure to hard drugs and alcohol (which is a common phenomenon among teens of this age group), living arrangements, house parties which increase the risk of contact contamination and available access to health care.

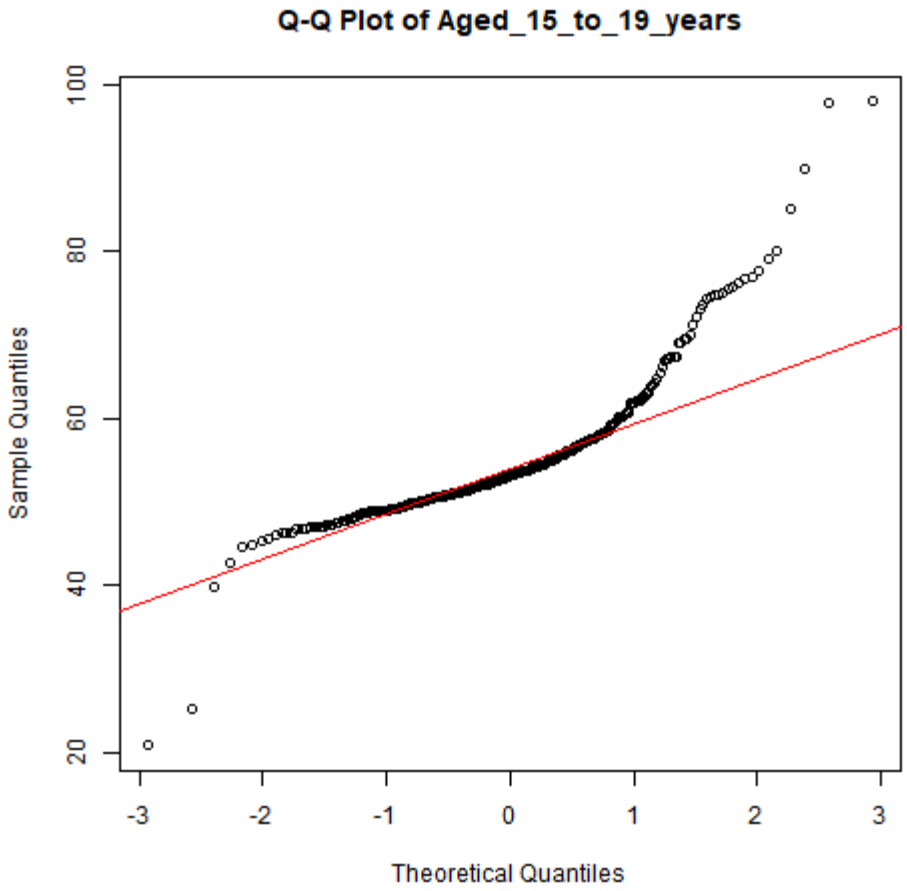

Figure 12 q-q plot of ages 15-19

The Q-Q plot of COVID deaths for ages 15-19 has so many upper outliers indicating increase in death rate which could be due to several factors ranging from underlying health conditions which could be due to exposure to hard drugs and alcohol (which is a common phenomenon among teens of this age group), living arrangements, house parties which increase the risk of contact contamination and available access to health care.

Figure 13 box plot of ages 20-24

The box plot of COVID deaths for ages 20-24 has so many upper outliers indicating increase in death rate which could be due to several factors ranging from underlying health conditions which could be due to exposure to hard drugs and alcohol (which is a common phenomenon among people of this age group), living arrangements, house parties which increase the risk of contact contamination and available access to health care.

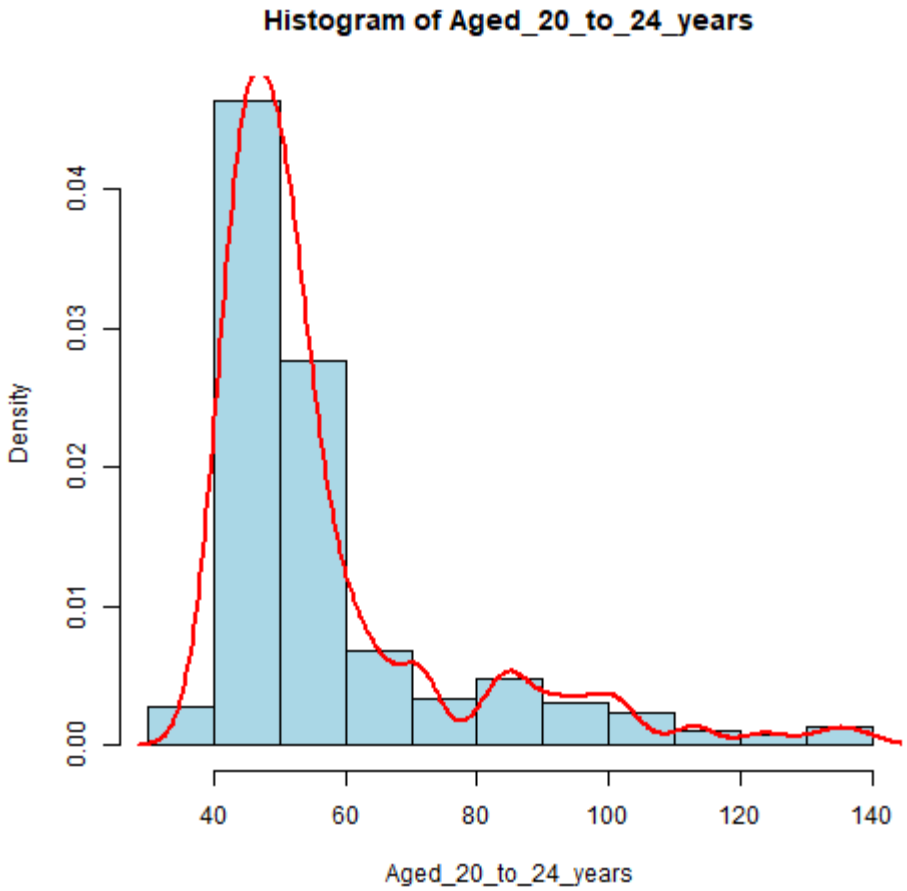

Figure 14 hist of ages 20-24

The histogram of COVID deaths for ages 20-24 has so many upper outliers indicating increase in death rate which could be due to several factors ranging from underlying health conditions which could be due to exposure to hard drugs and alcohol (which is a common phenomenon among people of this age group), living arrangements, house parties which increase the risk of contact contamination and available access to health care.

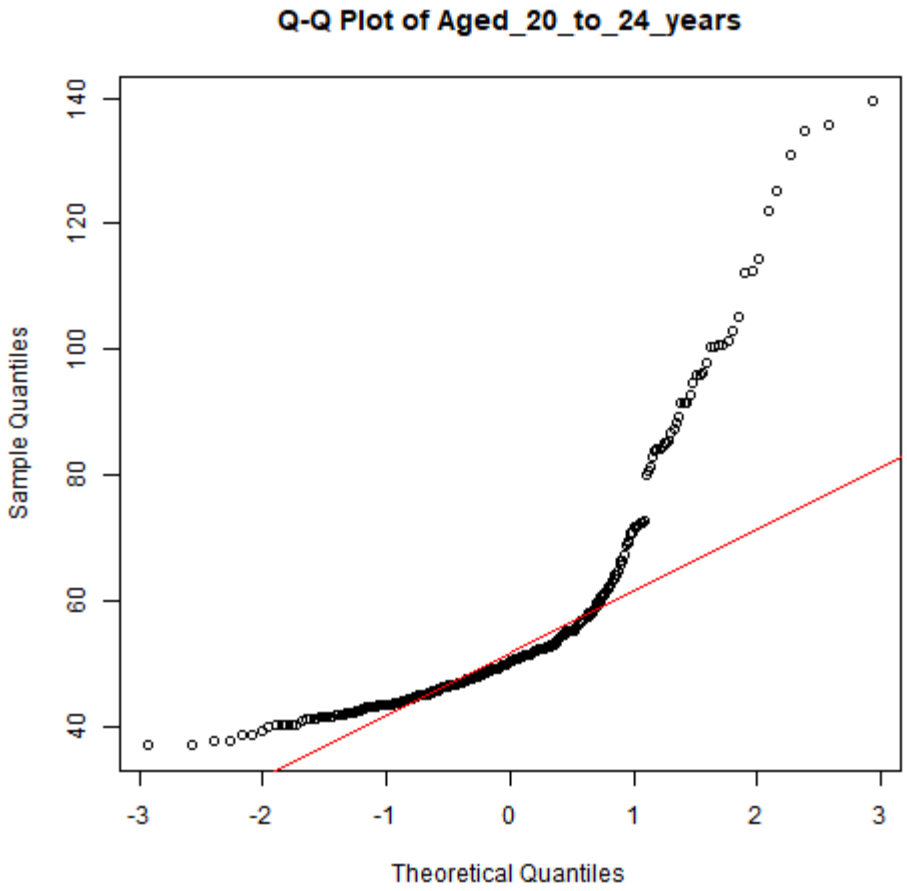

Figure 15 q-q plot of ages 20-24

The Q-Q of COVID deaths for ages 20-24 has so many upper outliers indicating increase in death rate which could be due to several factors ranging from underlying health conditions which could be due to exposure to hard drugs and alcohol (which is a common phenomenon among people of this age group), living arrangements, house parties which increase the risk of contact contamination and available access to health care.

Figure 16 box plot of ages 25-29

The box plot for Ages 25-29 shows several outliers suffering high mortality rates.

This age group is the most active and about age group in terms of social activity, workability and school life.

That could have been responsible for the much higher outliers.

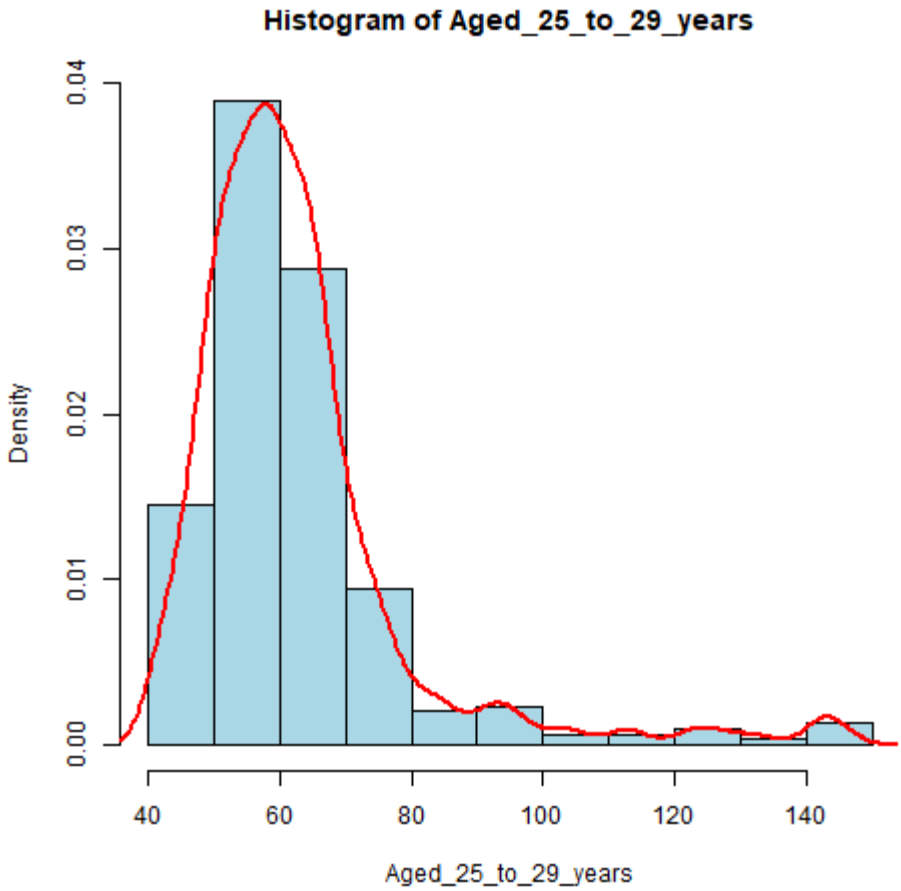

Figure 17 hist of ages 25-29

The histogram for Ages 25-29 shows several outliers suffering high mortality rates.

This age group is the most active and about age group in terms of social activity, workability and school life.

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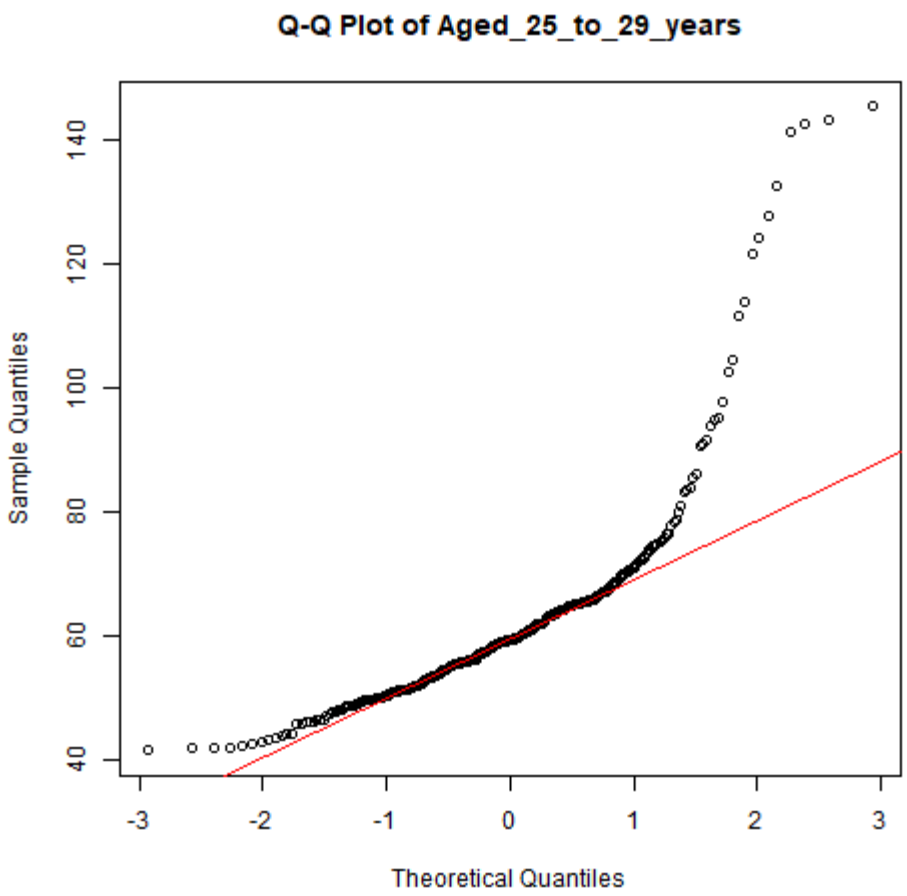

Figure 18 q-q plot of ages 25-29

The Q-Q plot for Ages 25-29 shows several outliers suffering high mortality rates.

This age group is the most active and about age group in terms of social activity, workability and school life.

That could have been responsible for the much higher outliers.

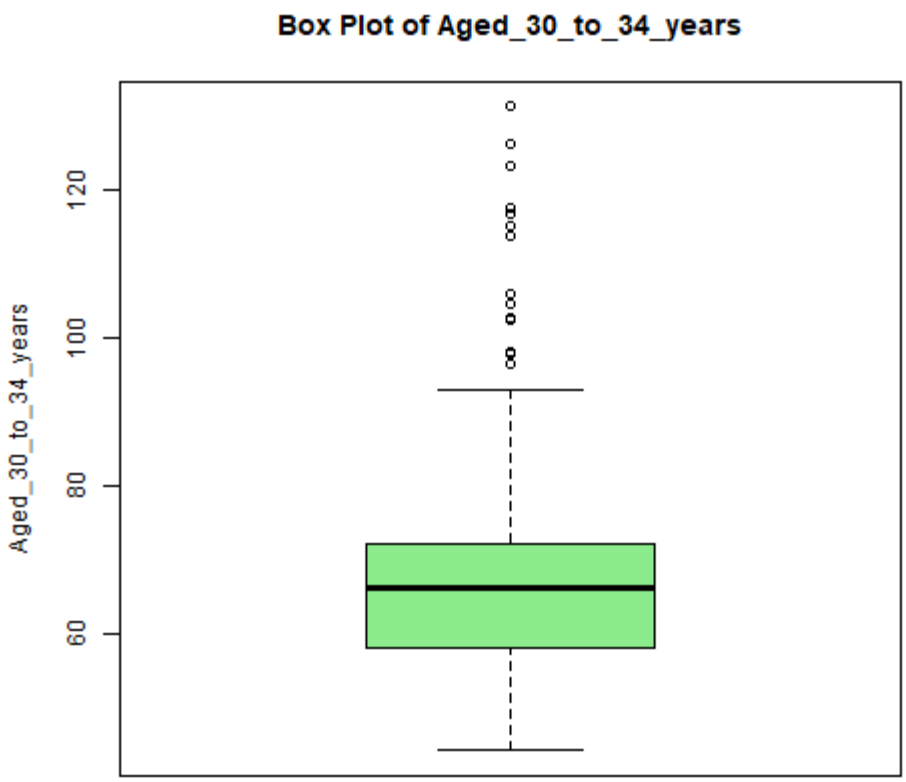

Figure 19 box plot of ages 30-34

The box plot for Ages 30-34 also shows several outliers suffering high mortality rates.

This age group is likely the most active and about age group in terms of workability and family life.

That could have been responsible for the much higher outliers.

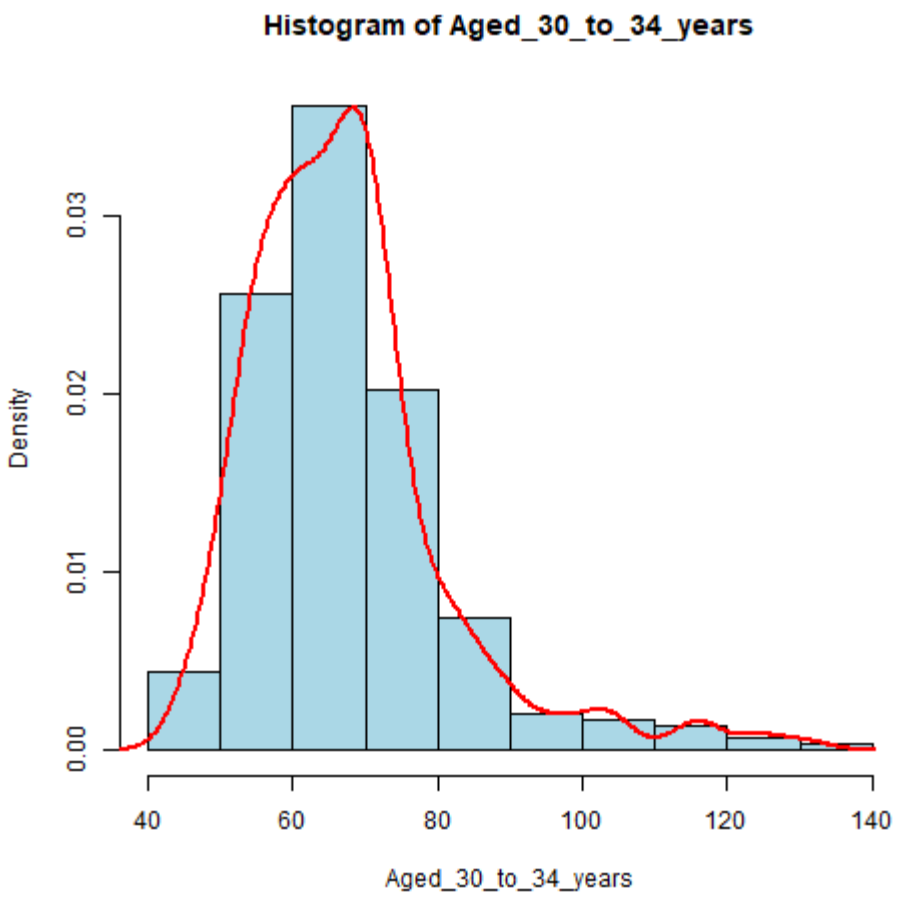

Figure 20 hist of ages 30-34

The histogram for Ages 30-34 also shows several outliers suffering high mortality rates.

This age group is likely the most active and about age group in terms of workability and family life.

That could have been responsible for the much higher outliers.

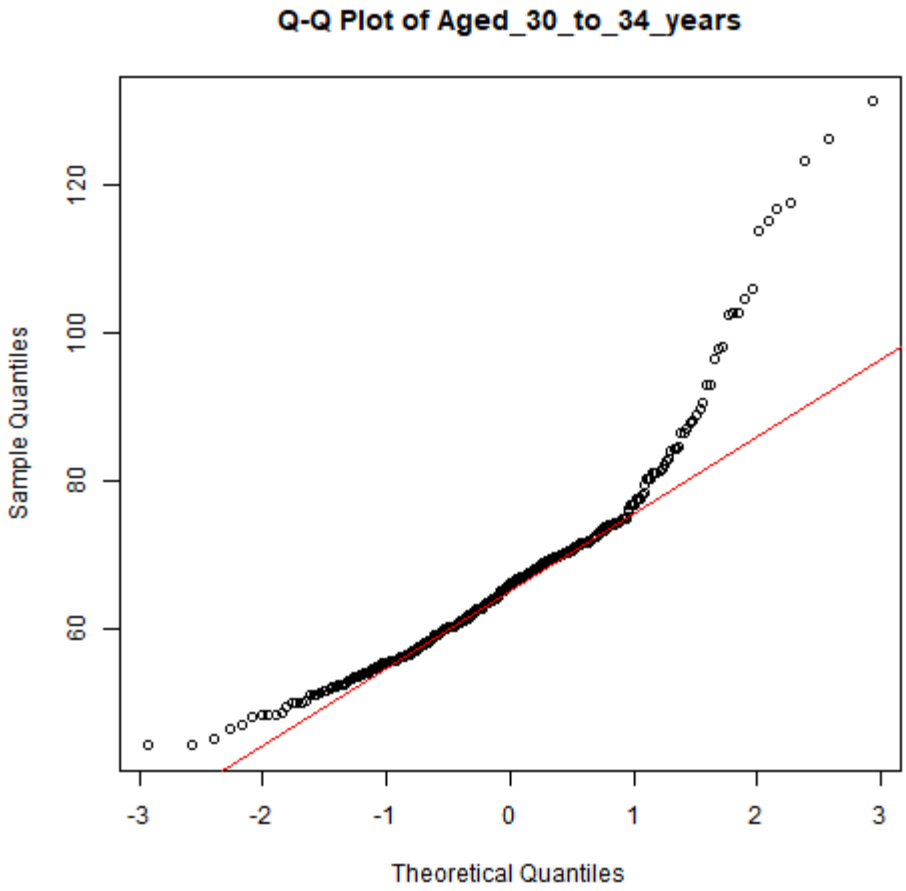

Figure 21 q-q plot of ages 30-34

The Q-Q plot for Ages 30-34 also shows several outliers suffering high mortality rates.

This age group is likely the most active and about age group in terms of workability and family life.

That could have been responsible for the much higher outliers.

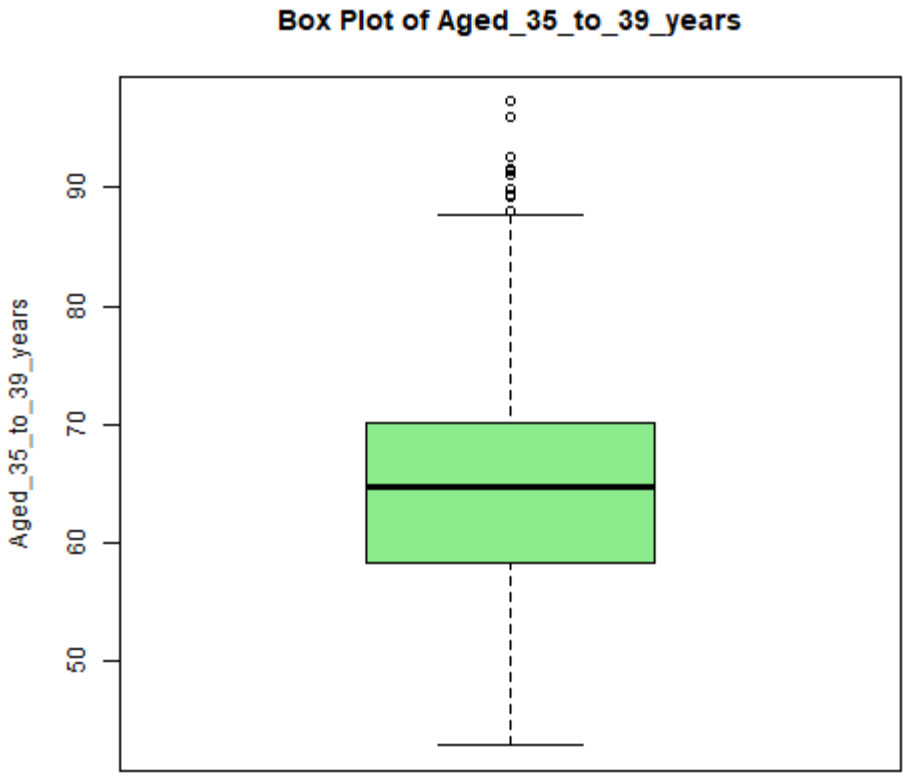

Figure 22 box plot of ages 35-39

The box plot for ages 35 – 39 shows several outliers indicating spike in mortality rates which could be due to work and family responsibilities which increases their social activity thereby exposing them to greater risk.

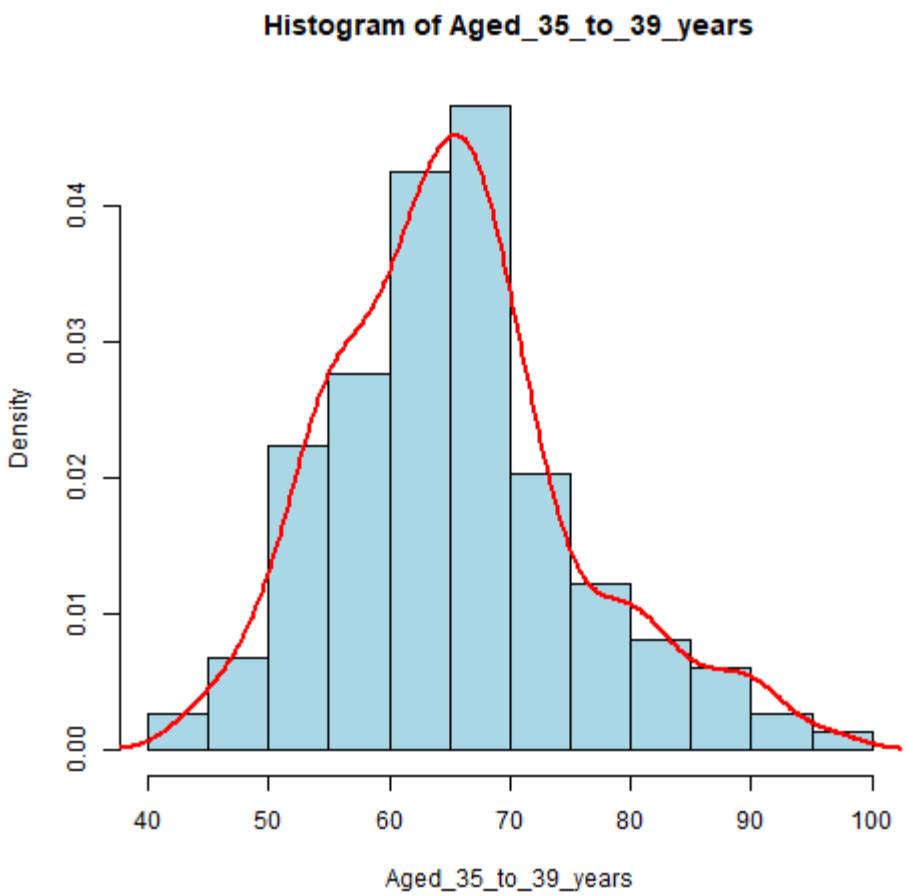

Figure 23 hist of ages 35-39

The histogram for ages 35 – 39 shows several outliers indicating spike in mortality rates which could be due to work and family responsibilities which increases their social activity thereby exposing them to greater risk.

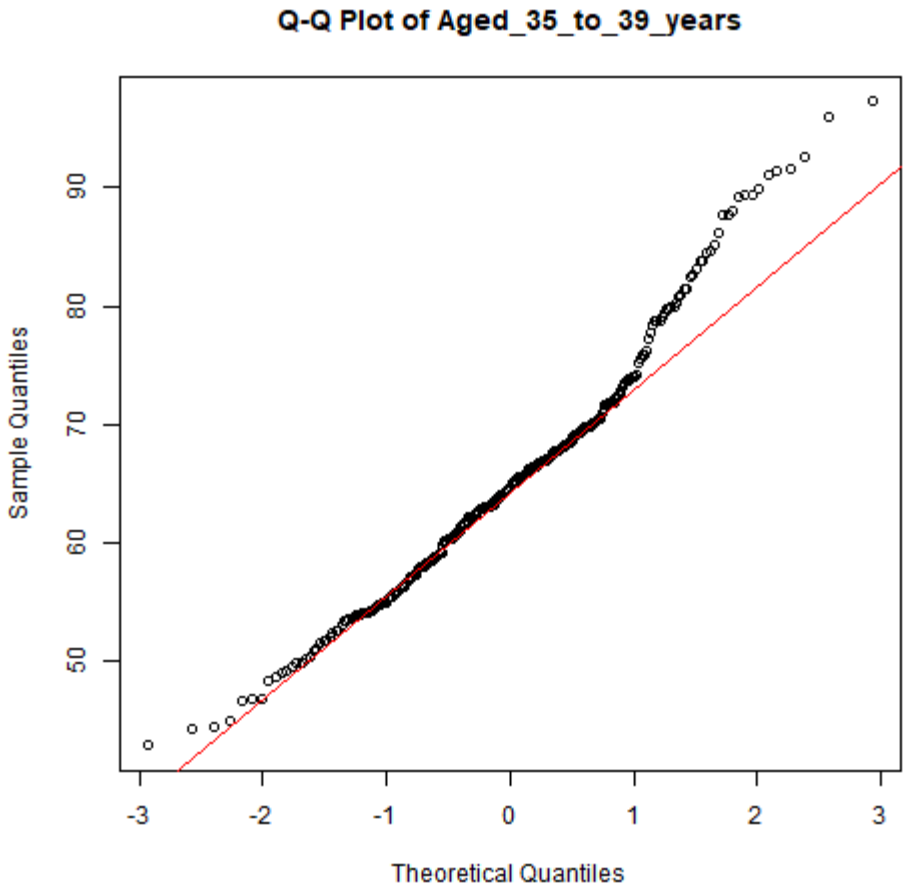

Figure 24 q-q plot of ages 35-39

The Q-Q plot for ages 35 – 39 shows several outliers indicating spike in mortality rates which could be due to work and family responsibilities which increases their social activity thereby exposing them to greater risk.

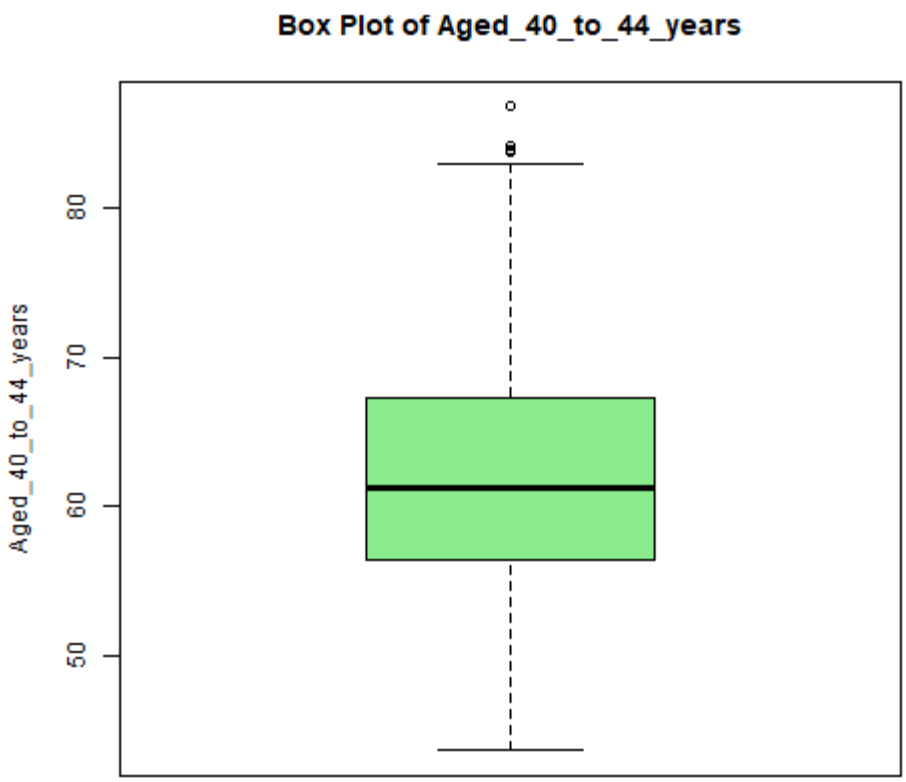

Figure 25 box plot of ages 40-44

The box plot of ages 40-44 shows a visible decline in outliers and a return to normal distribution.

This could be due to factors such as stabilised lifestyle (reduce in parties and unnecessary hang outs), increased health awareness leading to better and healthier lifestyles, regular medical check-ups, etc.

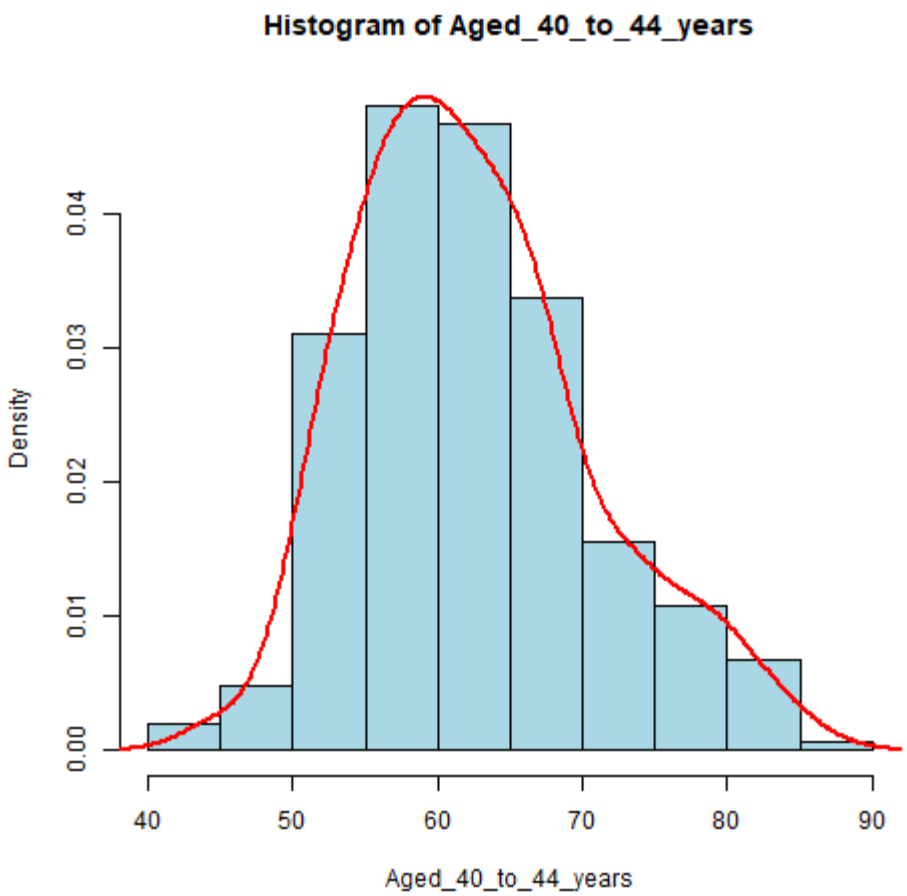

Figure 26 hist of ages 40-44

The histogram of ages 40-44 shows a visible decline in outliers and a return to normal distribution.

This could be due to factors such as stabilised lifestyle (reduce in parties and unnecessary hang outs), increased health awareness leading to better and healthier lifestyles, regular medical check-ups, etc.

Figure 27 q-q plot of ages 40-44

The Q-Q plot of ages 40-44 shows a visible decline in outliers and a return to normal distribution.

This could be due to factors such as stabilised lifestyle (reduce in parties and unnecessary hang outs), increased health awareness leading to better and healthier lifestyles, regular medical check-ups, etc.

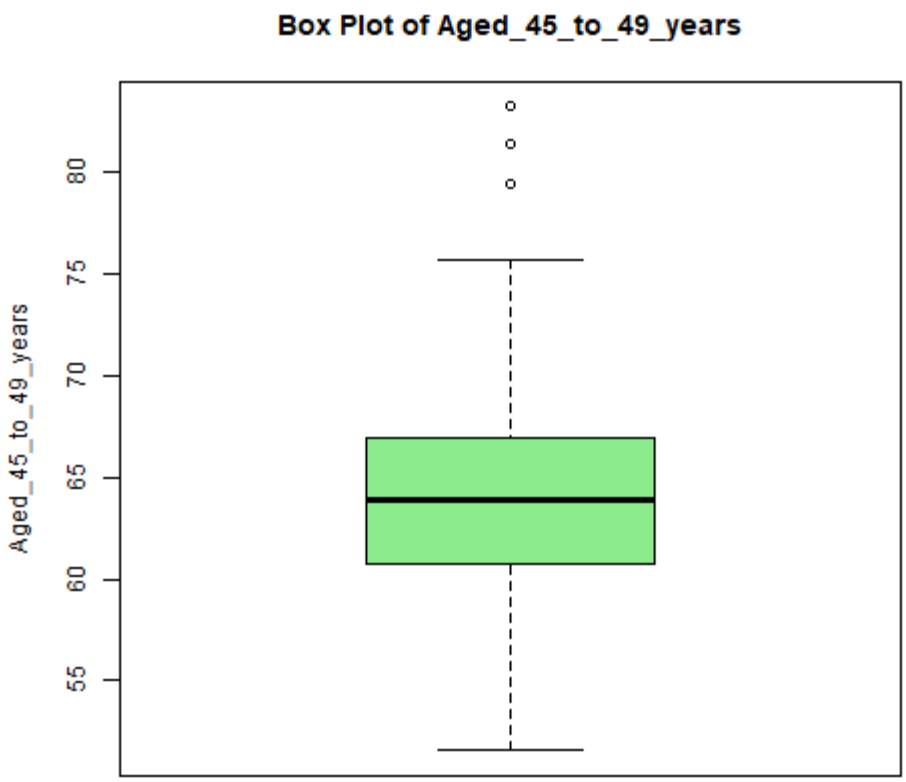

Figure 28 box plot of ages 45-49

The box plot of ages 45-49 shows a visible decline in outliers and a return to normal distribution.

This could be due to factors such as stabilised lifestyle (reduce in parties and unnecessary hang outs), increased health awareness leading to better and healthier lifestyles, regular medical check-ups, etc.

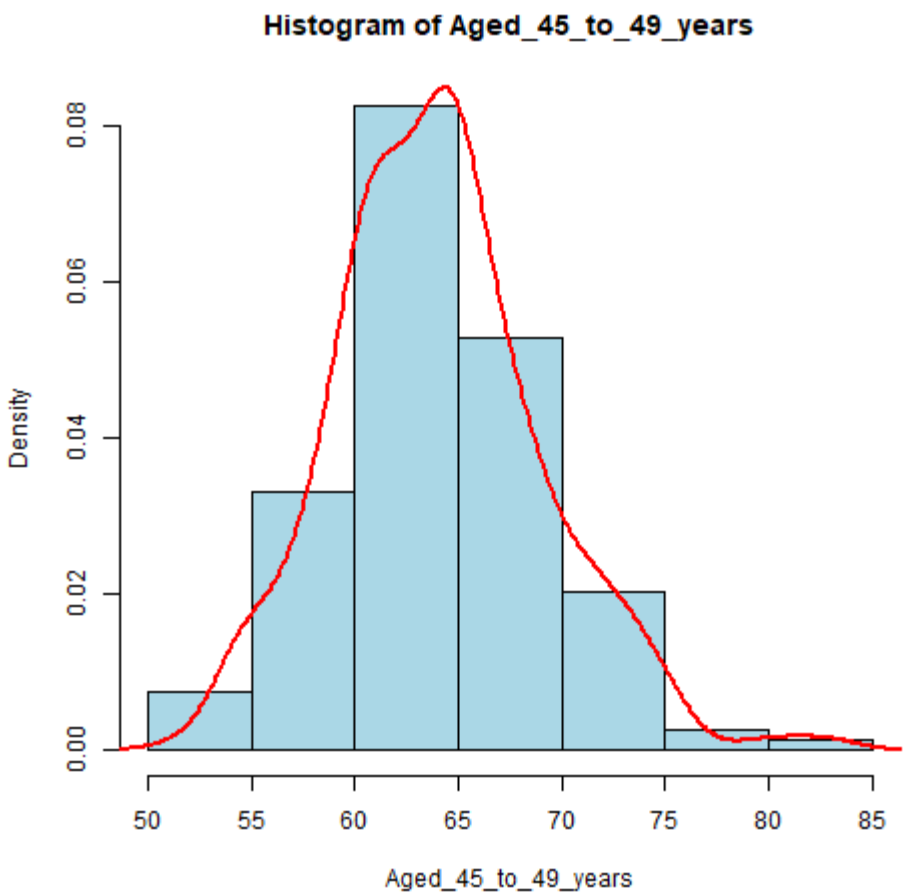

Figure 29 hist of ages 45-49

The histogram of ages 45-49 shows a visible decline in outliers and a return to normal distribution.

This could be due to factors such as stabilised lifestyle (reduce in parties and unnecessary hang outs), increased health awareness leading to better and healthier lifestyles, regular medical check-ups, etc.

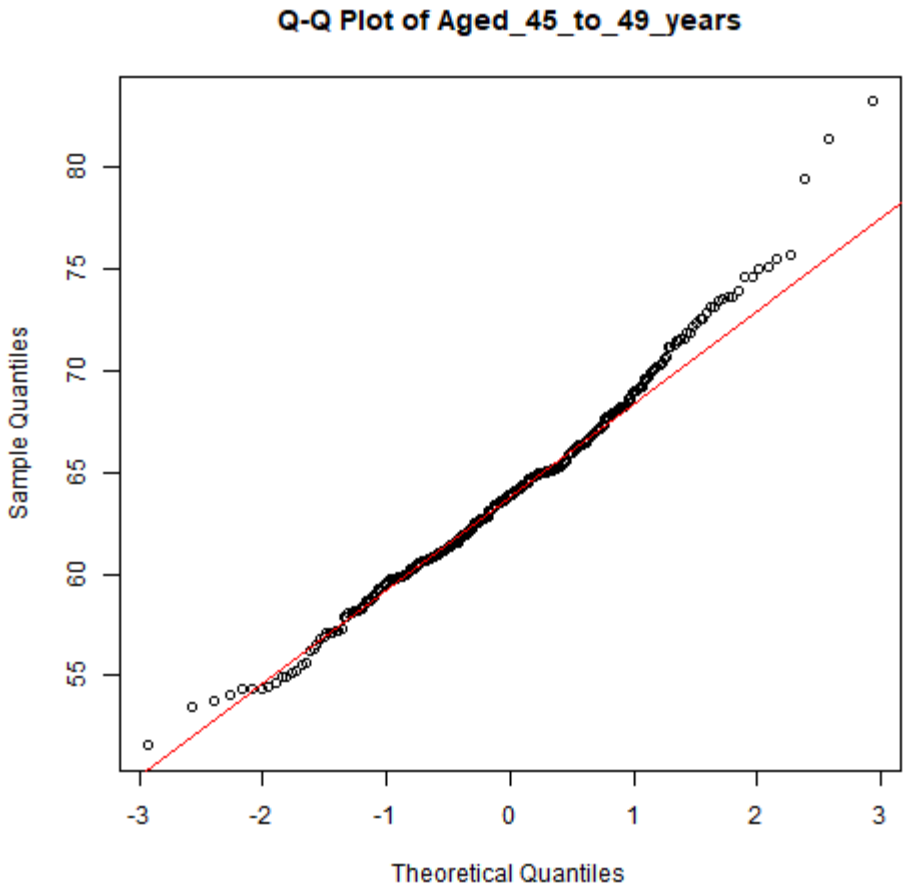

Figure 30 q-q plot of ages 45-49

The Q-Q plot of ages 45-49 shows a visible decline in outliers and a return to normal distribution.

This could be due to factors such as stabilised lifestyle (reduce in parties and unnecessary hang outs), increased health awareness leading to better and healthier lifestyles, regular medical check-ups, etc.

Figure 31 box plot of ages 50-54

The box plot for ages 50-54 shows outliers below which represents a drop in death rates which could be due to some factors such as effective protection due to vulnerable status as elder citizens, lower population density, higher vaccination, working from home or pensioners who stay home full time, better access to health care and reduction in smoking and drinking.

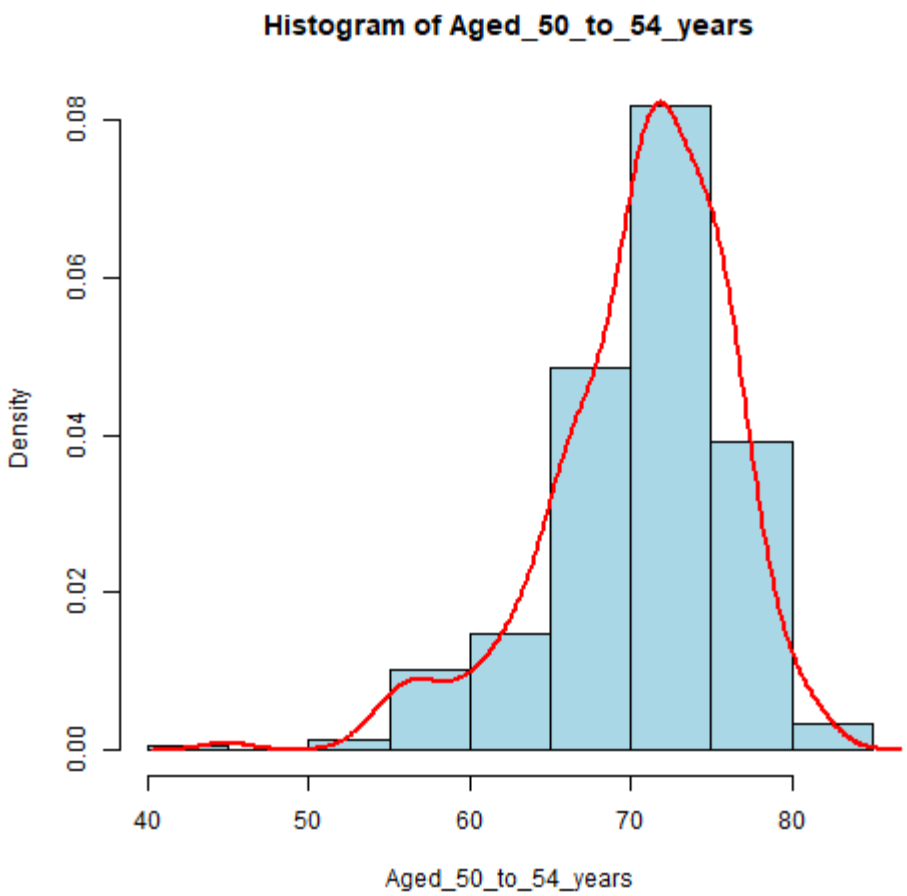

Figure 32 hist of ages 50-54

The histogram for ages 50-54 shows outliers below which represents a drop in death rates which could be due to some factors such as effective protection due to vulnerable status as elder citizens, lower population density, higher vaccination, working from home or pensioners who stay home full time, better access to health care and reduction in smoking and drinking.

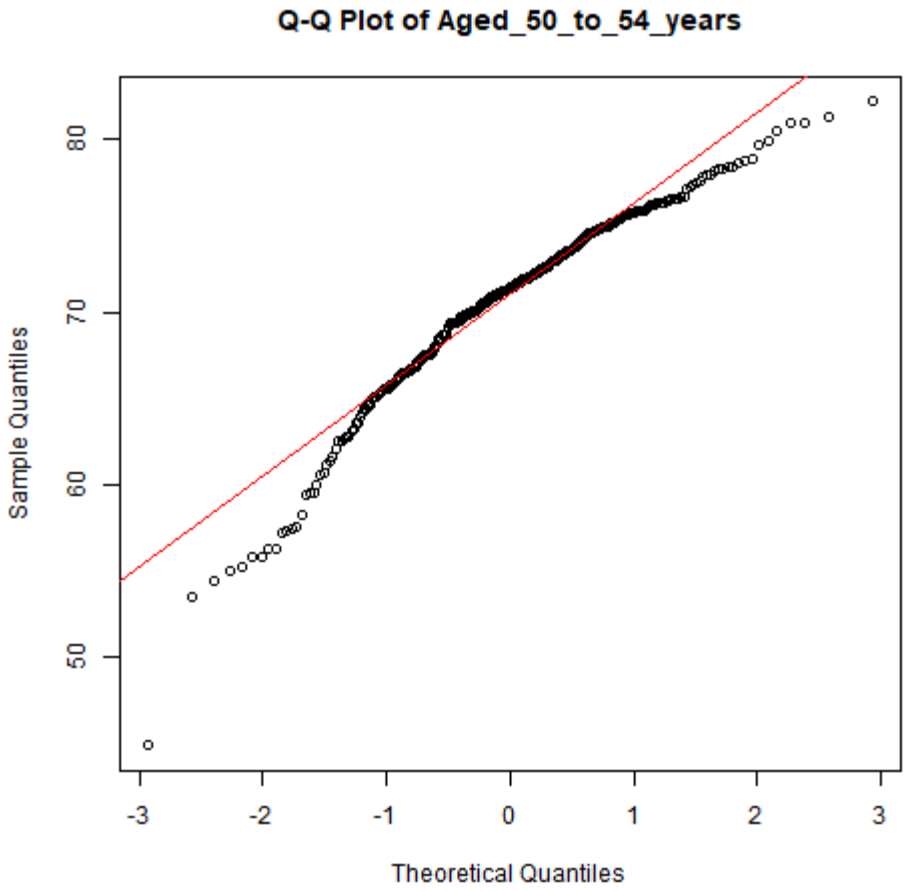

Figure 33 q-q plot of ages 50-54

The Q-Q plot for ages 50-54 shows outliers below which represents a drop in death rates which could be due to some factors such as effective protection due to vulnerable status as elder citizens, lower population density, higher vaccination, working from home or pensioners who stay home full time, better access to health care and reduction in smoking and drinking.

Figure 34 box plot of ages 55-59

The box plot for ages 55-59 shows outliers below which represents a drop in death rates which could be due to some factors such as effective protection due to vulnerable status as elder citizens, lower population density, higher vaccination, working from home or pensioners who stay home full time, better access to health care and reduction in smoking and drinking.

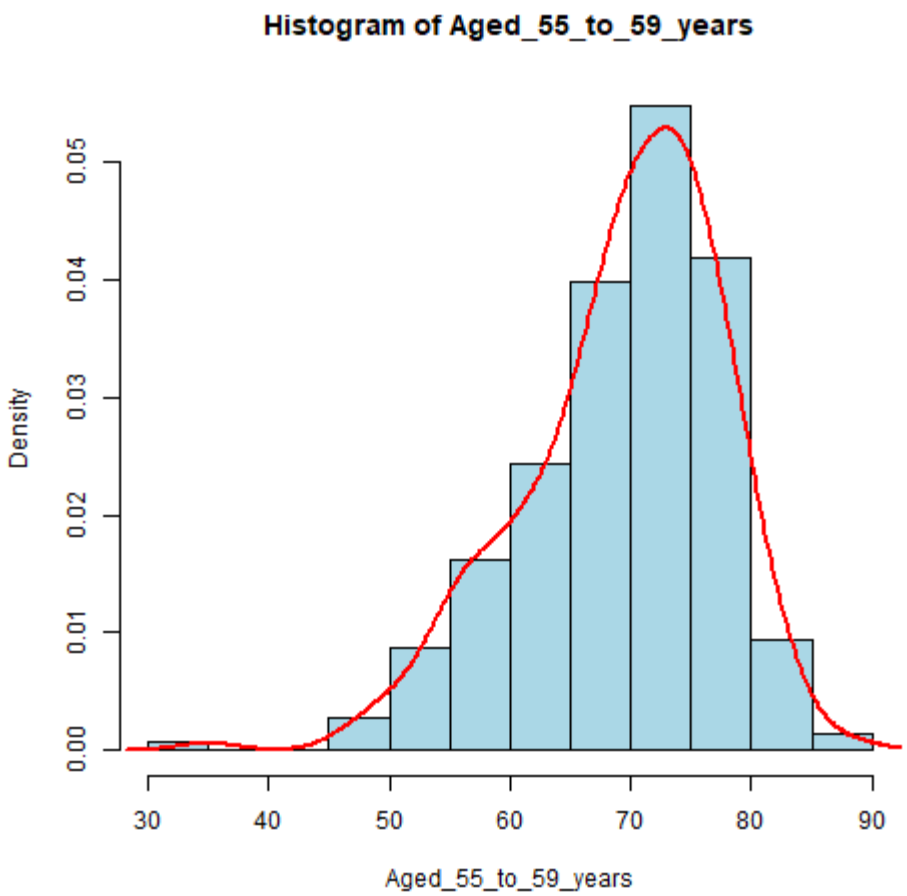

Figure 35 hist of ages 55-59

The histogram for ages 55-59 shows outliers below which represents a drop in death rates which could be due to some factors such as effective protection due to vulnerable status as elder citizens, lower population density, higher vaccination, working from home or pensioners who stay home full time, better access to health care and reduction in smoking and drinking.

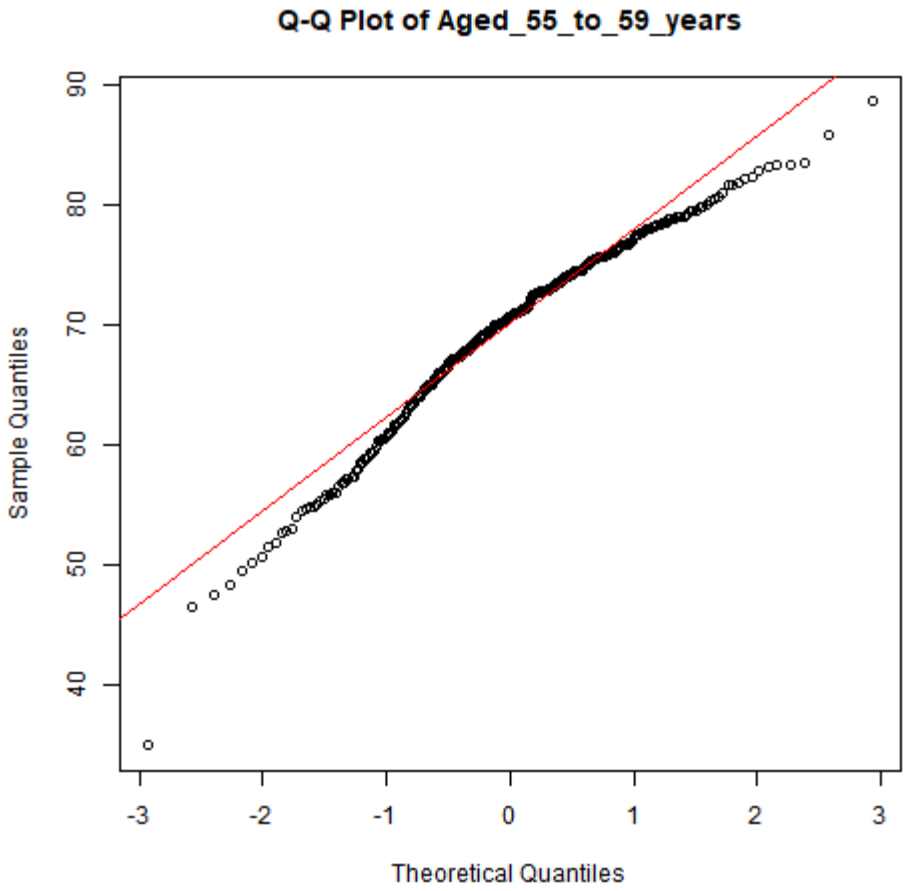

Figure 36 q-q plot of ages 55-59

The Q-Q plot for ages 55-59 shows outliers below which represents a drop in death rates which could be due to some factors such as effective protection due to vulnerable status as elder citizens, lower population density, higher vaccination, working from home or pensioners who stay home full time, better access to health care and reduction in smoking and drinking.

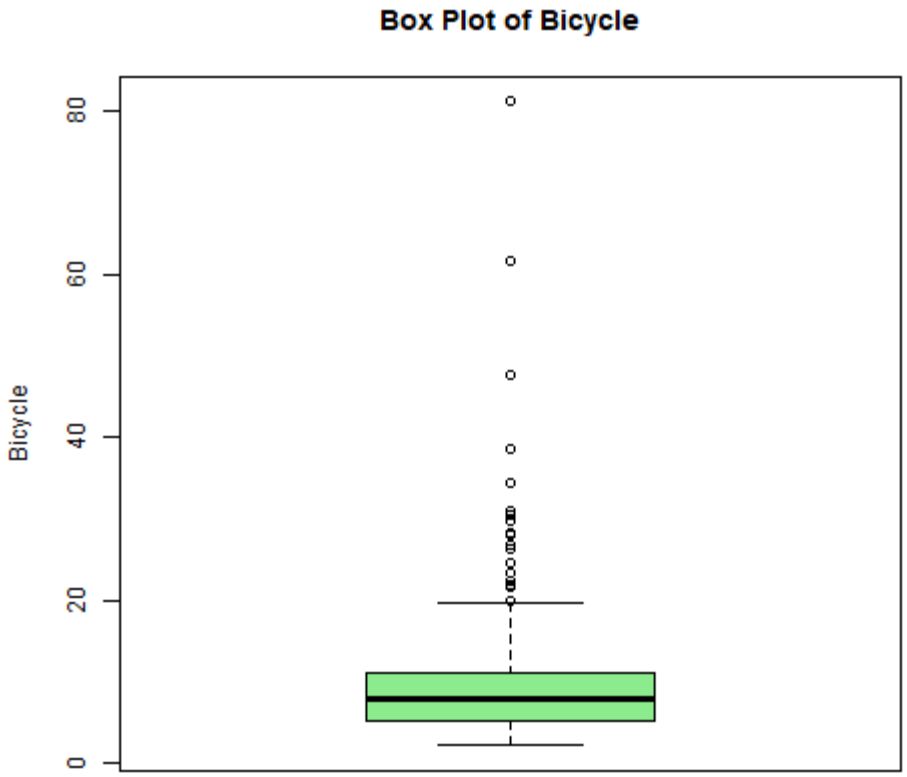

Figure 55 box plot of bicycle

The box plot of covid deaths for bicycle users shows so many upper outliers indicating increased mortality rates which could be due to exposure to contact and airborne viruses because bicycles are not protected with doors and windows.

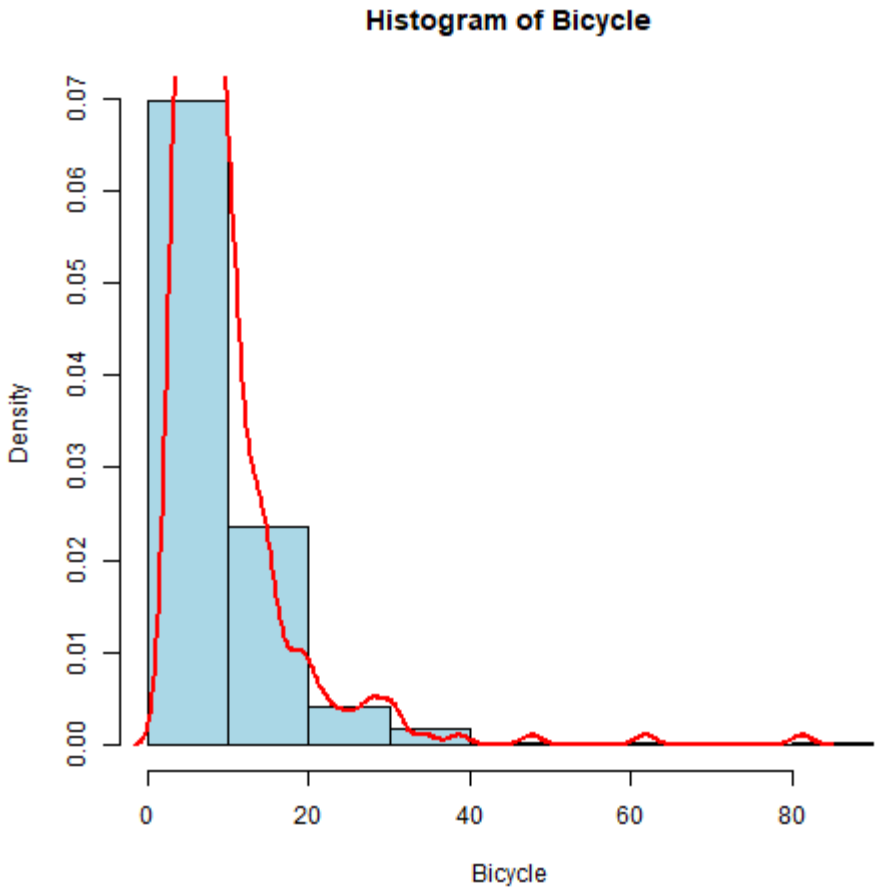

Figure 56 hist of bicycle

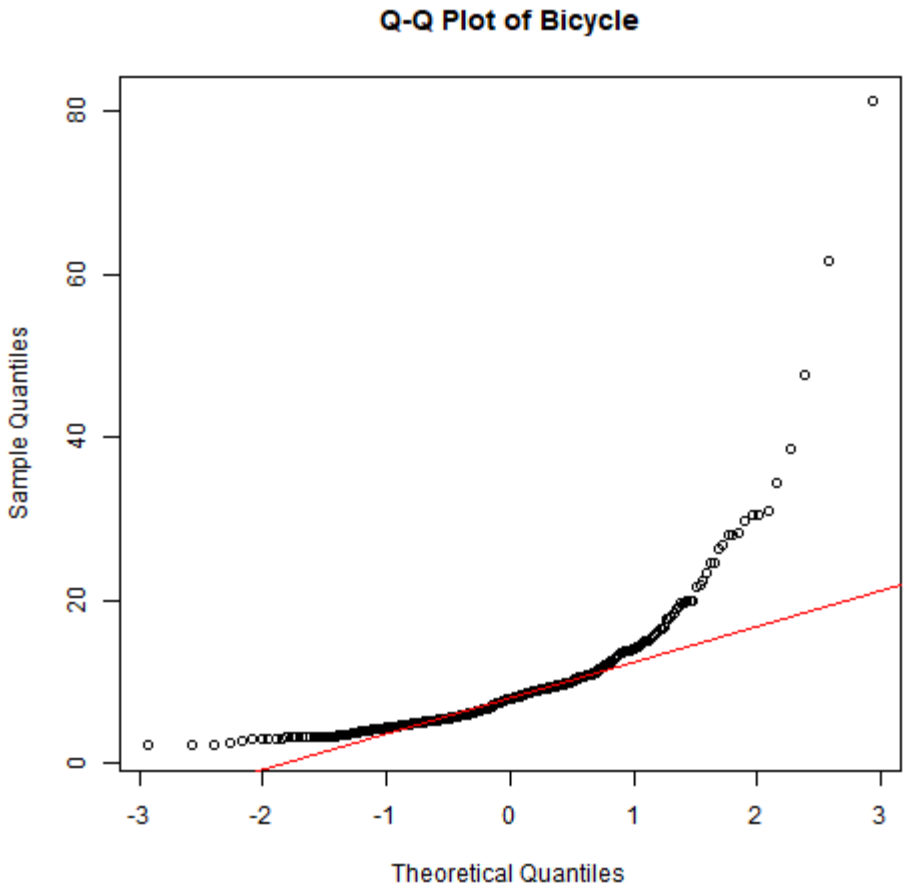

Figure 57 q-q plot of bicycle

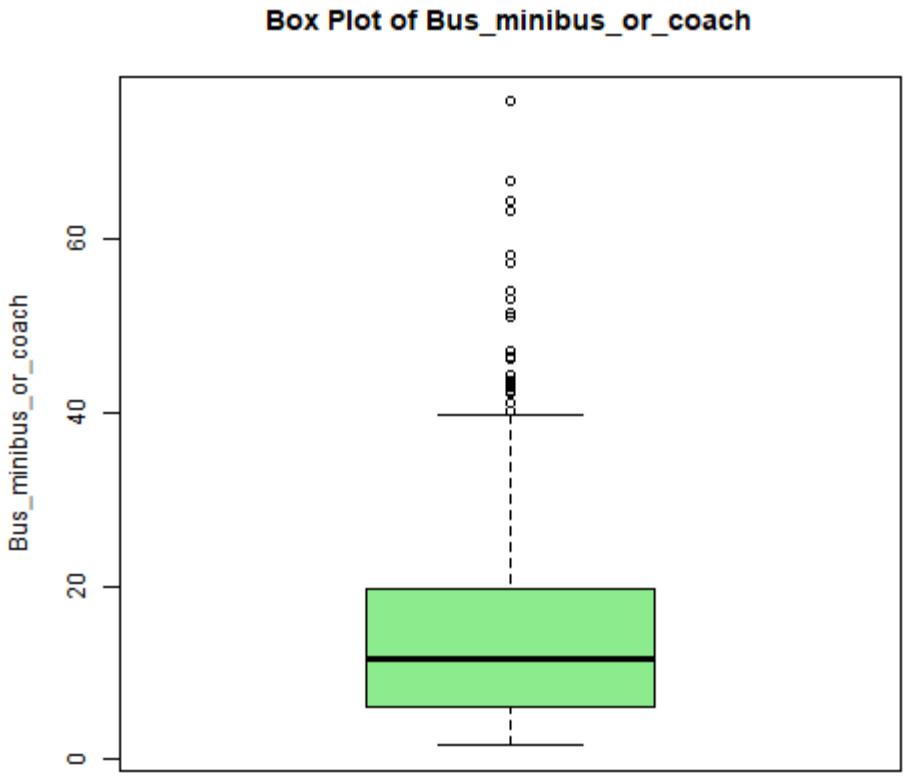

Figure 58 box plot of bus, minibus or coach

The box plot of covid deaths for minibuses or coaches shows many upper outliers indicating higher mortality rates which could be due to human-to-human contact in buses and coaches used for public transport.

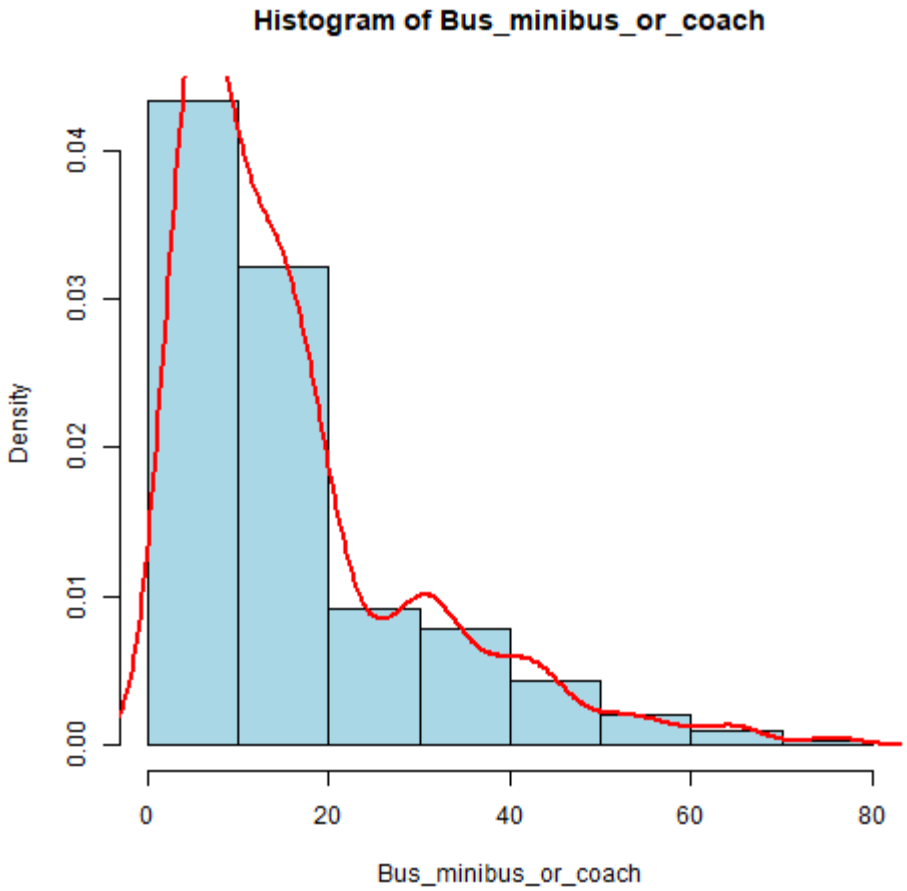

Figure 59 hist of bus, minibus or coach

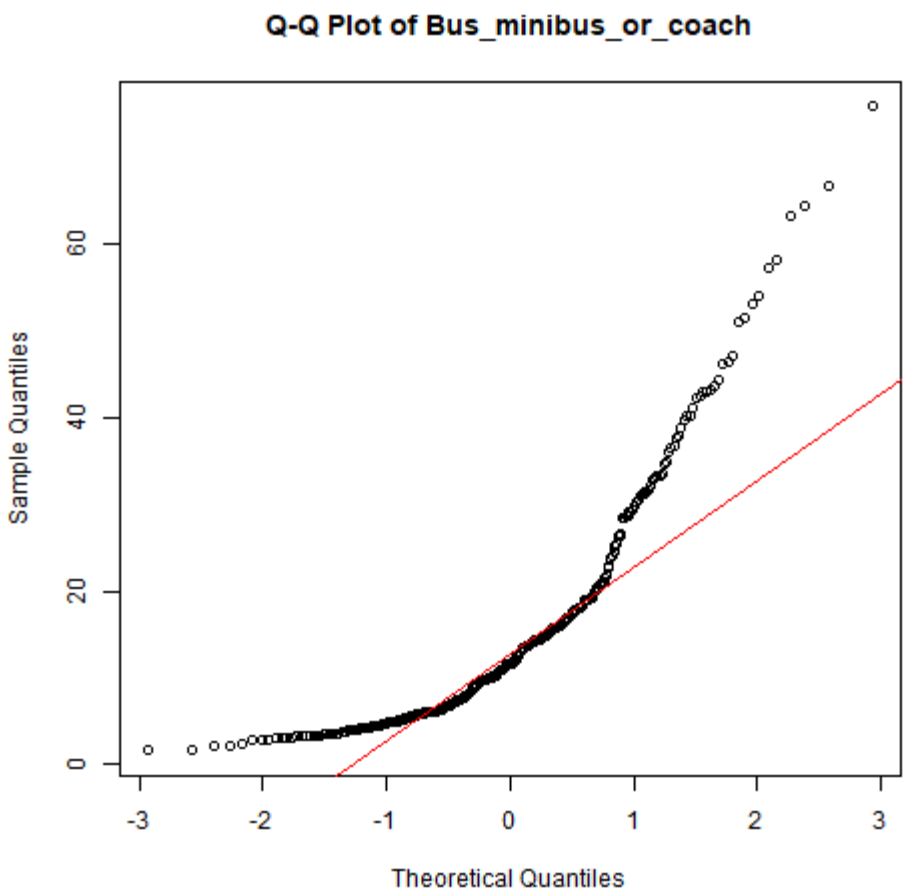

Figure 60 q-q plot of bus, minibus or coach

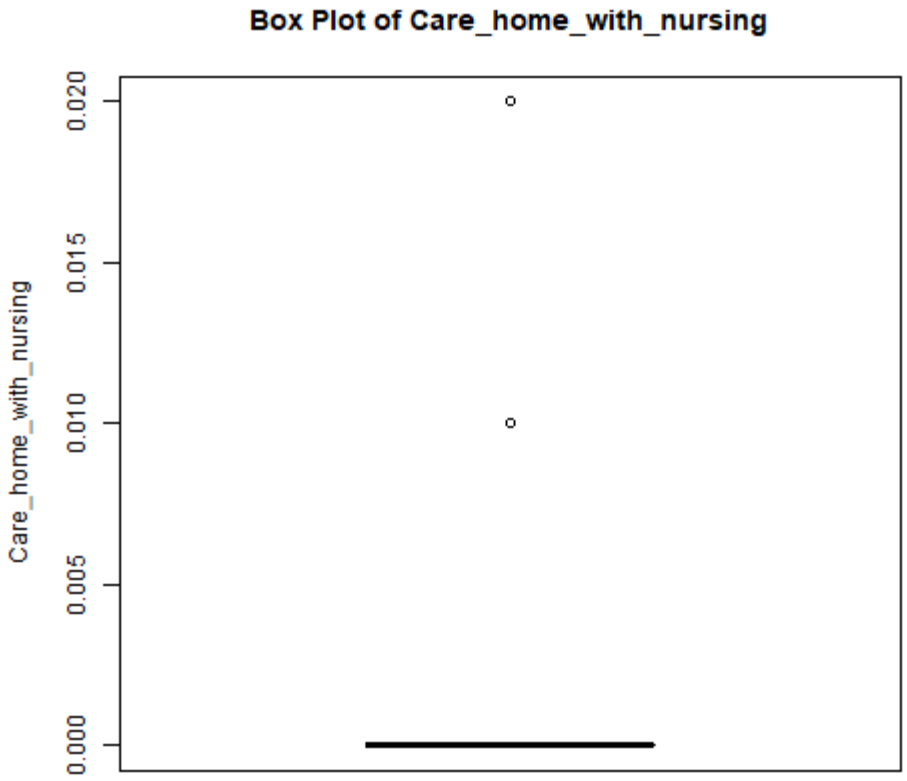

Figure 61 box plot of care home with nursing

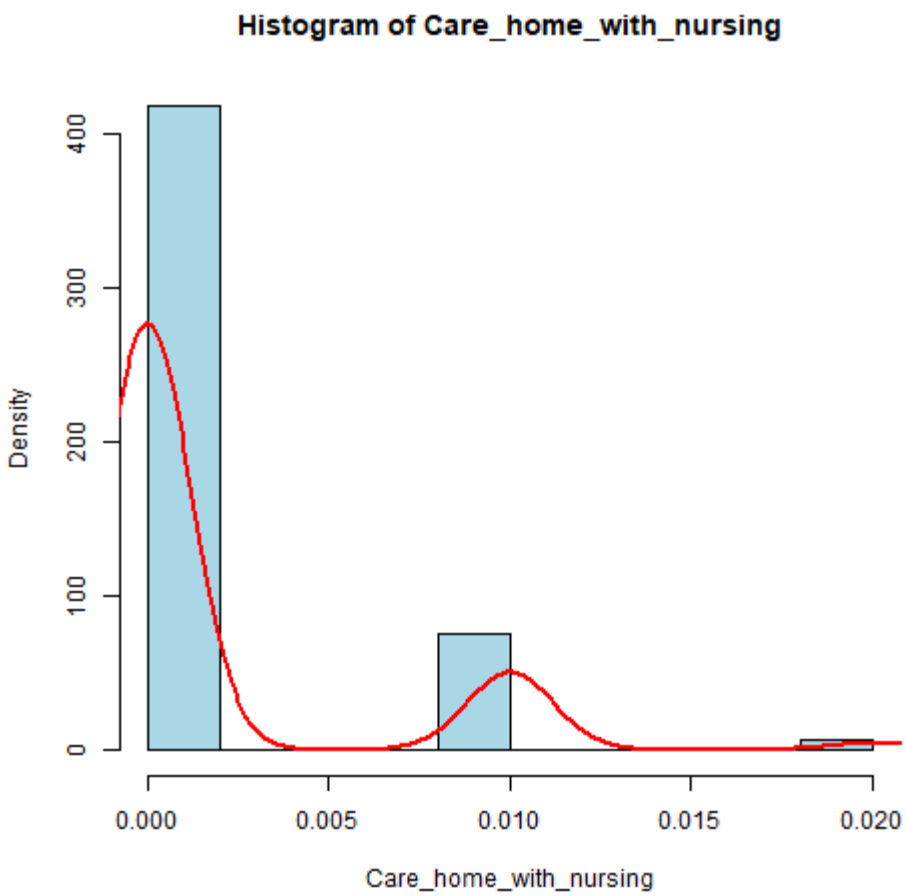

Figure 62 hist of care home with nursing

The covid death histogram of care home with nursing shows an uneven distribution which shows that in a care home, different factors affect different patients od different health backgrounds and medical and lifestyle history, thereby making death rates uneven.

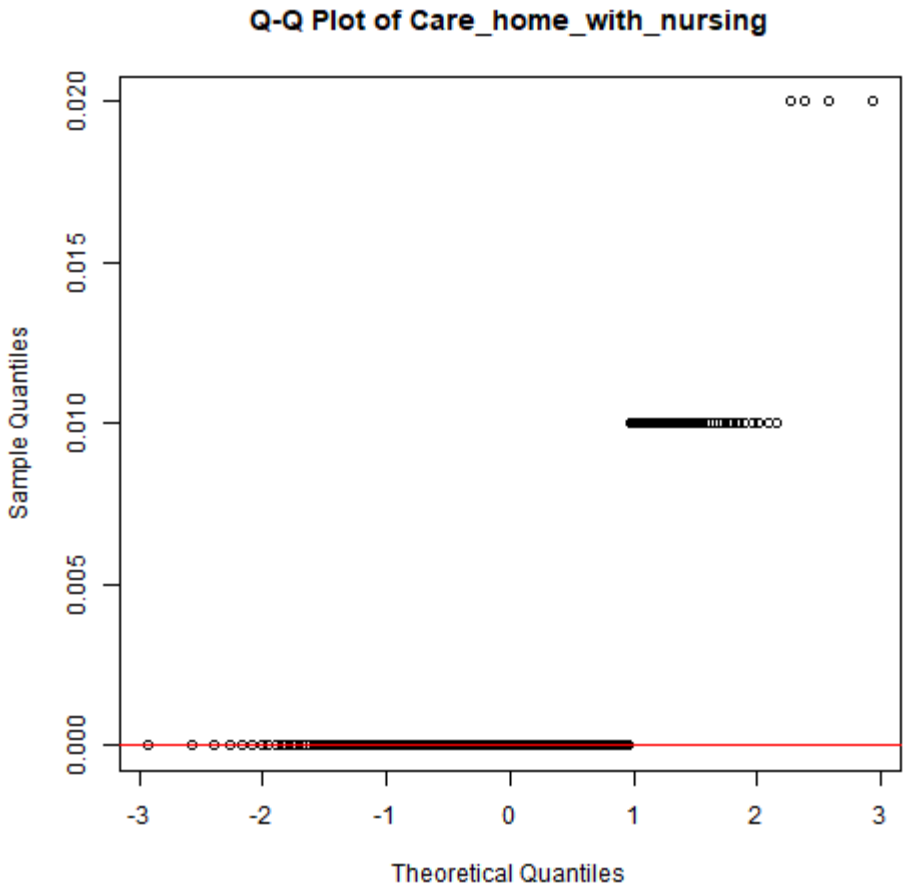

Figure 63 q-q plot of care home with nursing

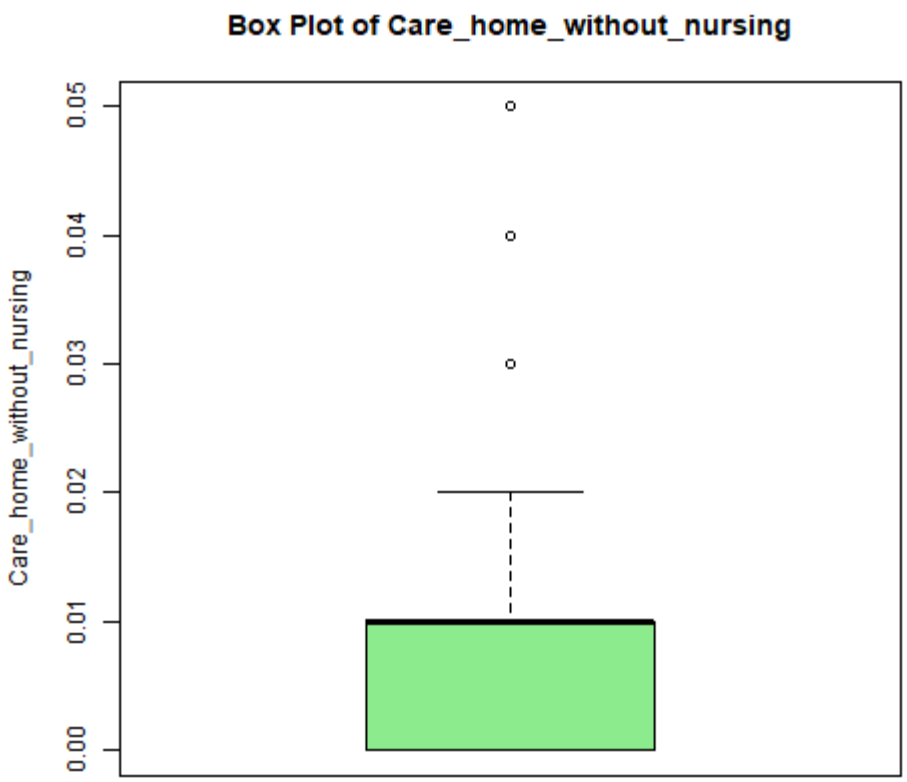

Figure 64 box plot of care home without nursing

The box plot of care home without nursing shows some higher outliers indicating increase in mortality rate which means that the absence of nurses in care homes during the covid outbreak led to increased death rates.

Nurses are very important in the preservation of life.

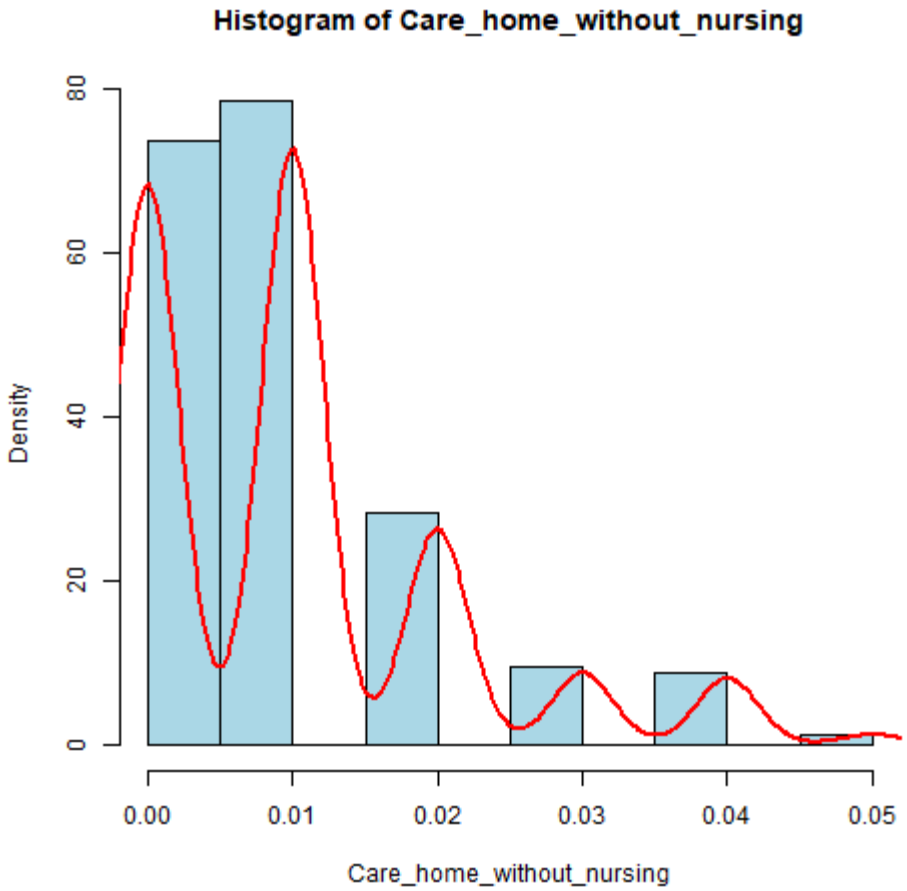

Figure 65 hist of care home without nursing

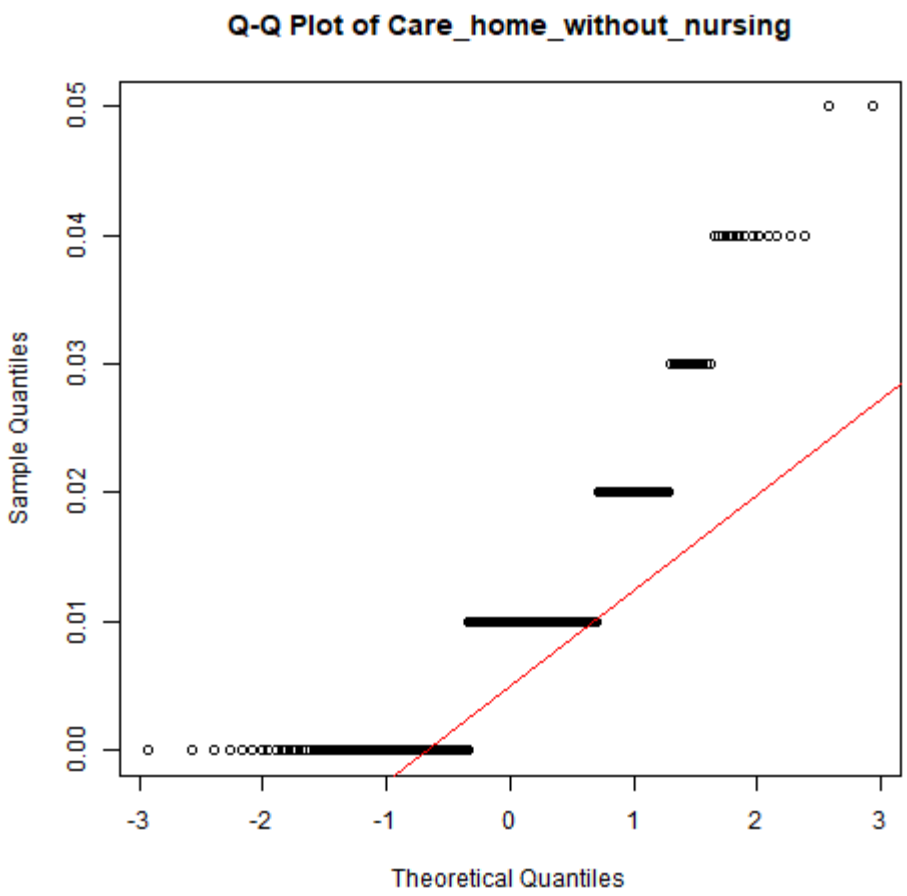

Figure 66 q-q plot of care home without nursing

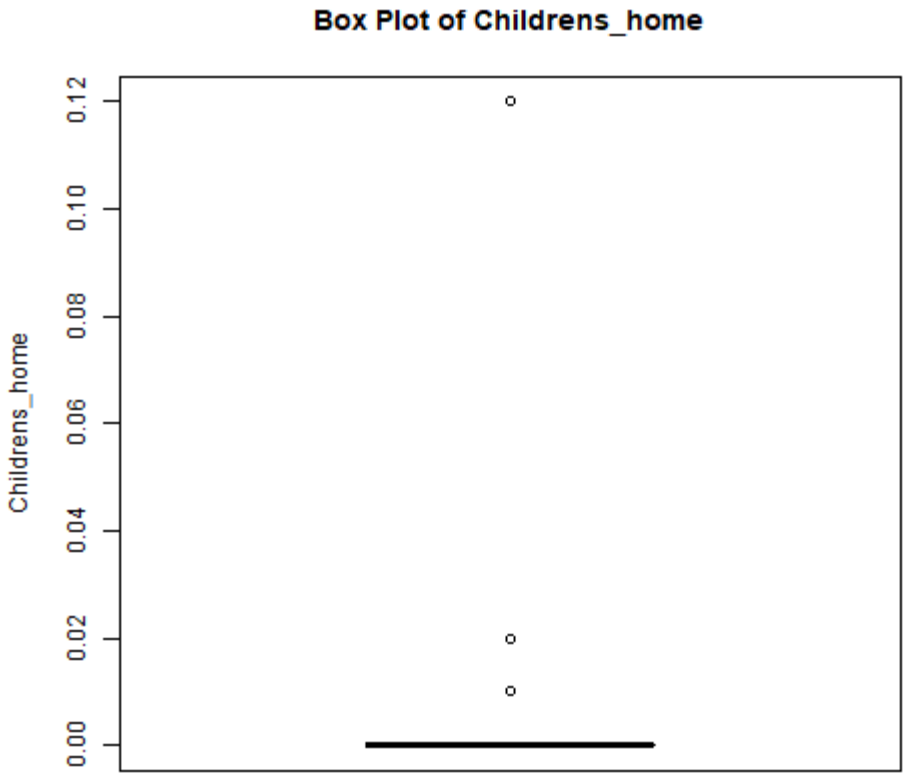

Figure 67 box plot of children's home

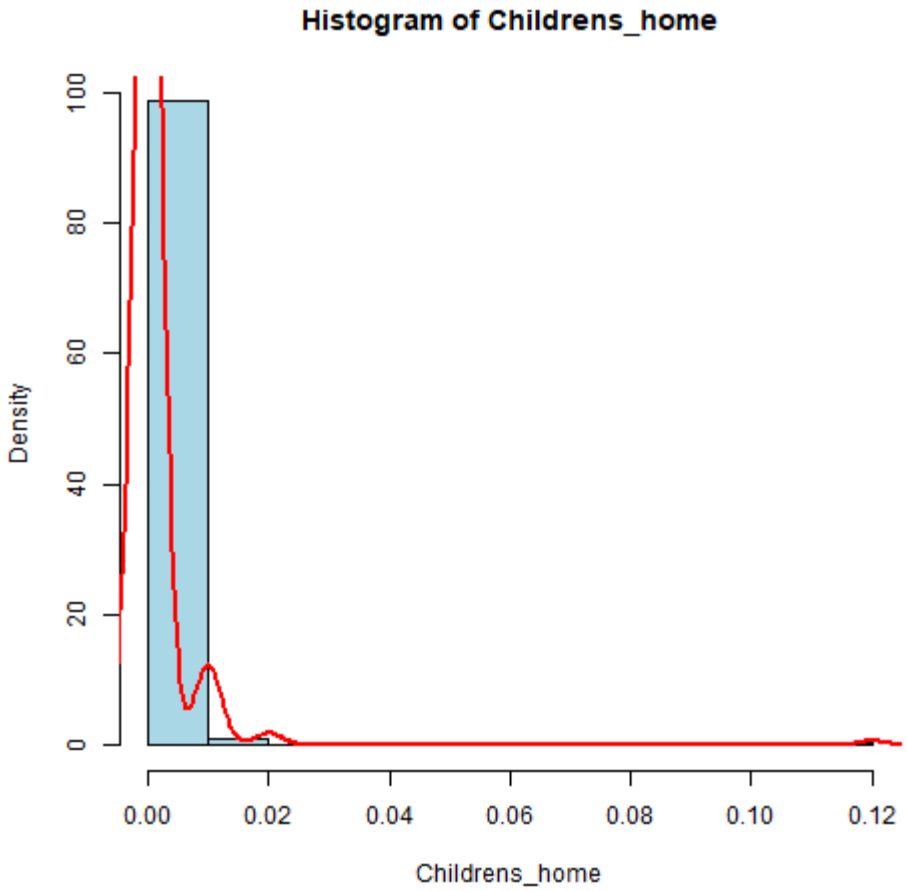

Figure 68 hist of children's home

The histogram of covid death in children's home shows a sharp spike in mortality at early childhood followed by a plateau reduction.

This is understandable because early age children are very vulnerable during disease outbreaks because their immune systems are yet to fully develop.

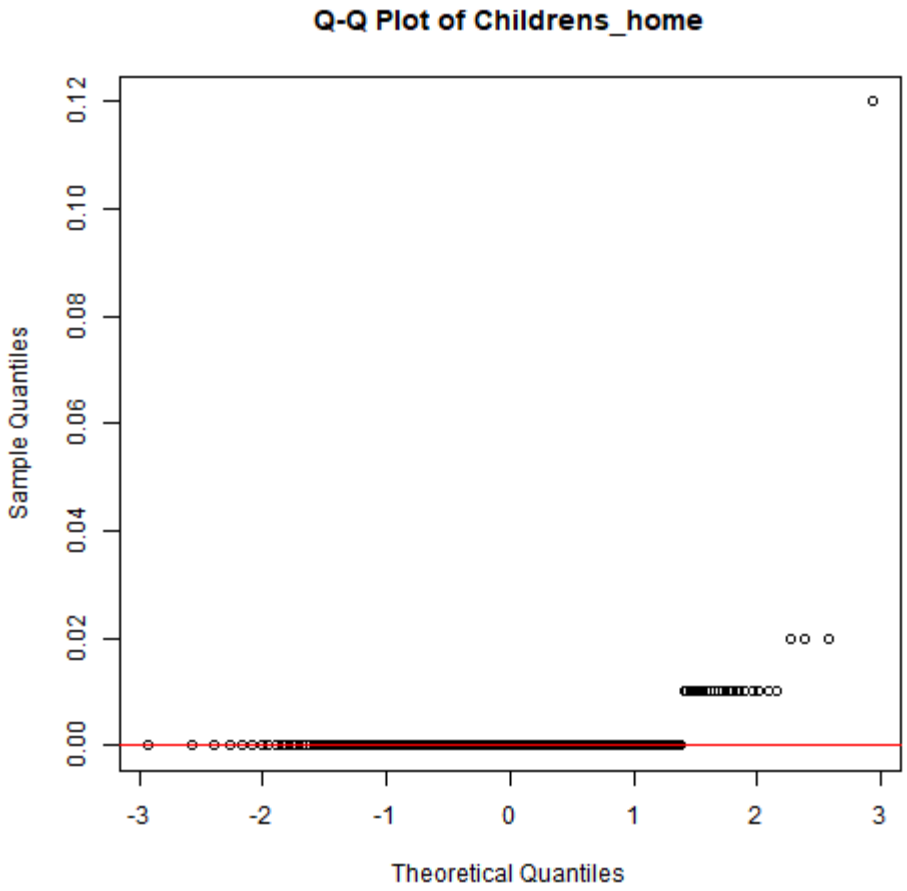

Figure 69 q-q plot of children's home

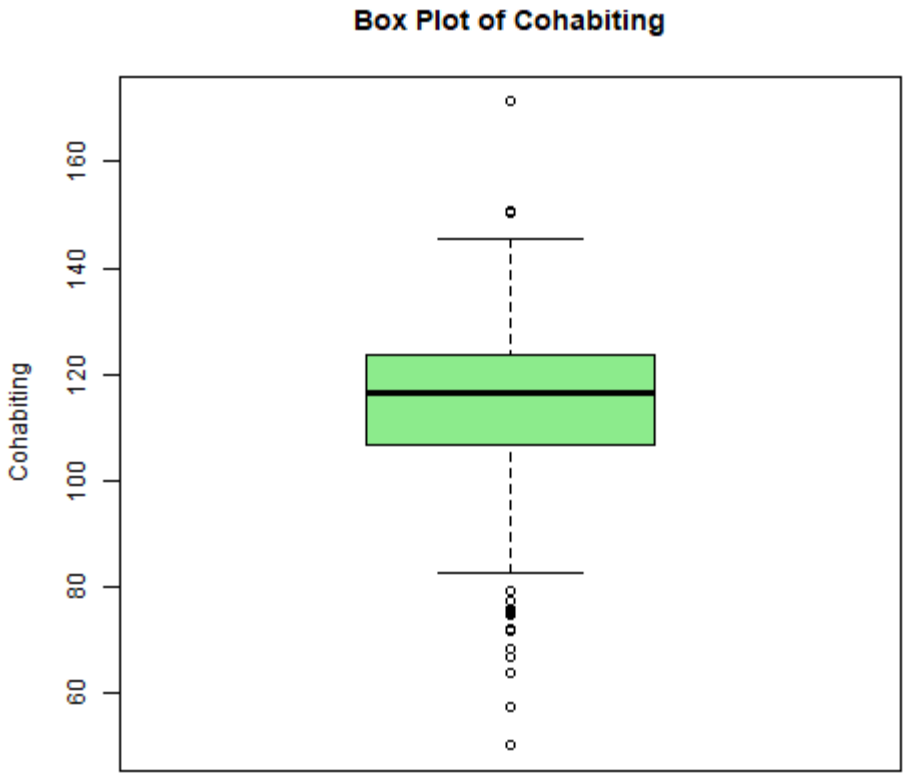

Figure 70 box plot of cohabiting

The box plot of covid deaths for cohabiting couples shows lower outliers which means a decrease in mortality rate due to some factors such as the couples supporting each other and taking care of one another, ensuring they take their medications and working from home. Also, the emotional support they give one another cannot be overlooked.

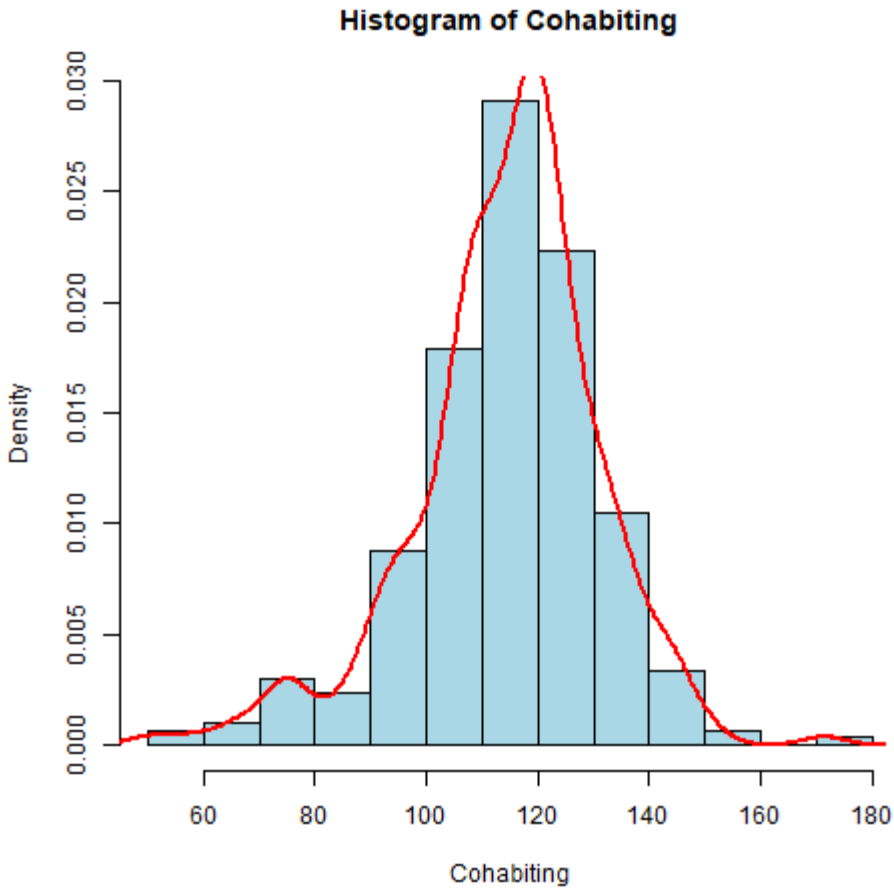

Figure 71 hist of cohabiting

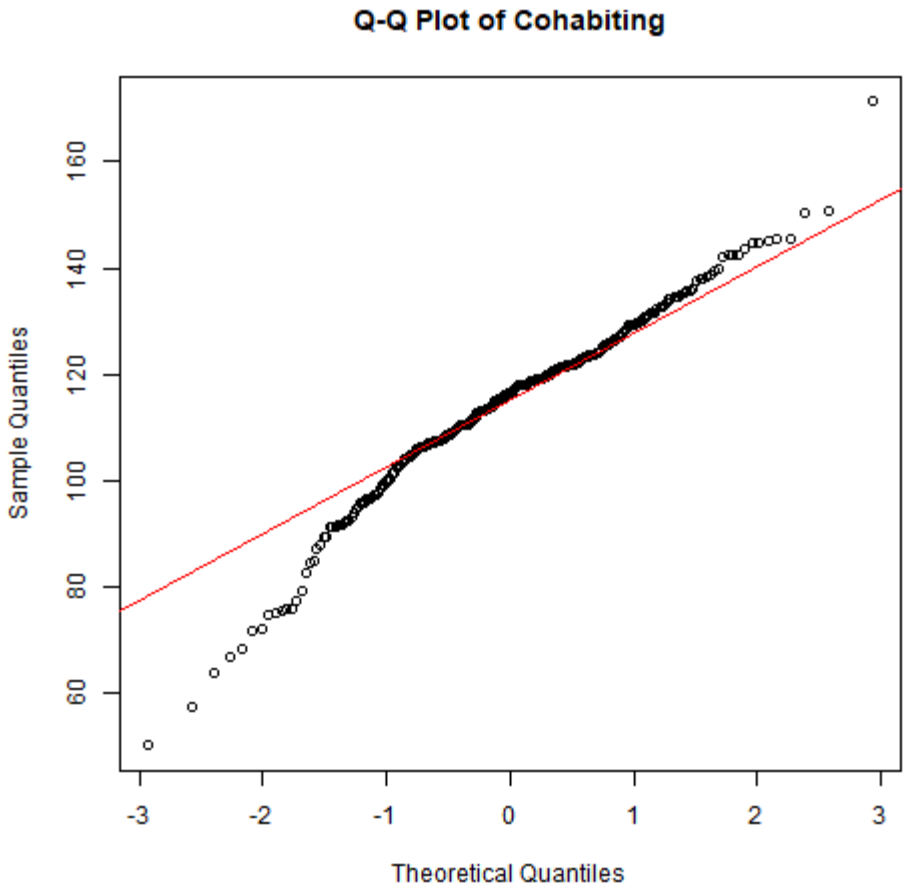

Figure 72 q-q plot of cohabiting

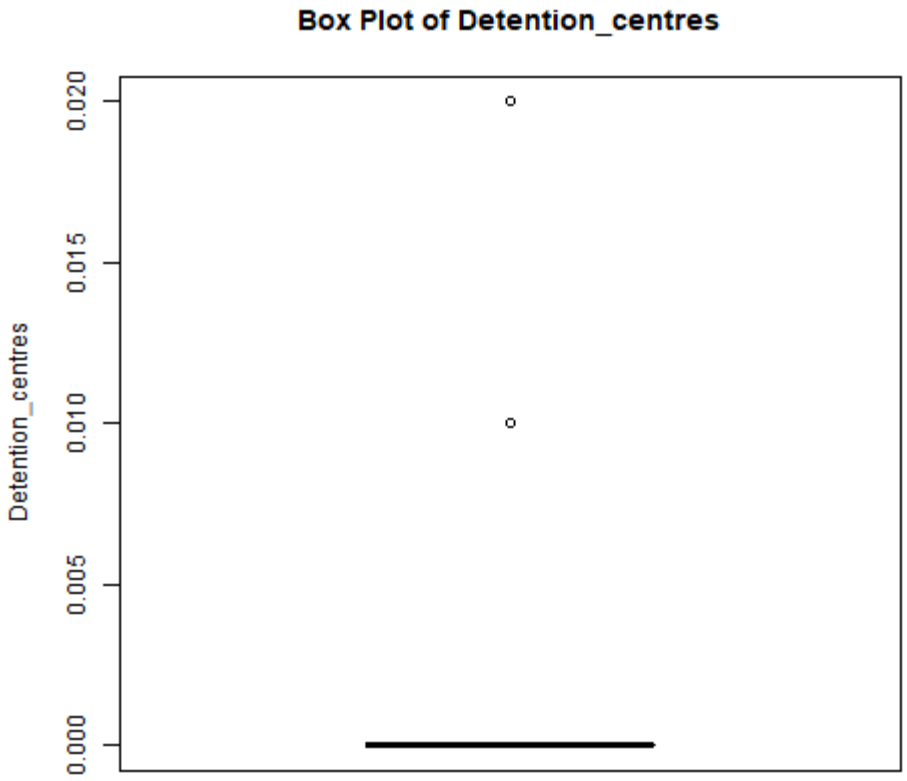

Figure 73 box plot of detention centres

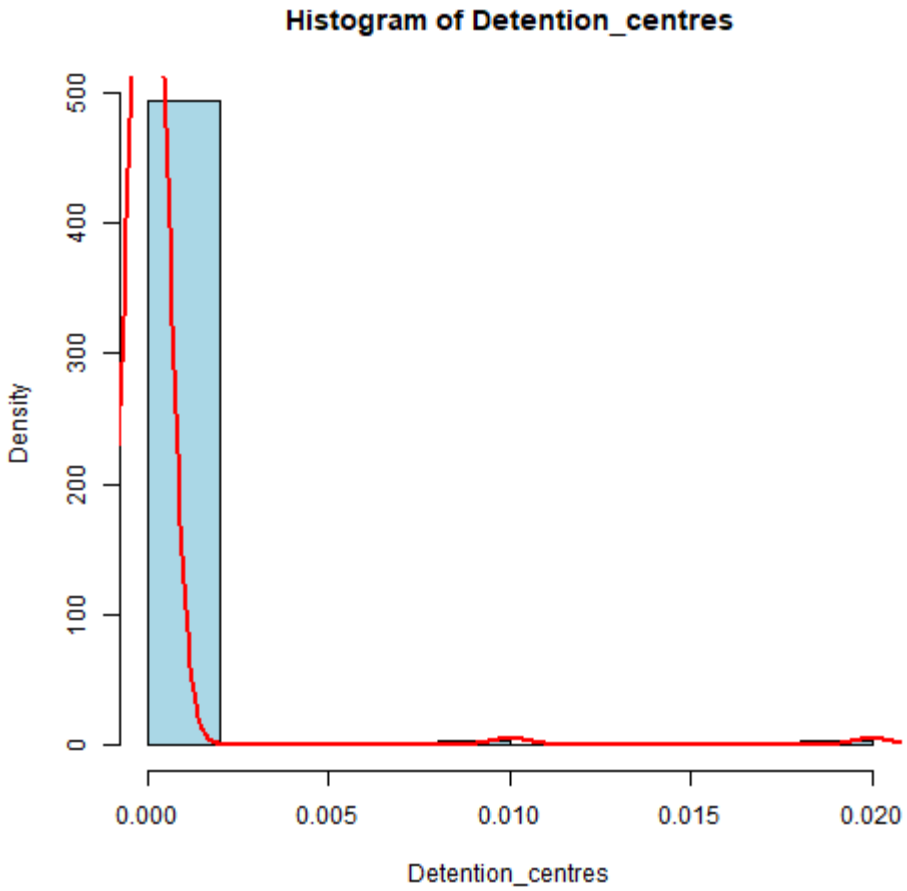

Figure 74 hist of detention centres

The histogram of covid deaths for people in detention centres shows a sharp spike in mortality rate followed by a plateaued reduction. The high mortality rate belongs to a class of people with distinct health and lifestyle patterns and different immune systems (they could be young or elderly inmates).

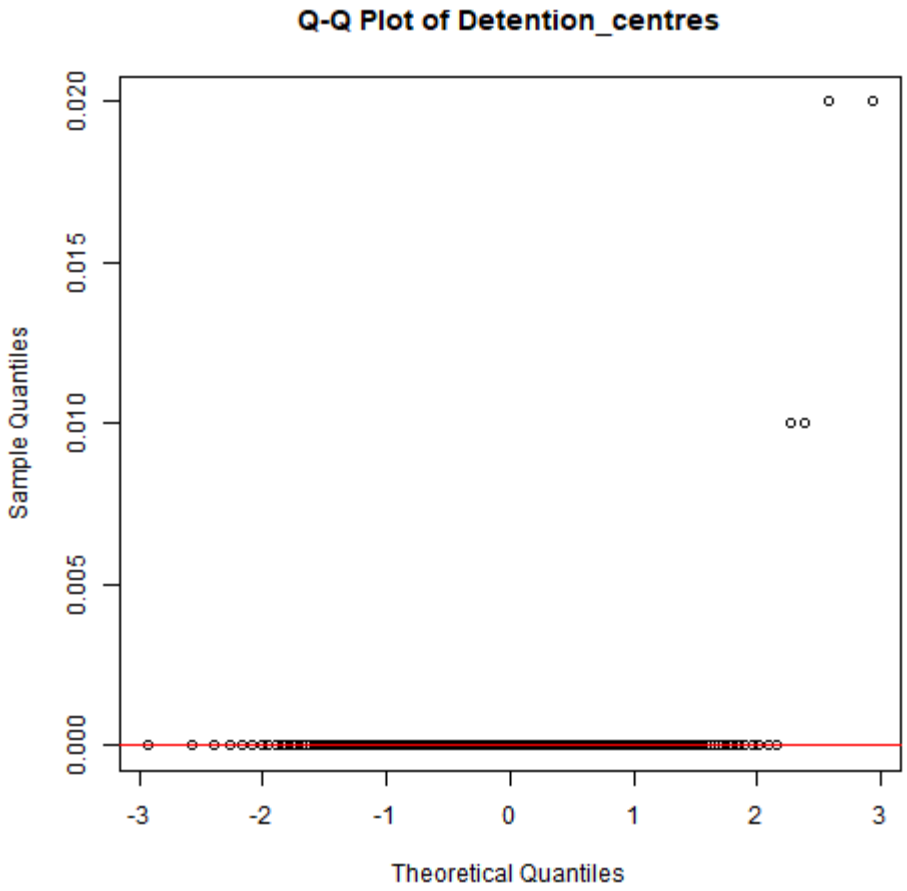

Figure 75 q-q plot of detention centres

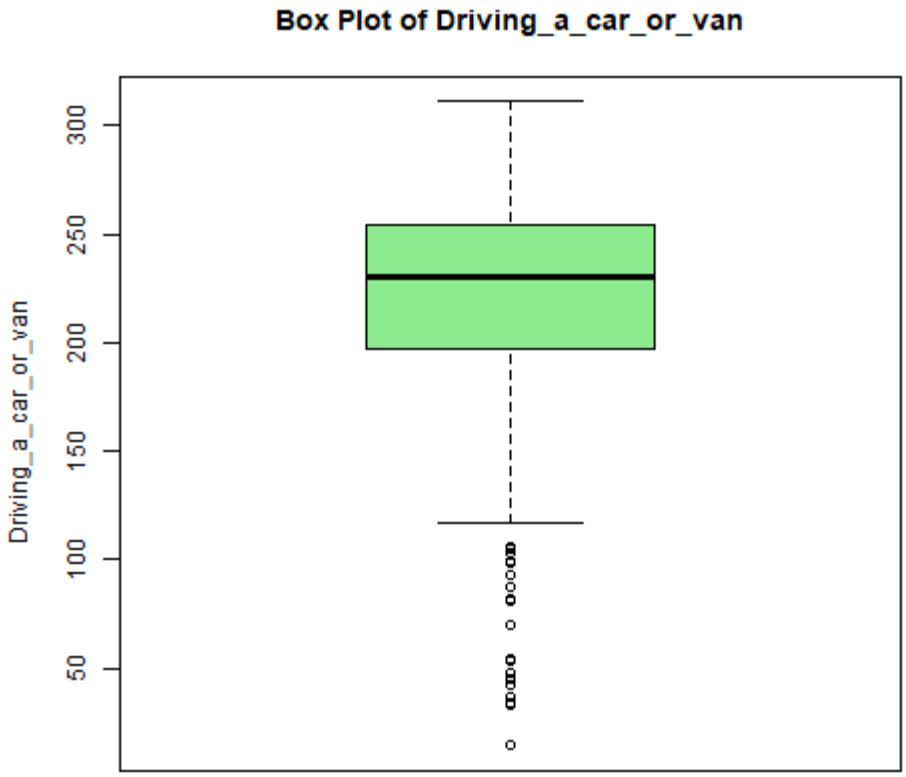

Figure 76 box plot of driving a car or van

The box plot of covid deaths for people driving a car or van shows downward outliers which indicate a decrease in mortality which could be due to the health advantages that come with private vehicular ownership which means reduced social interaction, reduced stranger body contact and privacy. Also, people with private vehicles are likely to earn a good income that will enable them to afford quality healthcare.

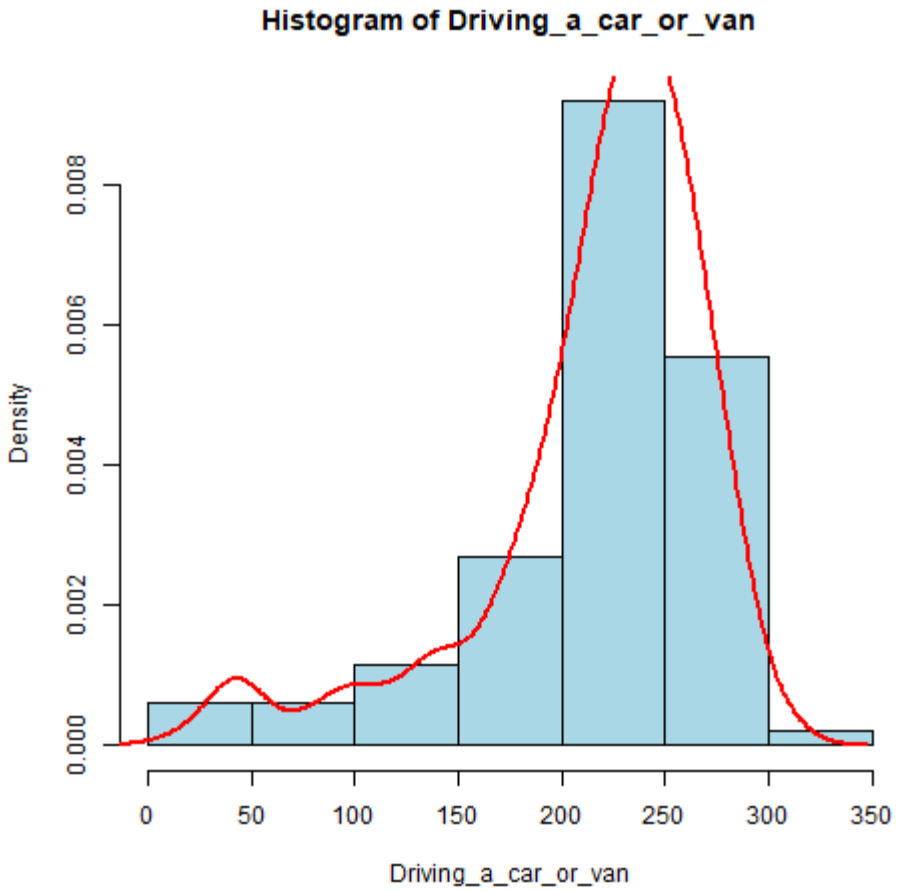

Figure 77 hist of driving a car or van

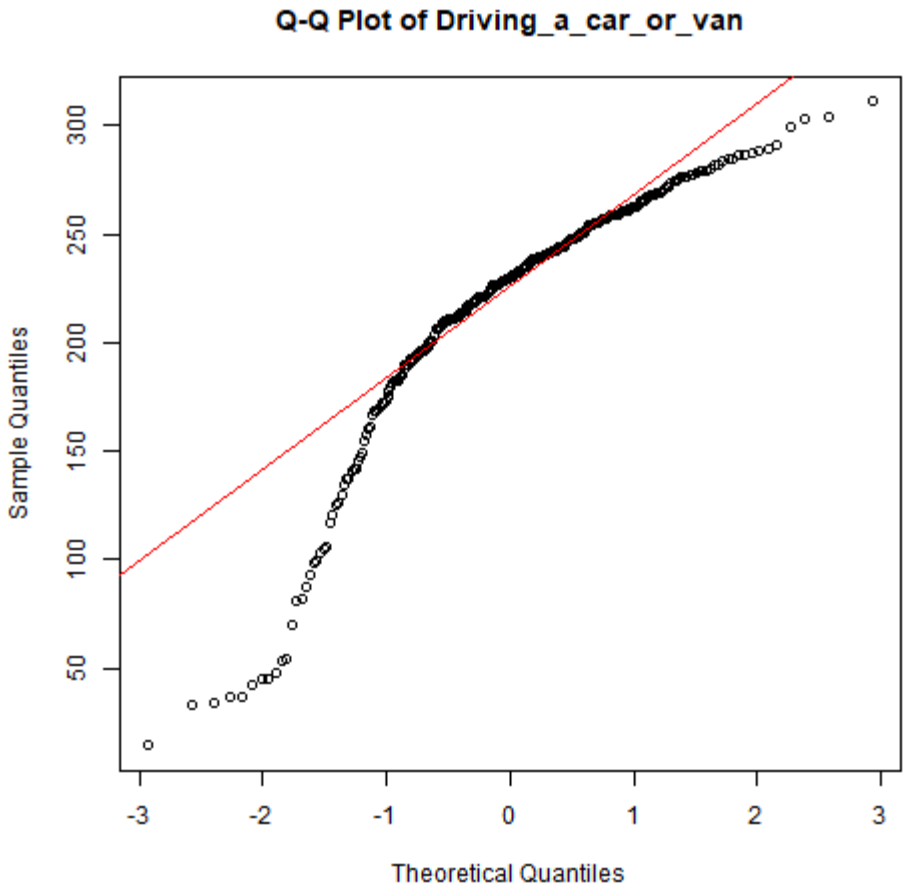

Figure 78 q-q plot of driving a car or van

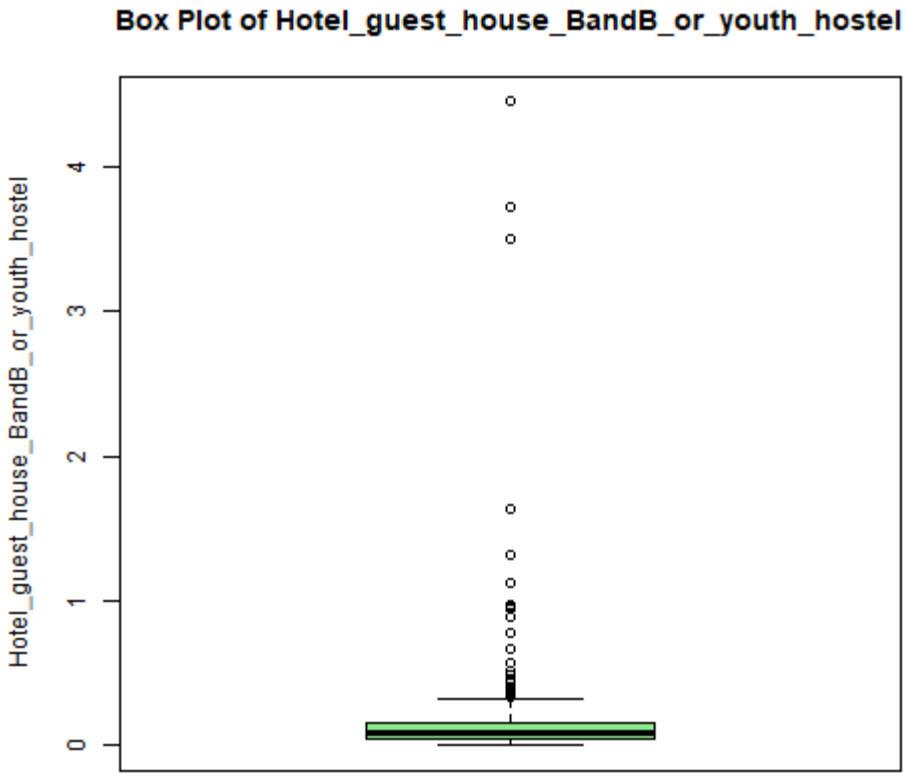

Figure 83 box plot of hotel, guest house or bnb

The box plot of covid deaths for people living in guest houses or youth hostels show several upper outliers indicating spikes in mortality rates due to factors like poor hygienic conditions, overcrowded facilities leading to stranger contacts and transmission of the virus.

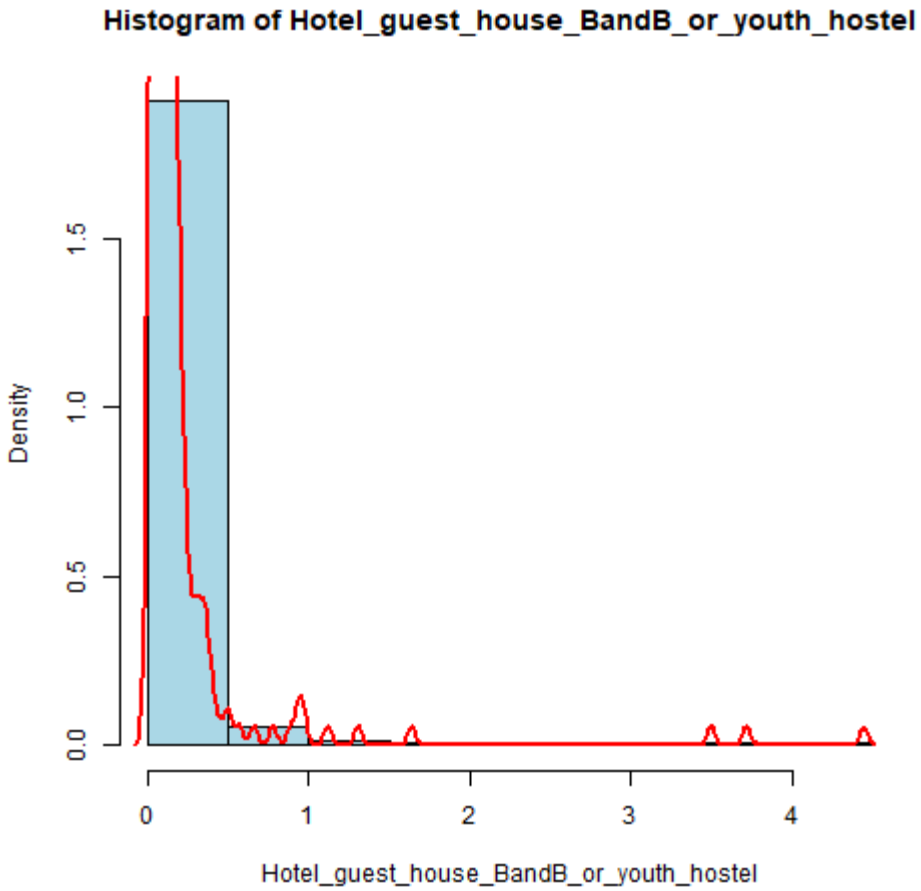

Figure 84 hist of hotel, guest house or bnb

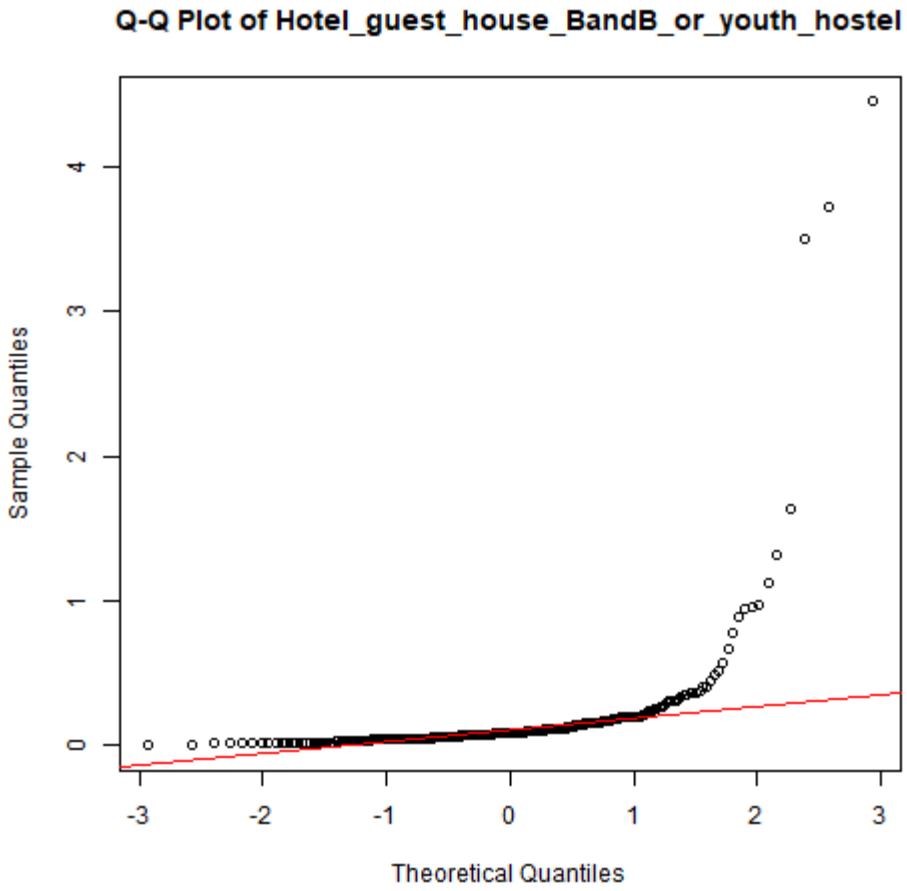

Figure 85 q-q plot of hotel, guest house or bnb

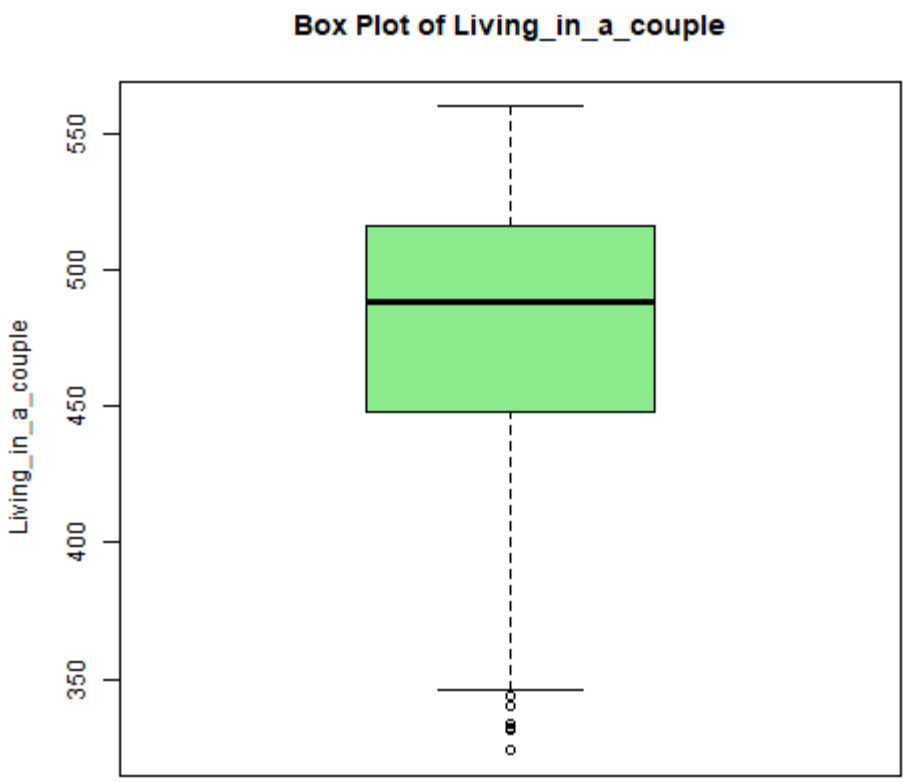

Figure 86 box plot of living in a couple

The box plot of covid deaths for people living as a couple shows downward outliers which means decrease in mortality rates due to factors such as support and care system couples give one another, limited outside exposure and work from home.

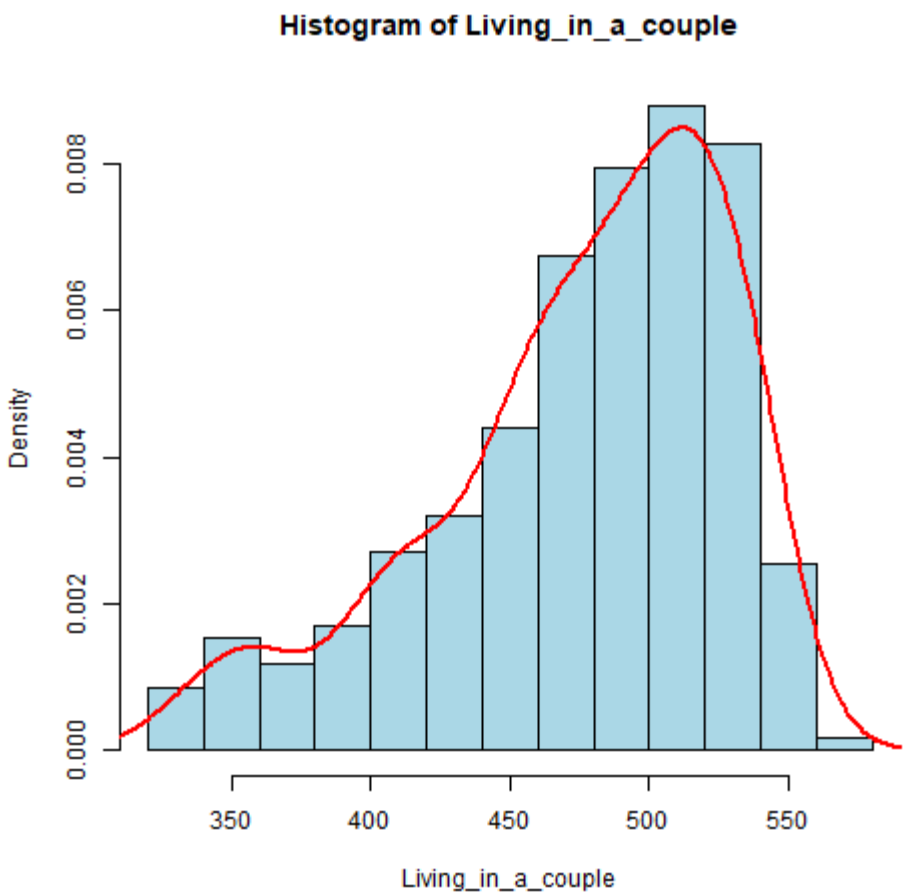

Figure 87 hist of living in a couple

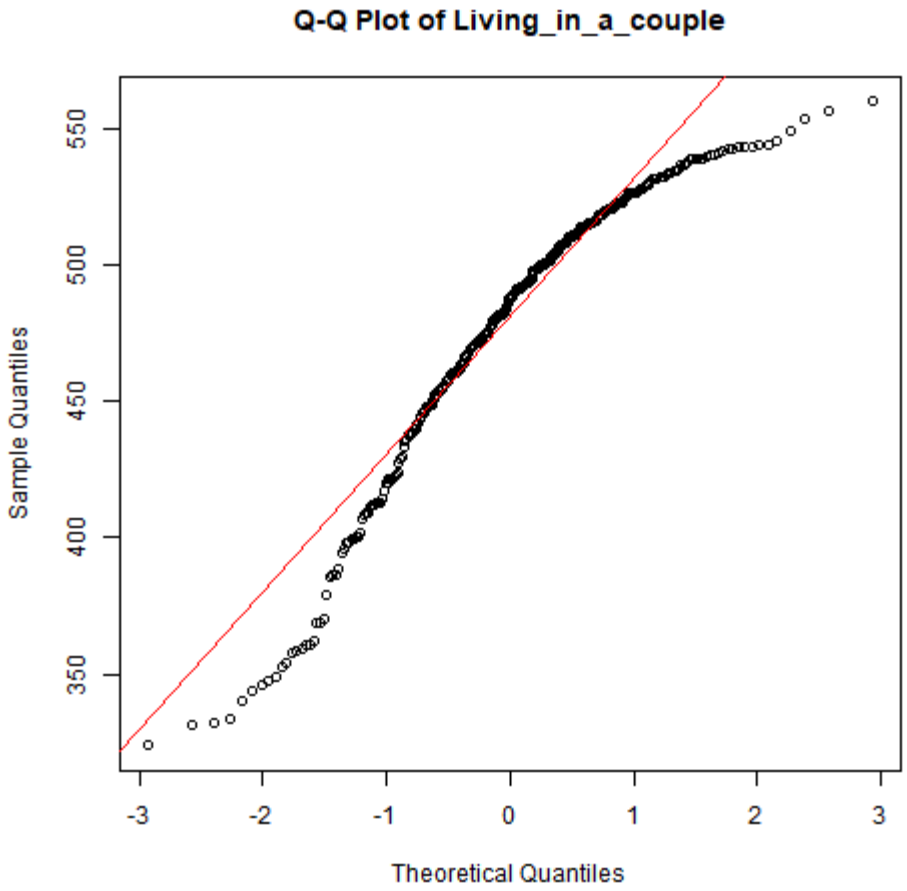

Figure 88 q-q plot of living in a couple

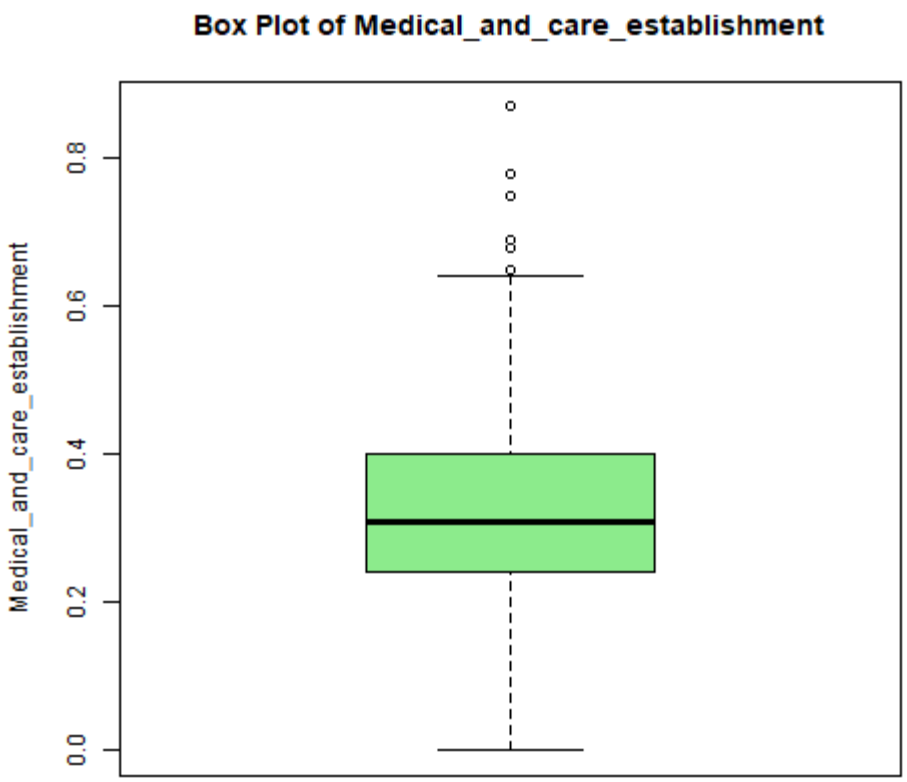

Figure 89 box plot of medical and care establishment

The box plot of covid deaths in medical and care establishment shows several upward outliers which indicate spikes in mortality rates which could be due to overcrowded emergency centres during the peak covid season, over stretched medical facilities, doctors and nurses dying due to contact with diseased persons and a host of other plausible reasons.

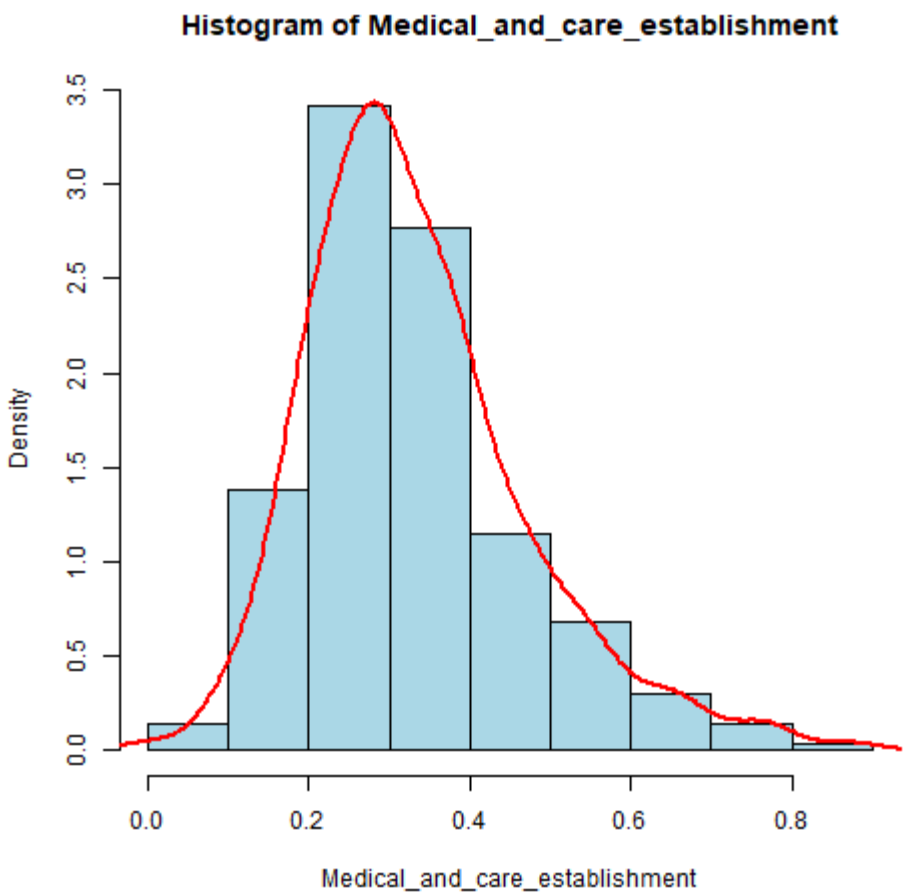

Figure 90 hist of medical and care establishment

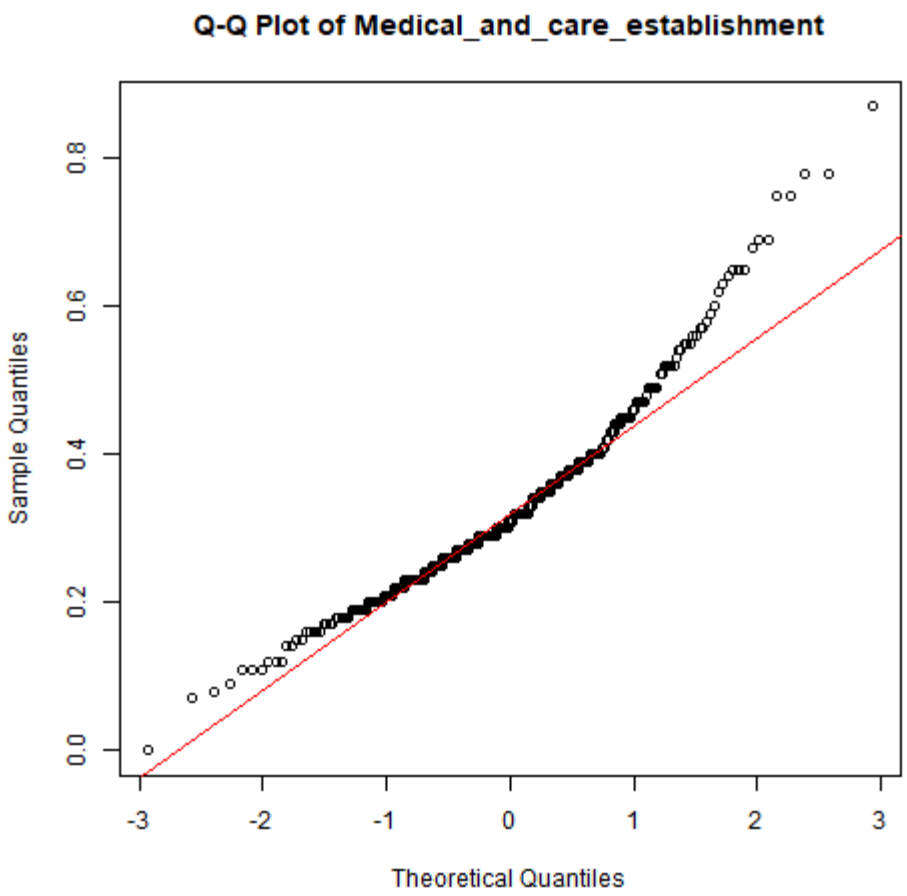

Figure 91q-q plot of medical and care establishment

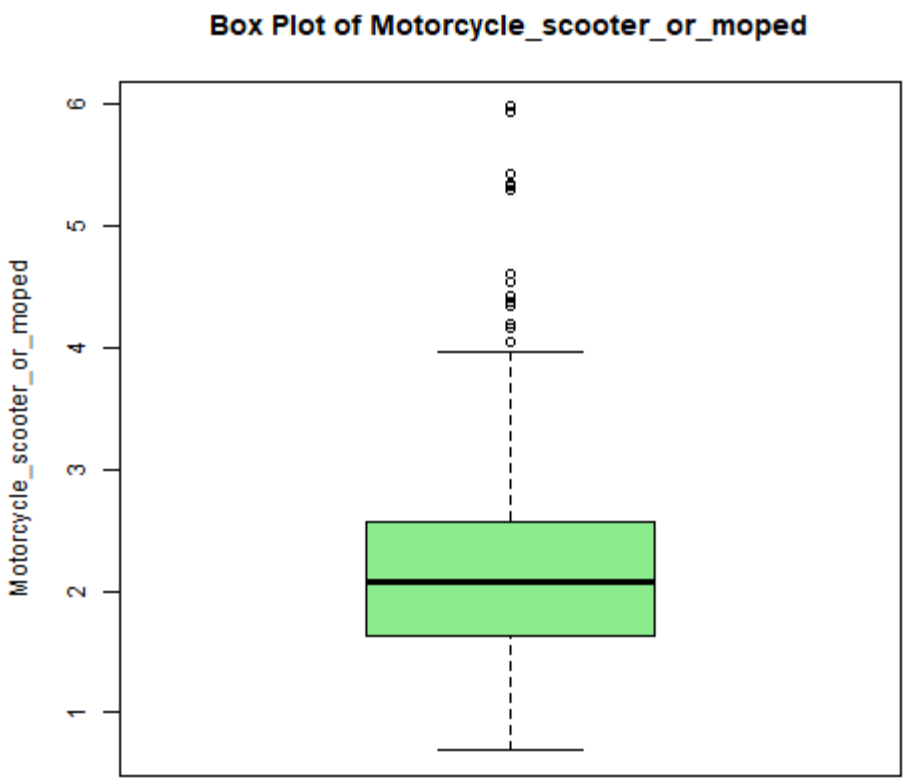

Figure 92 box plot of motorcycle, scooter or moped

Box plot of covid deaths for motorcycle users and scoter users shows several upper outliers which indicate increased mortality due to factors such as social stranger contact due to the exposed nature of motorcycles and scooters.

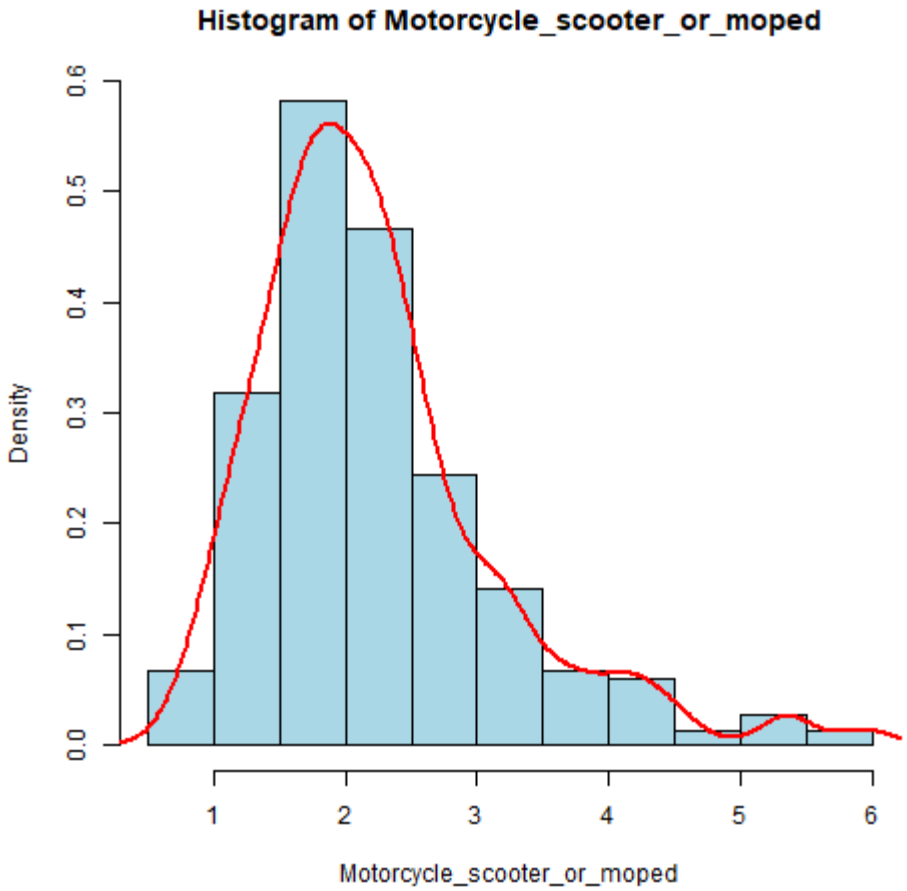

Figure 93 hist of motorcycle, scooter or moped

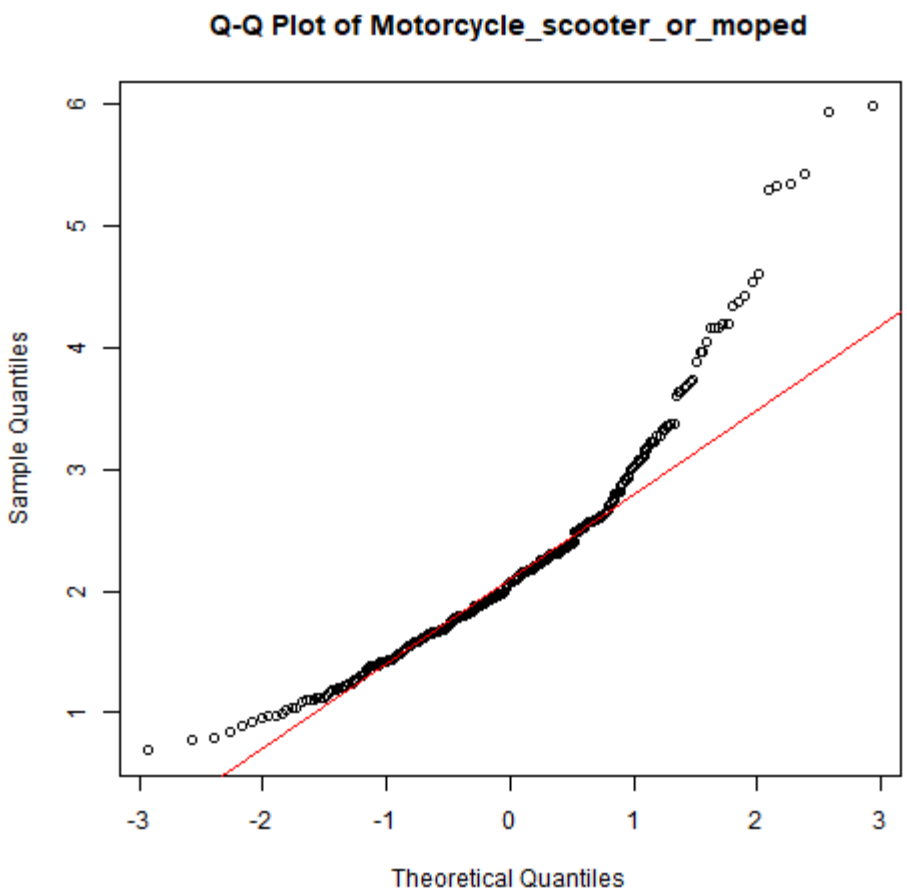

Figure 94 q-q plot of motorcycle, scooter or moped

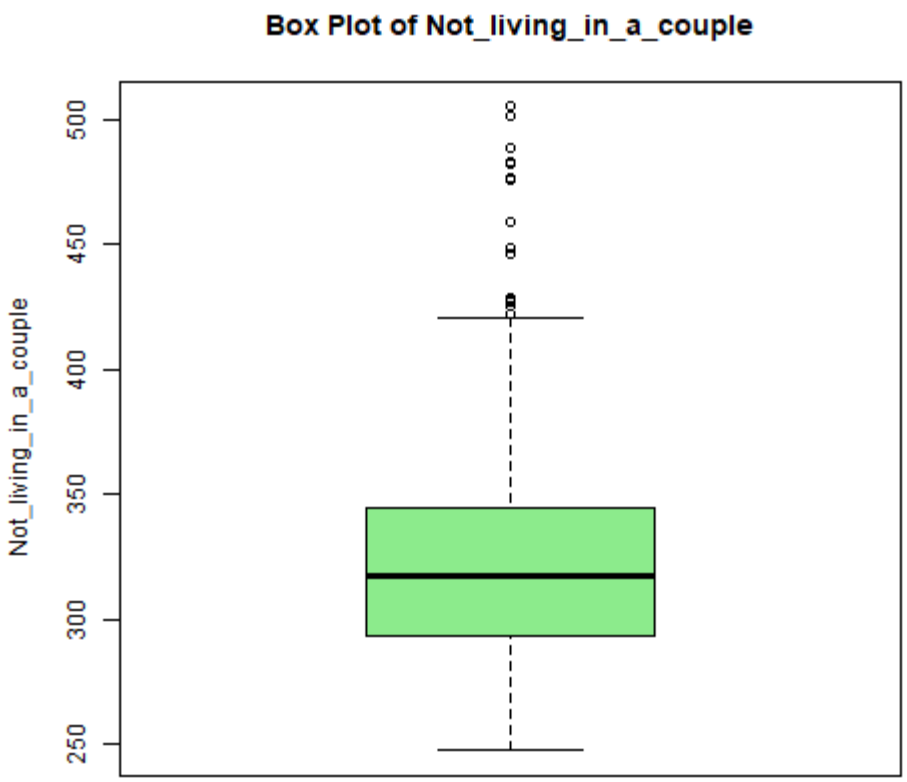

Figure 98 box plot of not living in a couple

The box plot of covid deaths for people not living as a couple shows so many upper outliers which indicate increase in mortality rate which is because people living alone experience difficulties in taking care of themselves, taking their medications as at when due and they generally lack emotional support.

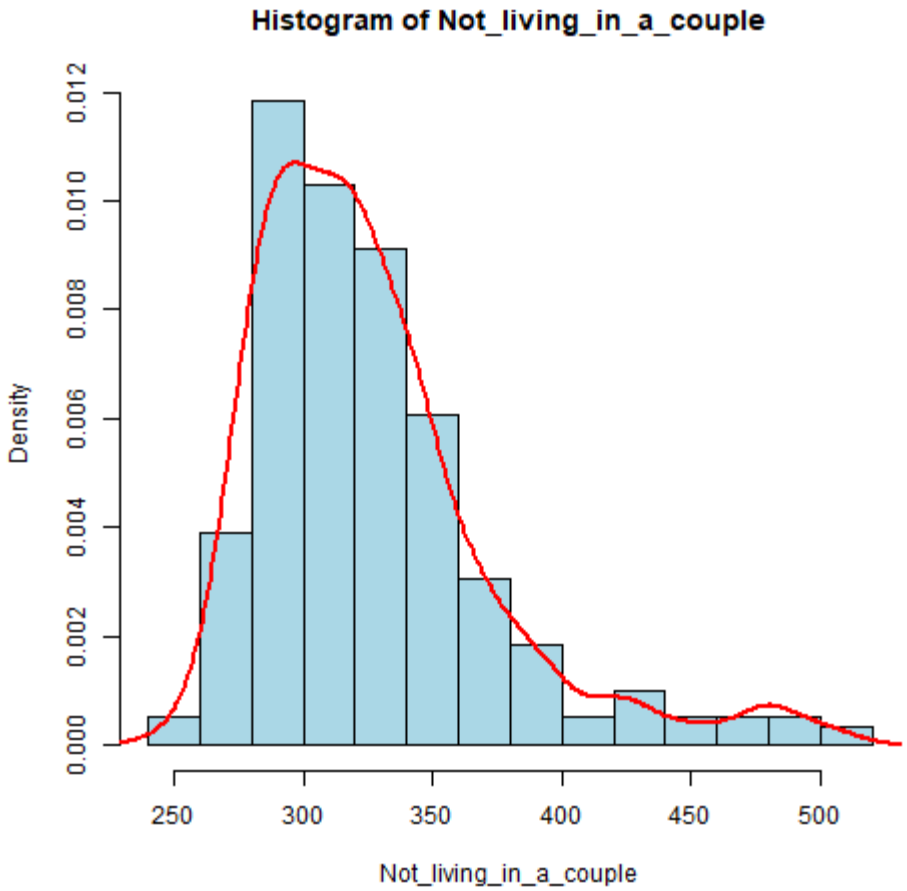

Figure 99 hist of not living in a couple

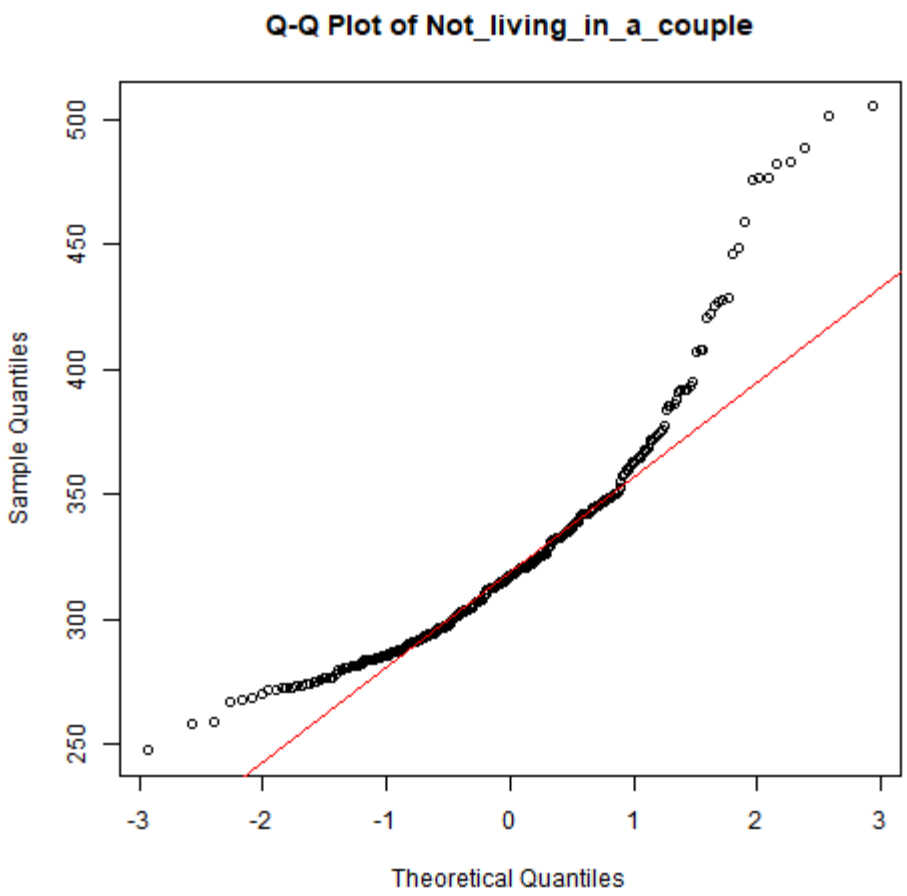

Figure 100 q-q plot of not living in a couple

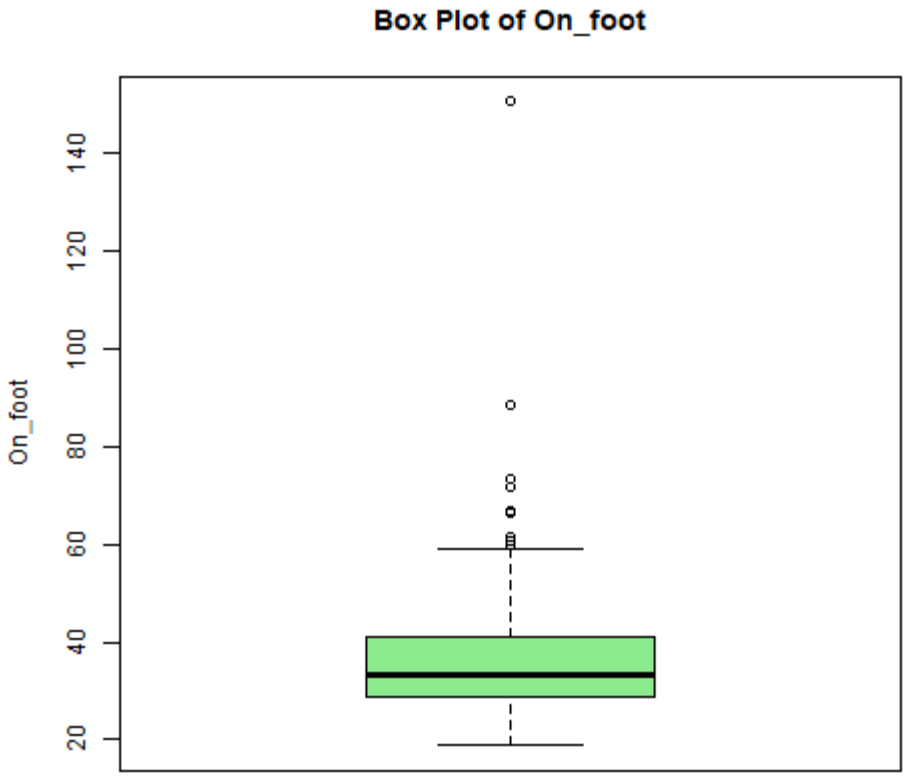

Figure 101 box plot of on foot

The box plot of covid deaths for people going about on foot show so many upward outliers which could be due to increased exposure to social stranger contact.

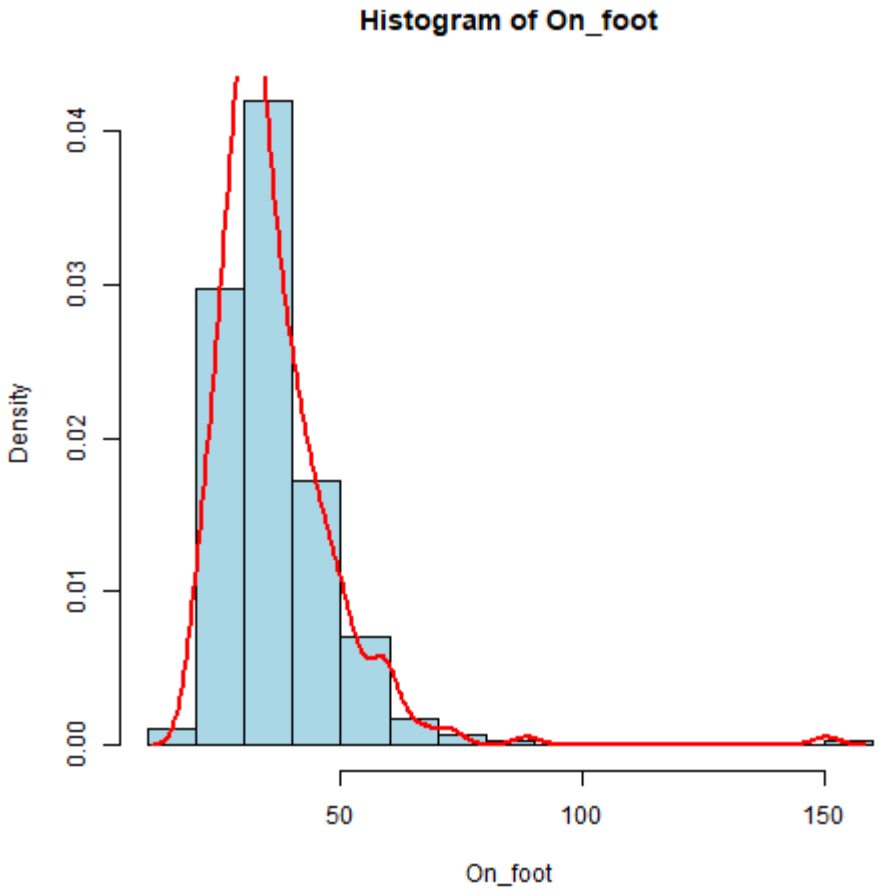

Figure 102 hist of on foot

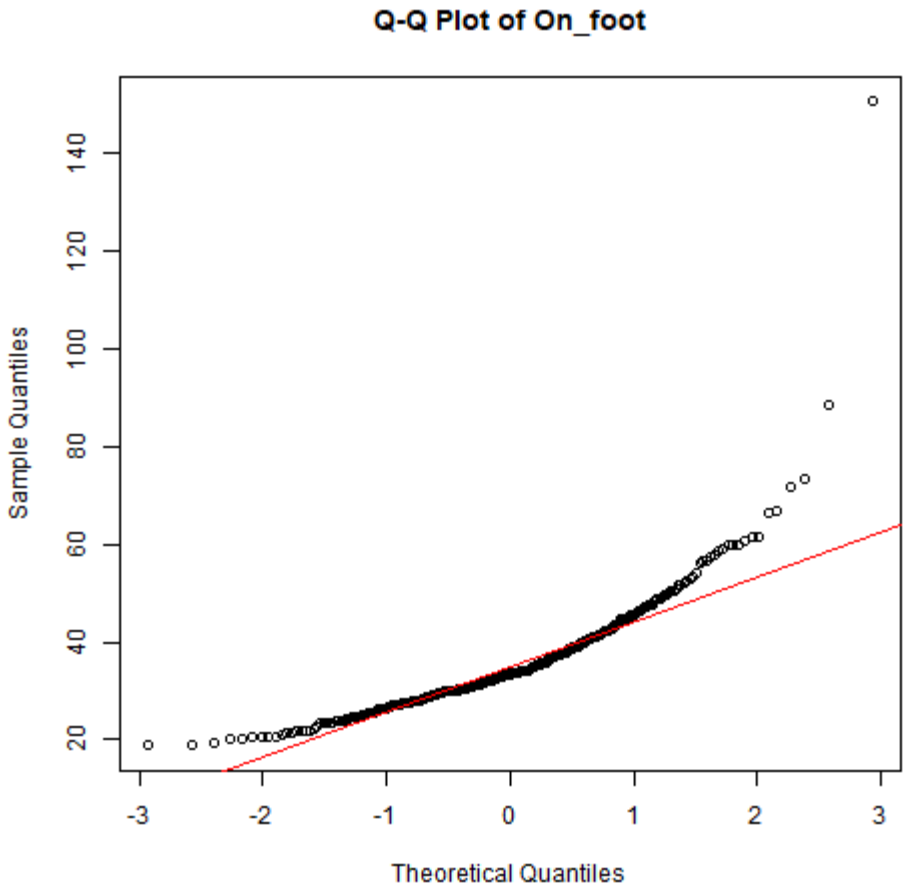

Figure 103 q-q plot of on foot

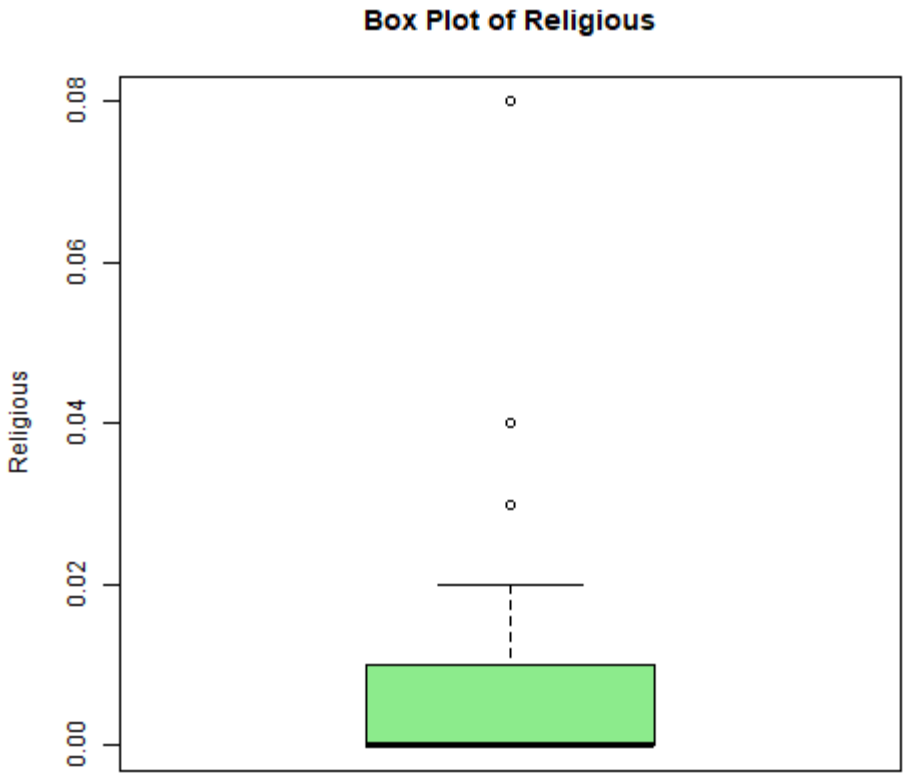

Figure 107 box plot of religious

The box plot of covid deaths for religious dwellers shows few upward outliers which could be due to social contact during communal worship.

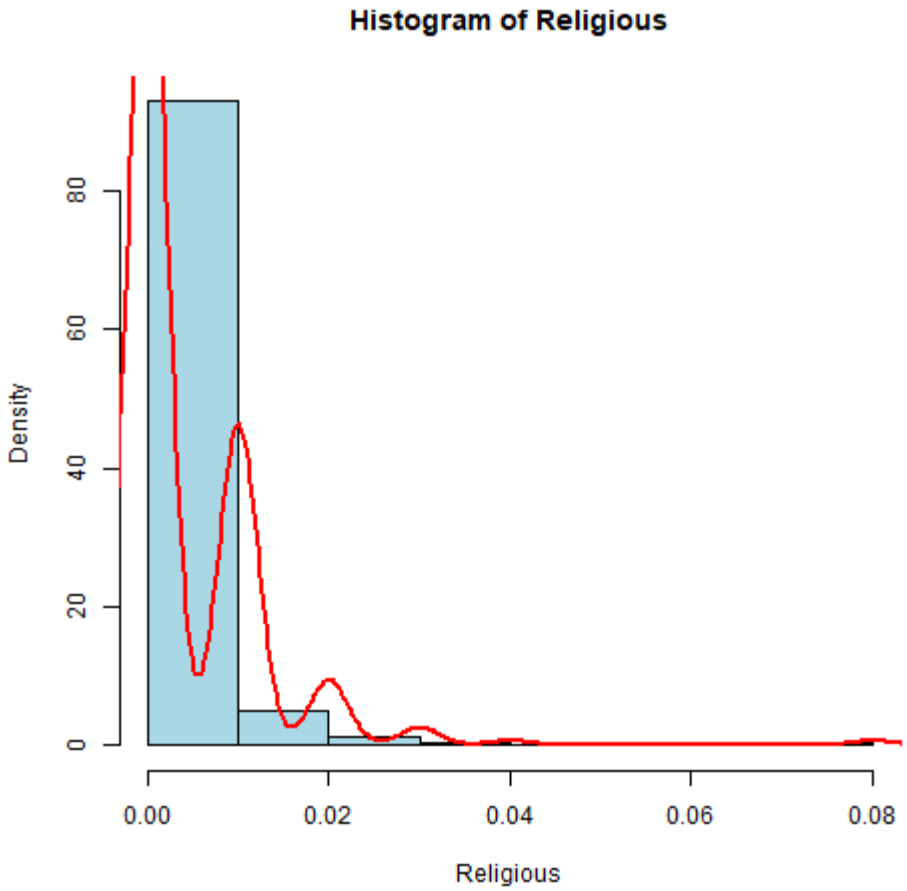

Figure 108 hist of religious

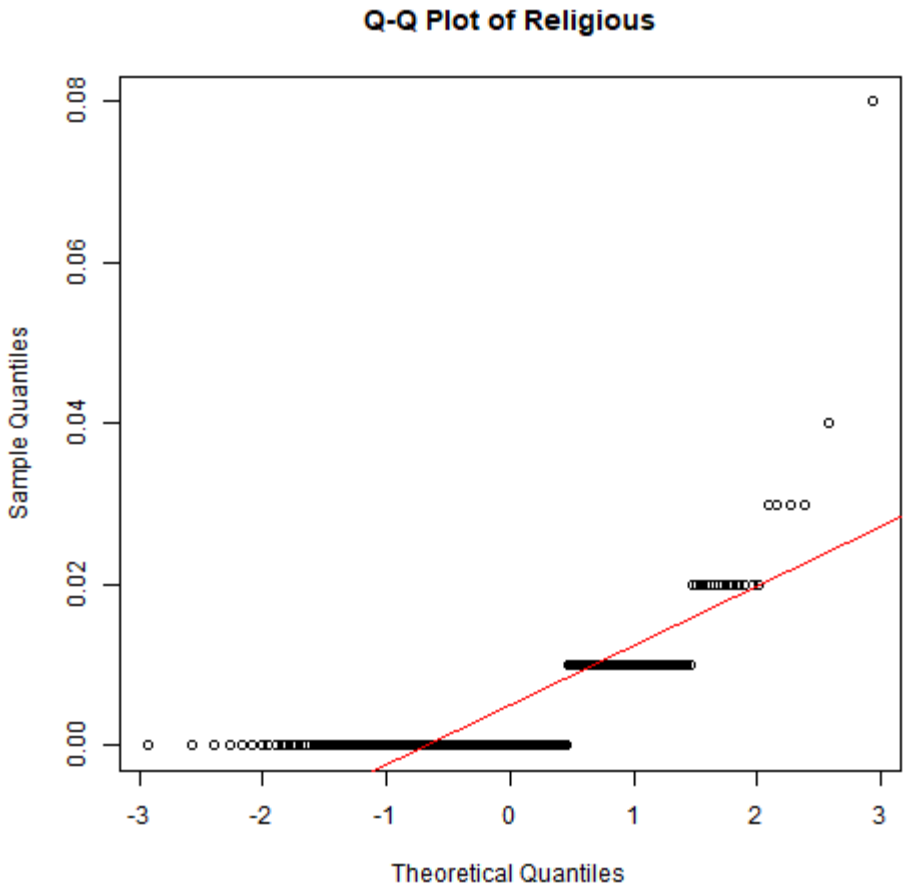

Figure 109 q-q plot of religious

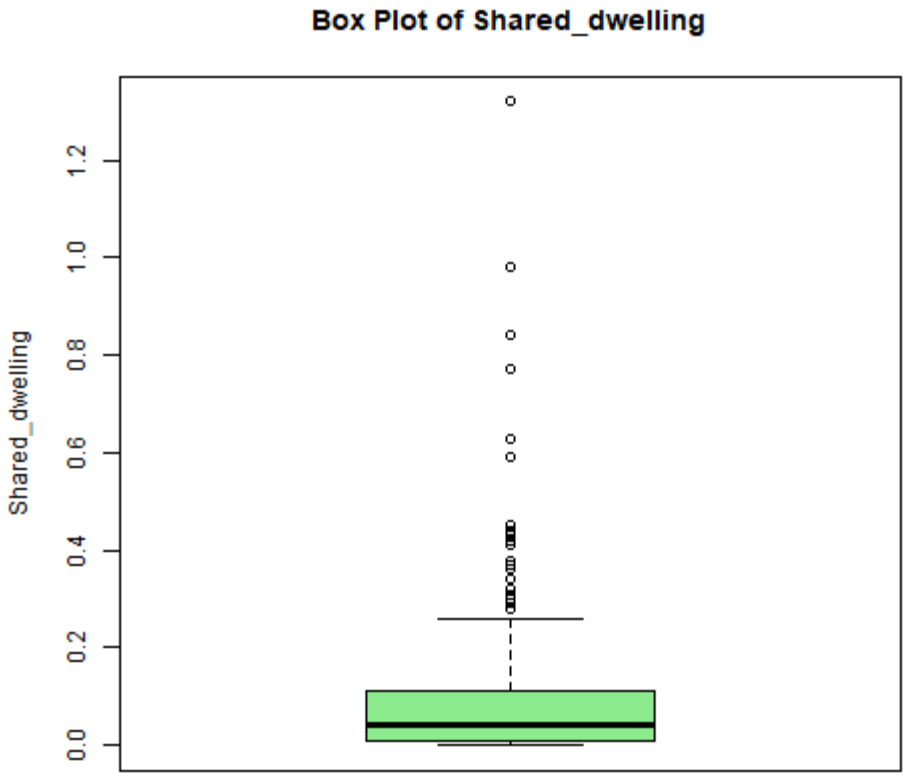

Figure 110 box plot of shared dwelling

The box plot of covid deaths for people in shared dwelling shows several upward outliers and death spikes due to social contact, unhygienic facilities and possibility of disease transmission.

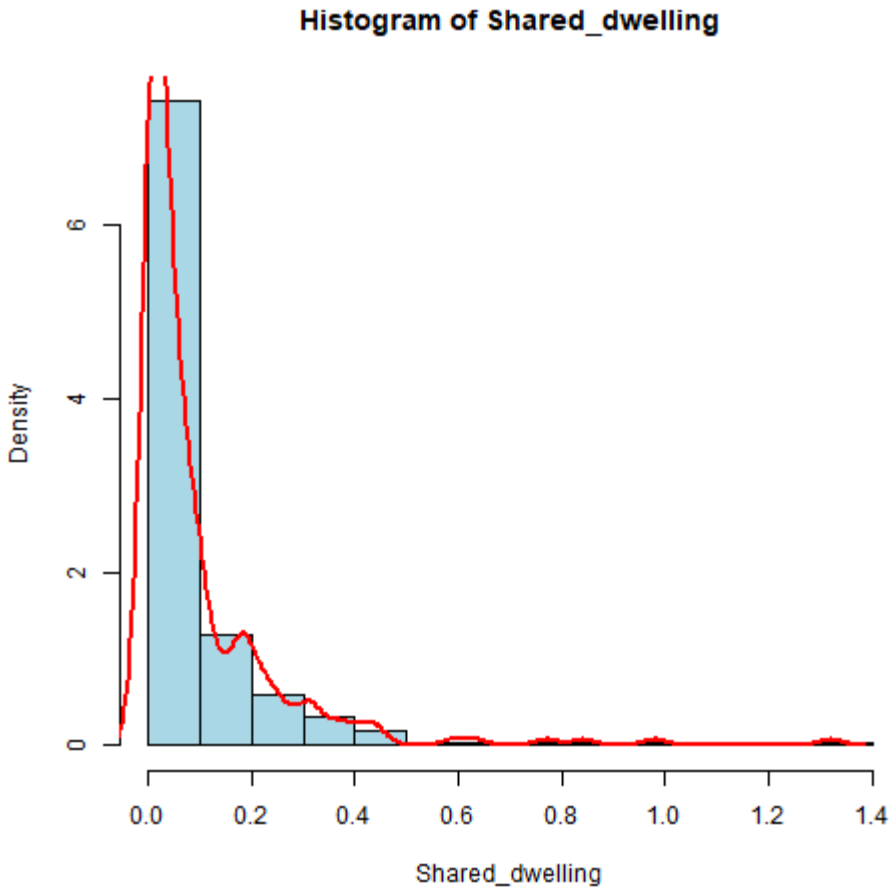

Figure 111 hist of shared dwelling

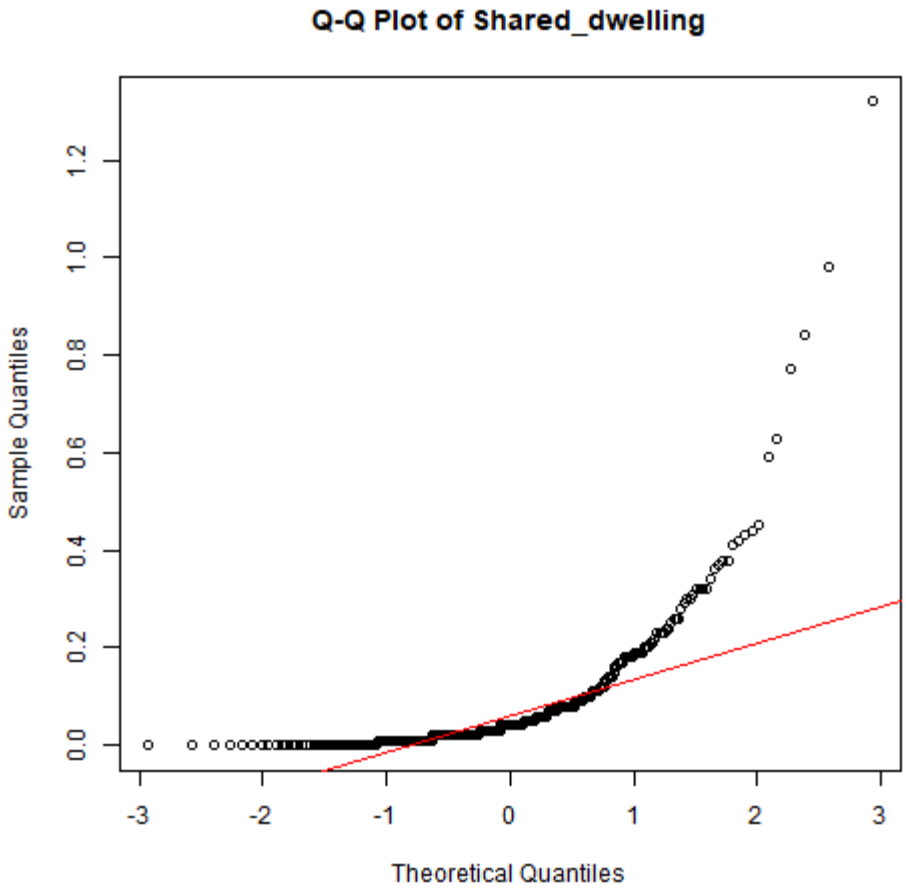

Figure 112 q-q plot of shared dwelling

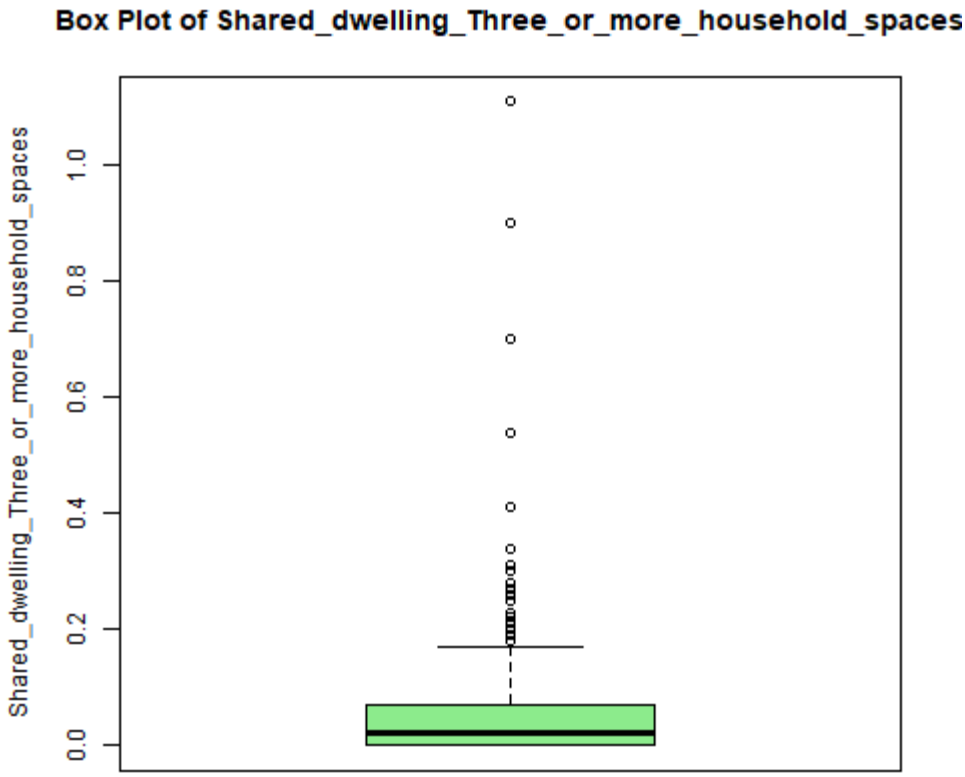

Figure 113 box plot of shared dwelling: 3 or more households

The box plot of covid deaths for people in shared dwelling and three or more household spaces shows several upper outlier spikes in mortality which could be due to social contact and unhygienic facilities.

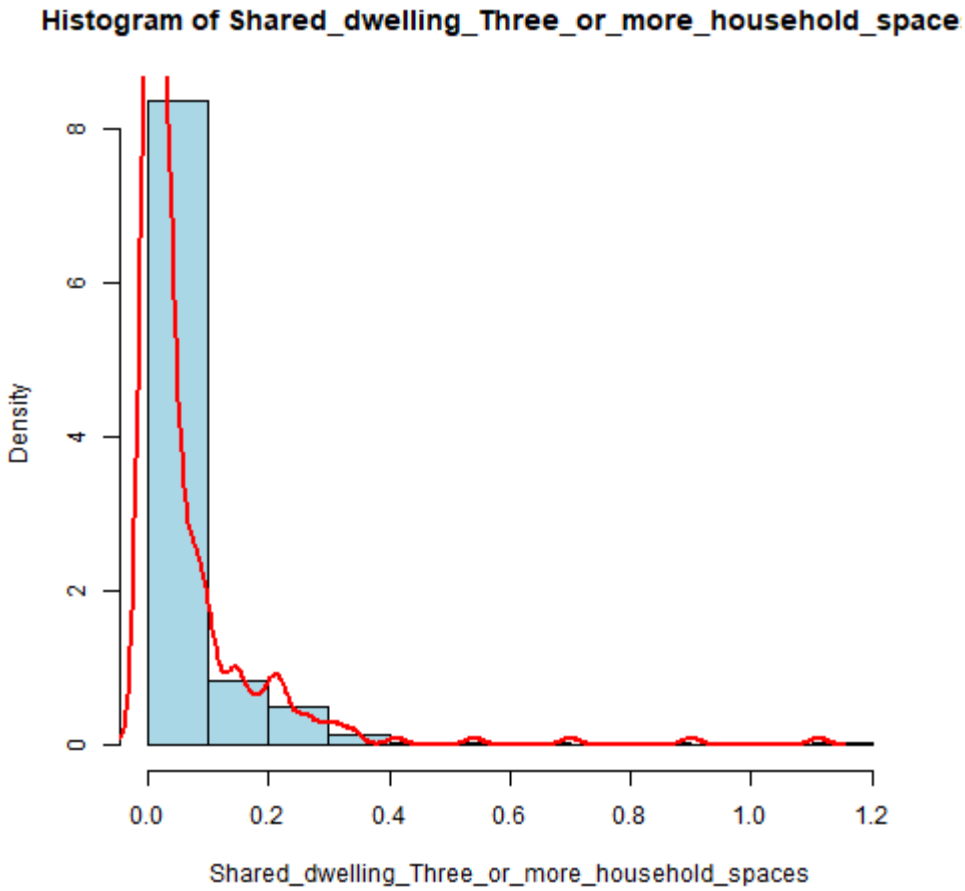

Figure 114 hist of shared dwelling 3 or more households

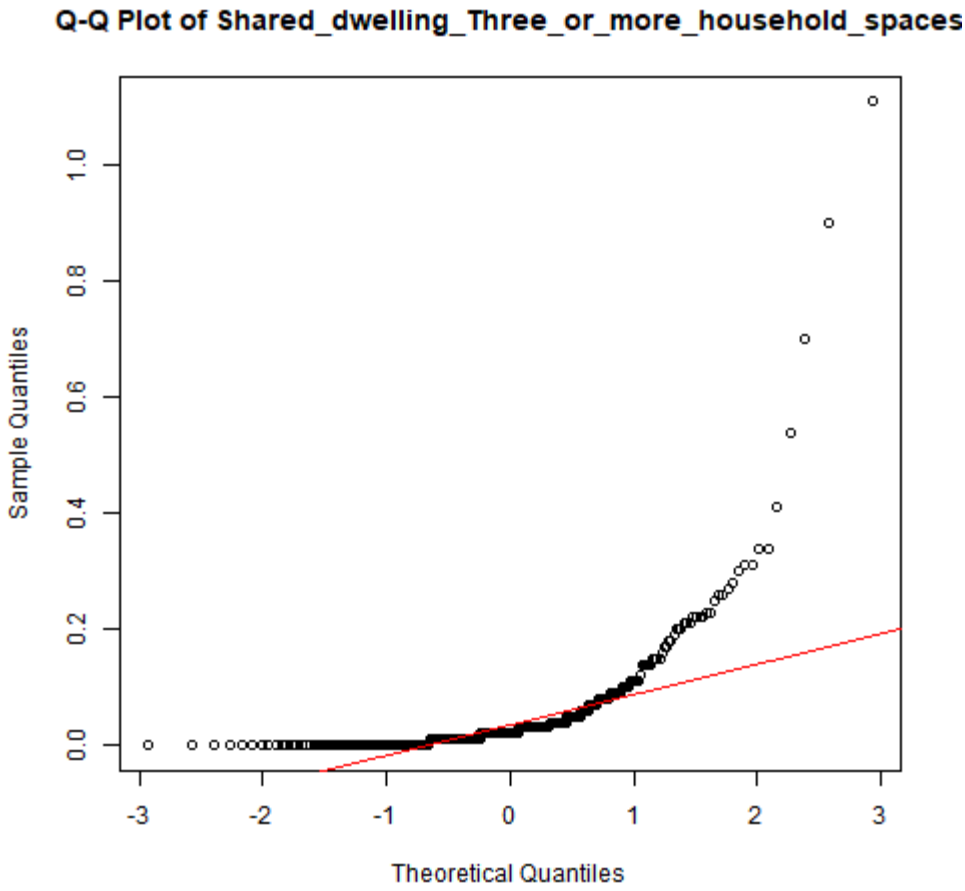

Figure 115 q-q plot of shared dwelling 3 or more households

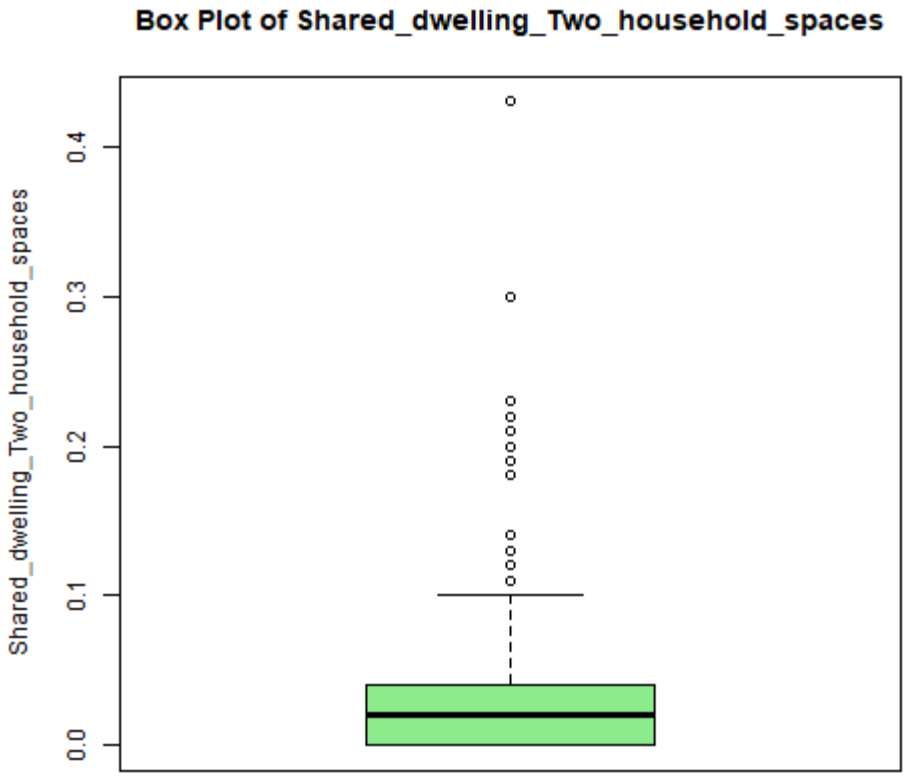

Figure 116 box plot of shared dwelling 2 household spaces

The box plot of covid deaths for shared dwelling or two household spaces shows several upward outliers indicating spike in mortality rate which could be due to increased social contact and unhygienic facilities.

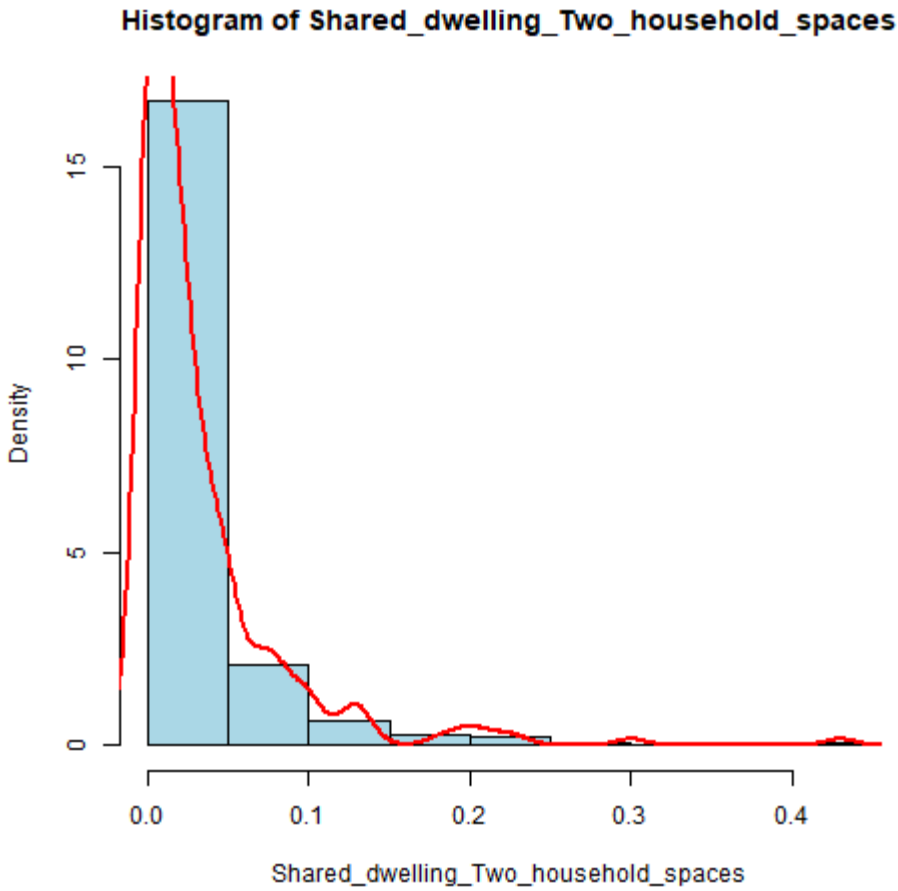

Figure 117 hist of shared dwelling 2 household spaces

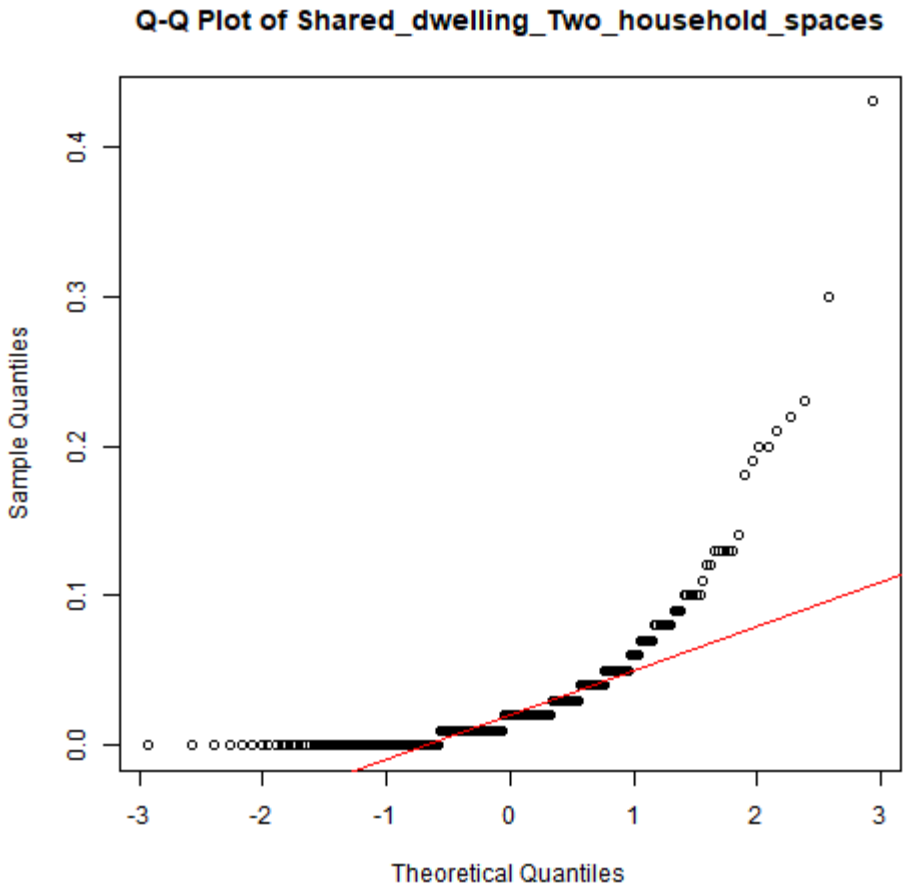

Figure 118 q-q plot of shared dwelling 2 household spaces

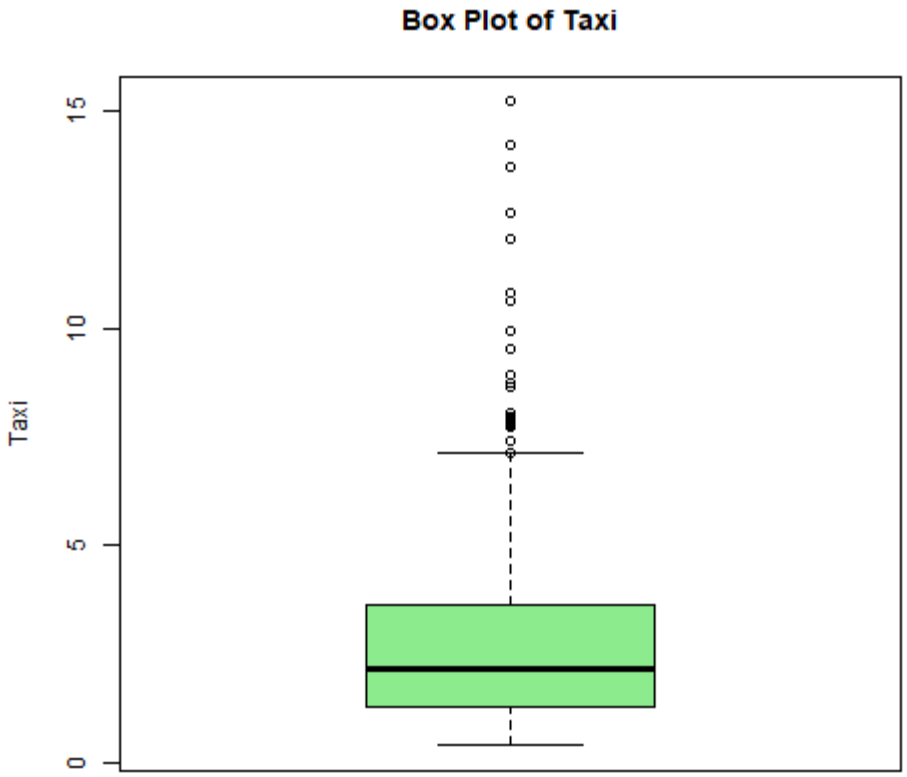

Figure 119 box plot of taxi

The box plot of covid deaths for people using taxis shows several spikes and upward outliers indicating increased mortality due to increased stranger social contact in public transport.

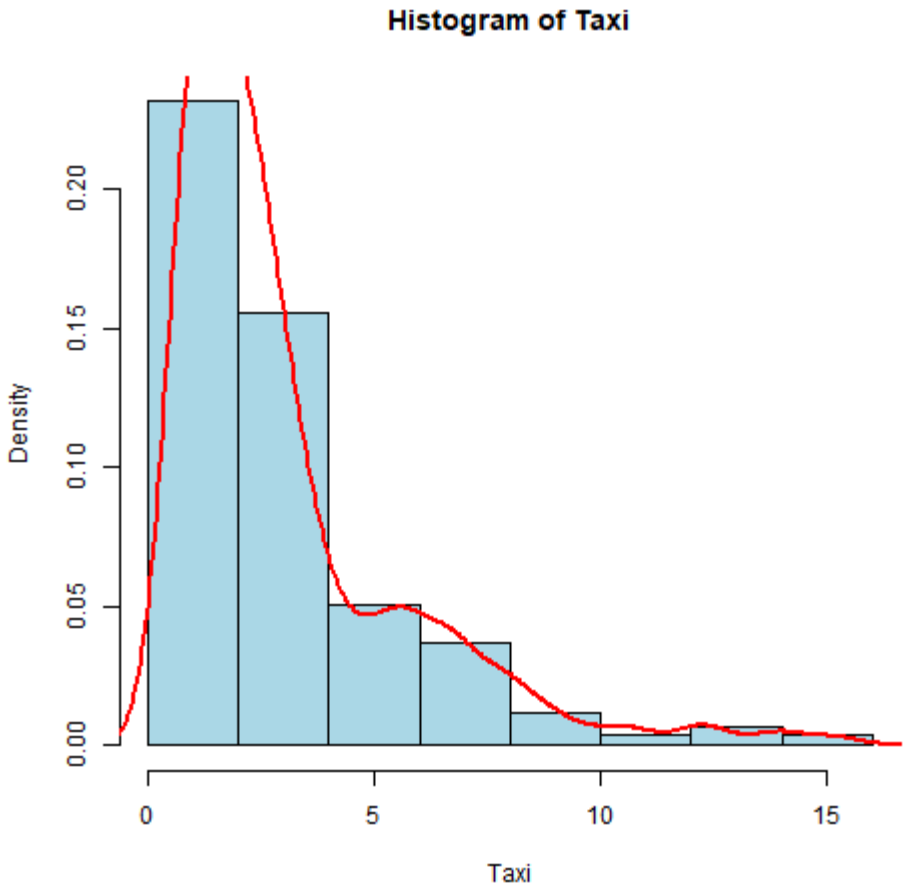

Figure 120 hist of taxi

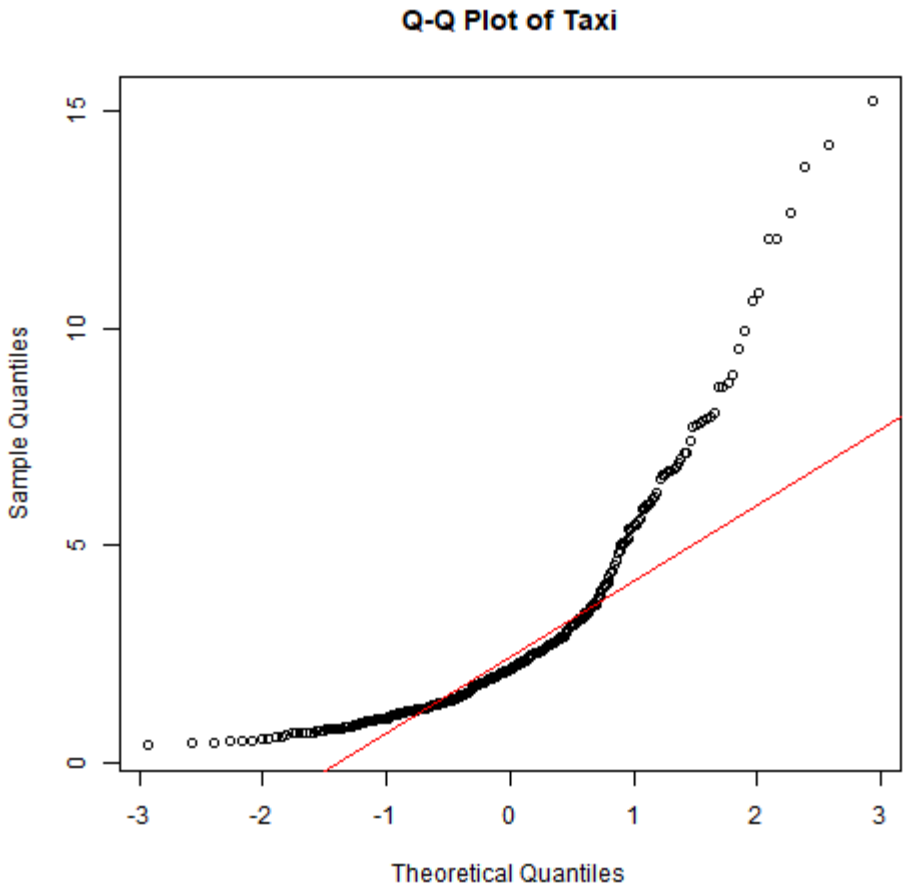

Figure 121 q-q plot of taxi

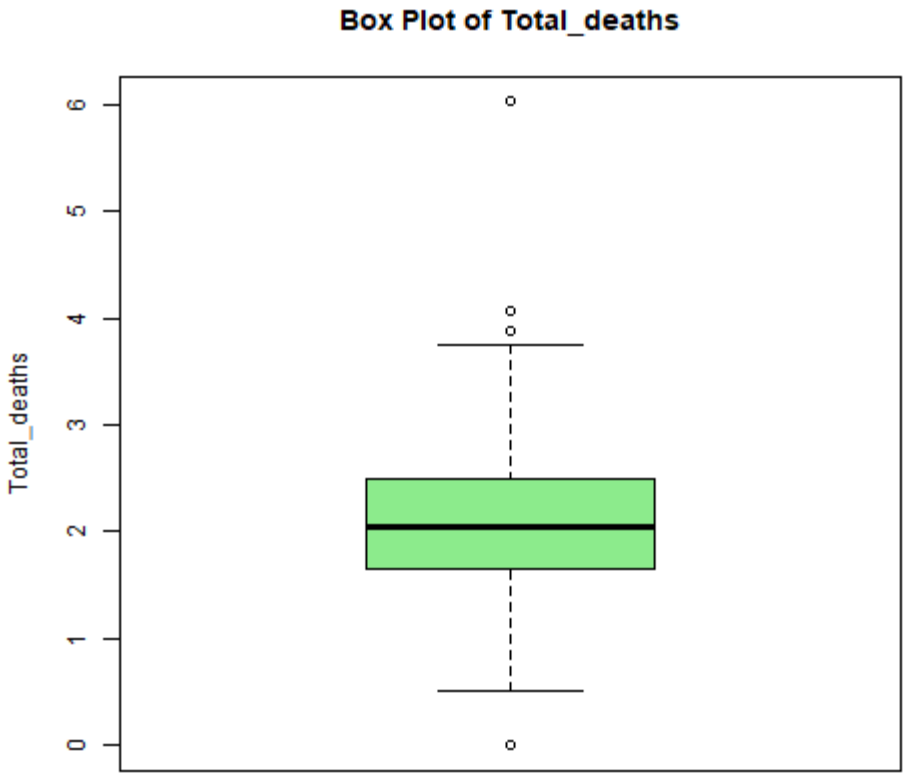

Figure 122 box plot of total deaths

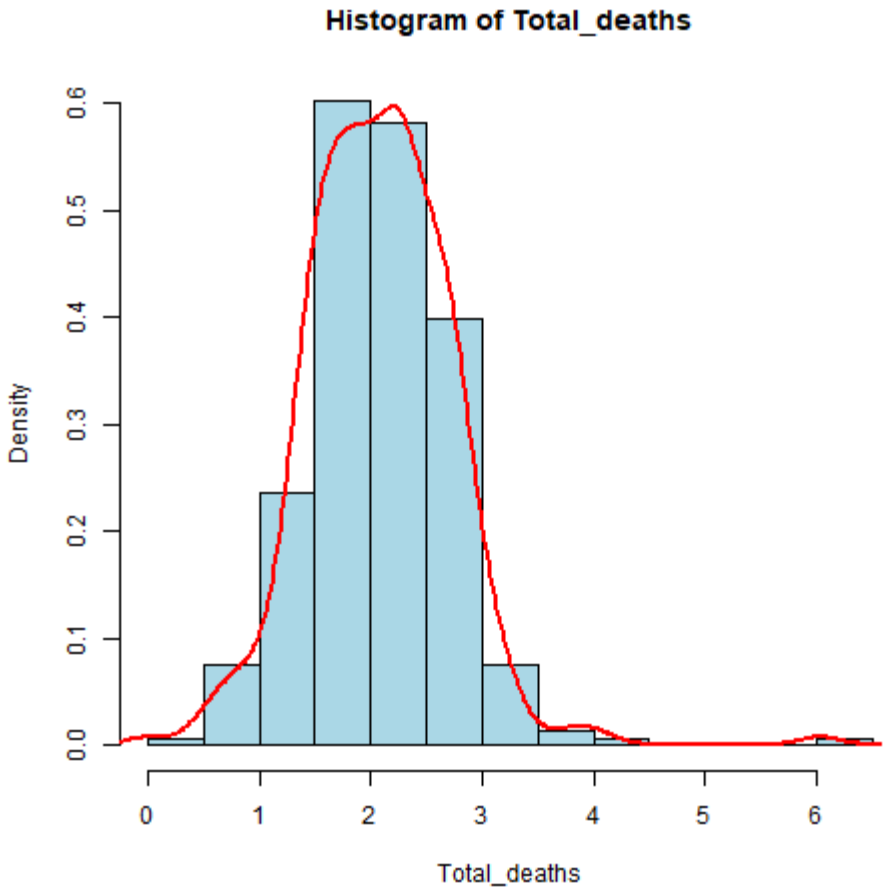

Figure 123 hist of total deaths

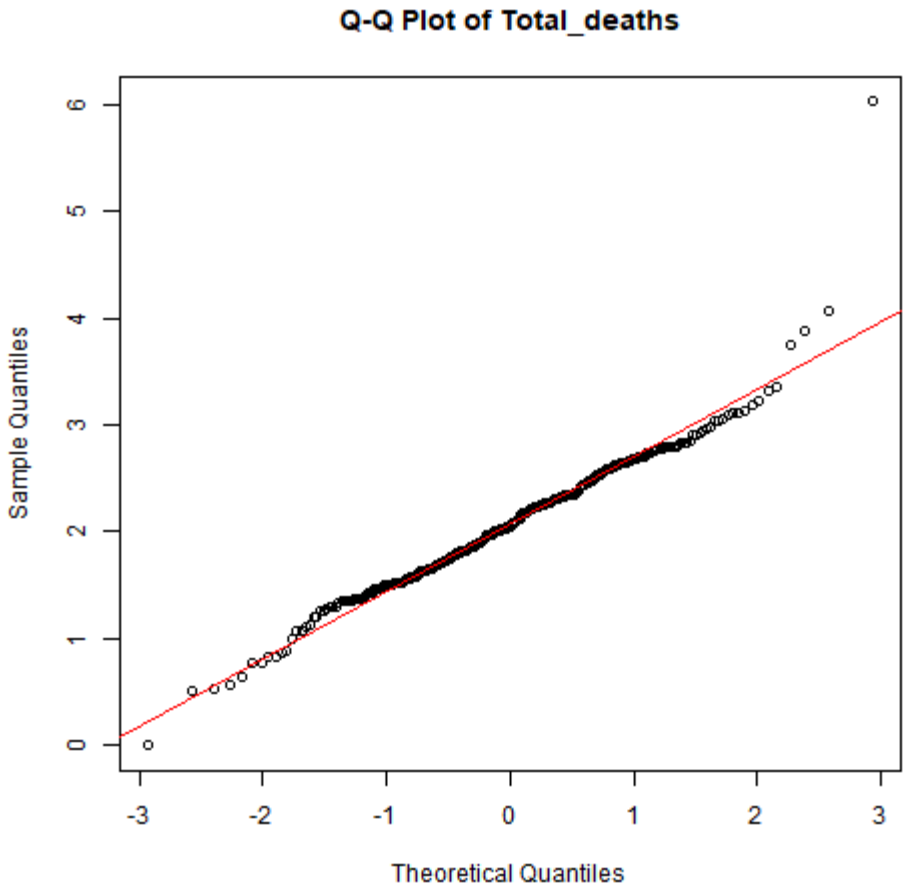

Figure 124 q-q plot of total deaths

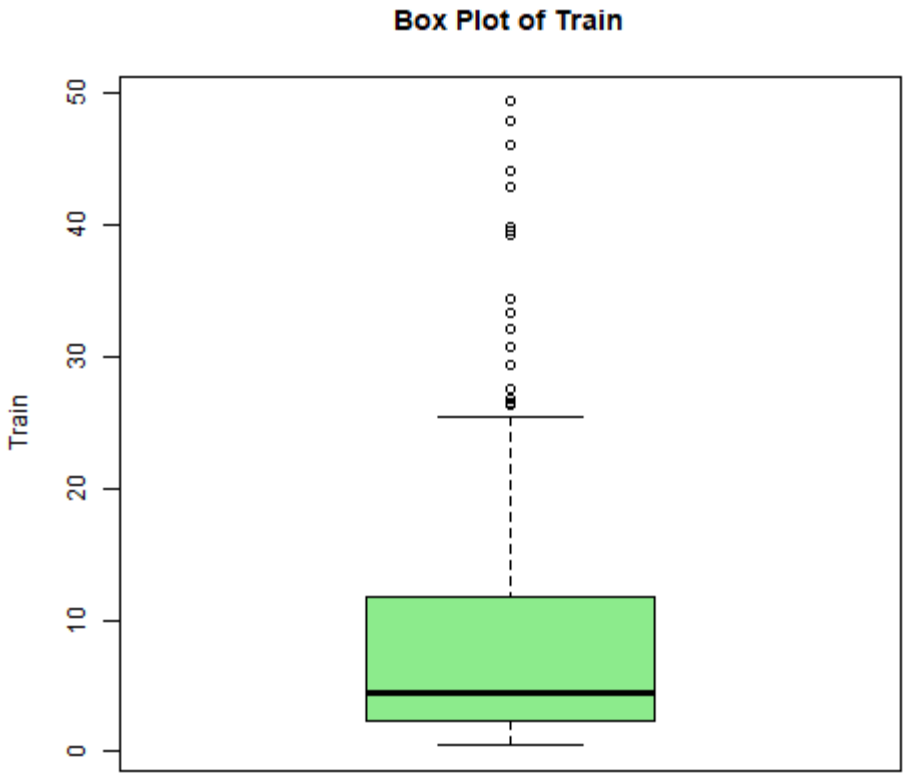

Figure 125 box plot of train

The box plot of covid deaths for people commuting with trains shows several upper outliers indicating spikes in mortality rate which is due to stranger social body contact.

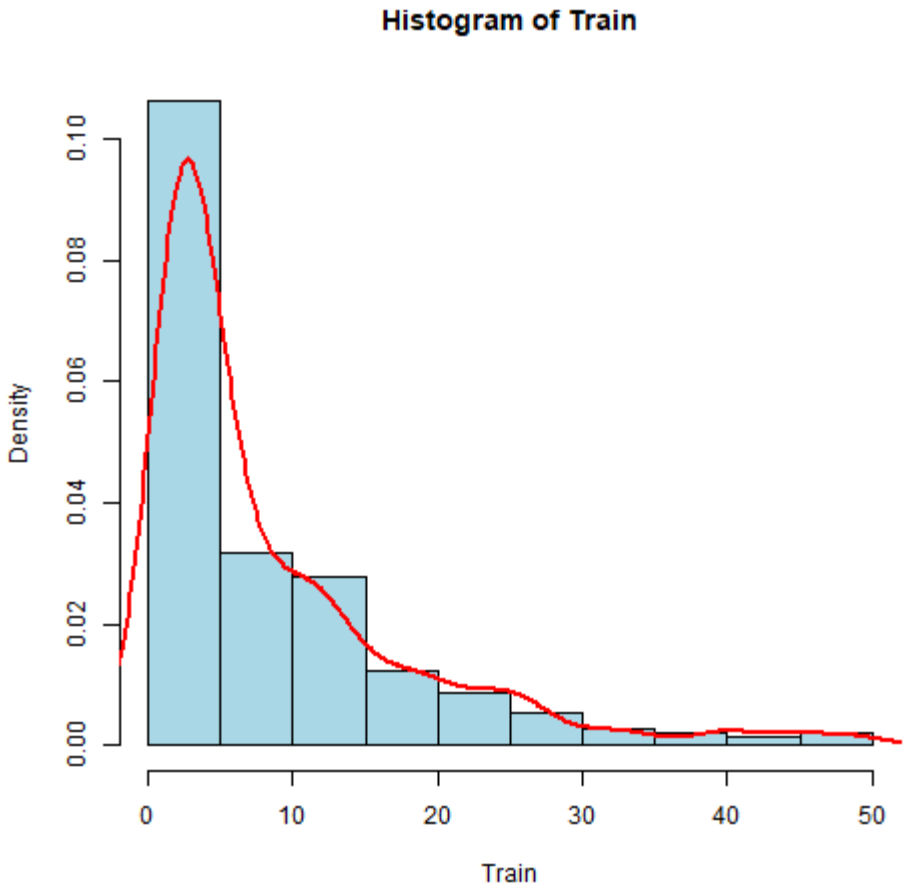

Figure 126 hist of train

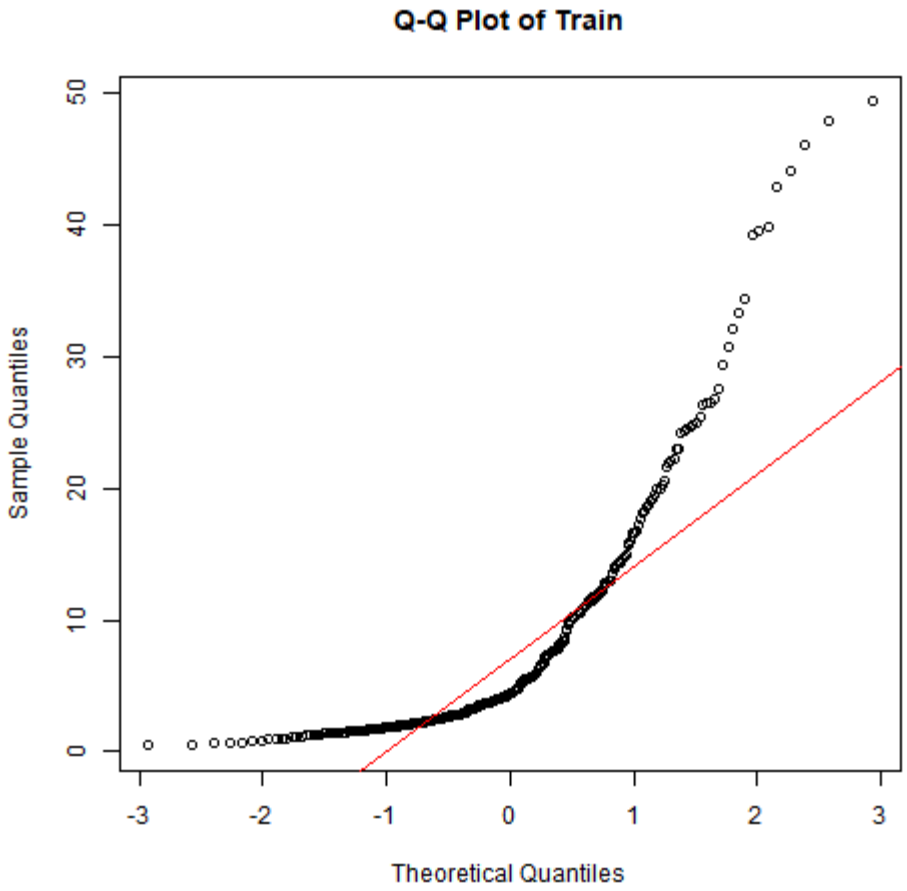

Figure 127 q-q plot of train

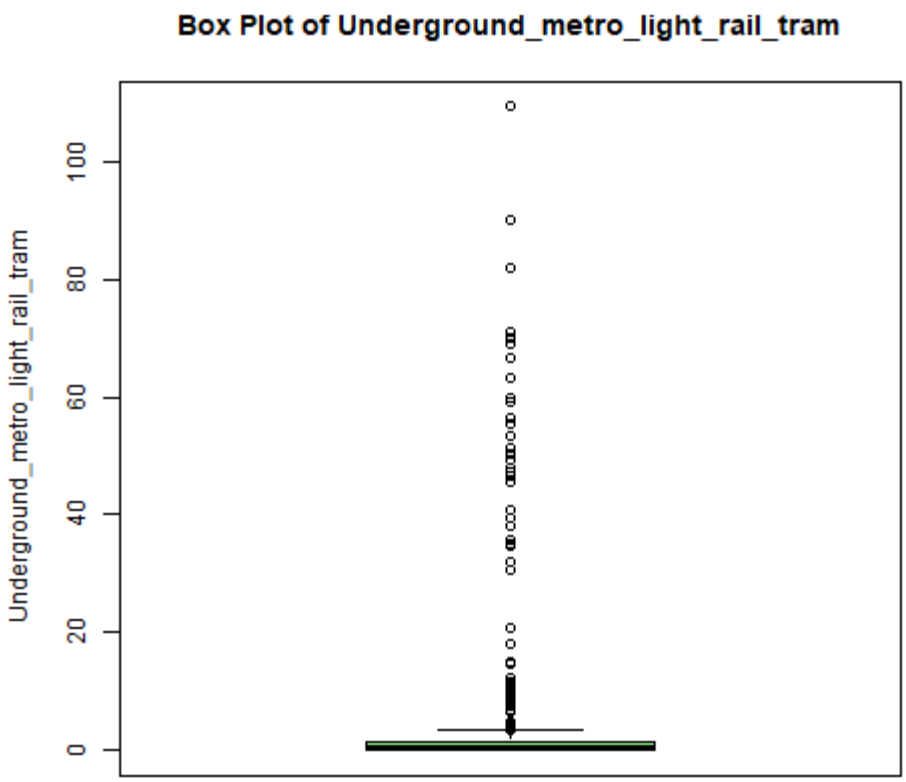

Figure 128 box plot of underground, metro, light rail

The box plot of covid deaths for people commuting with underground metro or light rail or tram shows several upper outliers indicating spikes in mortality rate which is due to stranger social body contact.

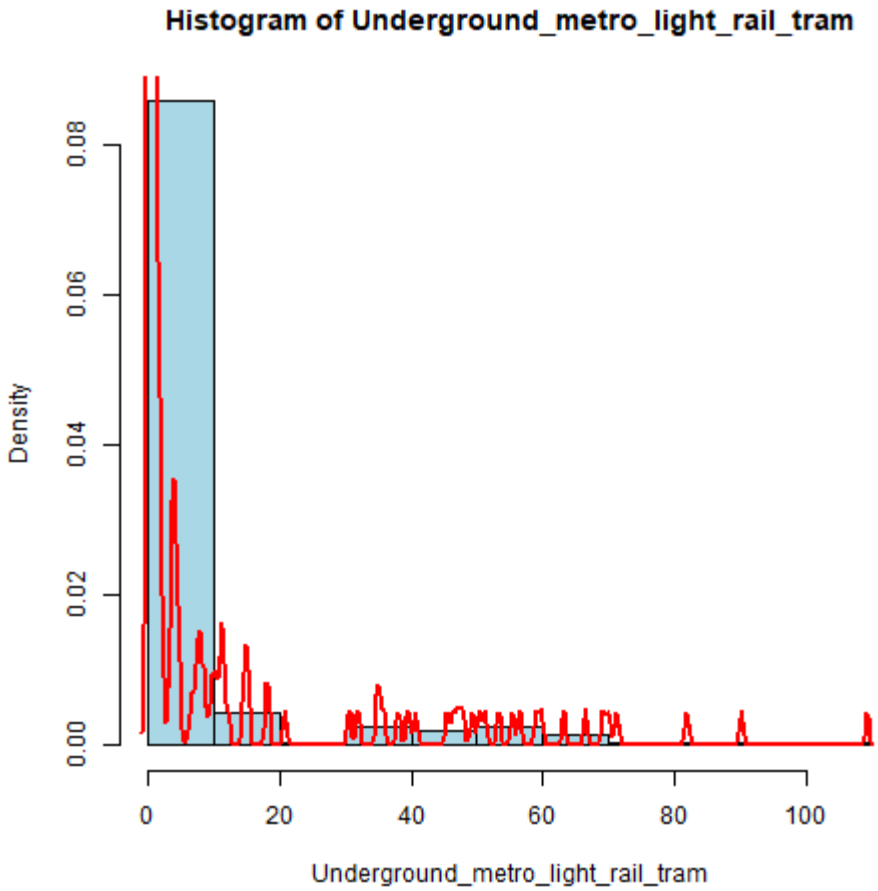

Figure 129 hist of underground, metro, light rail

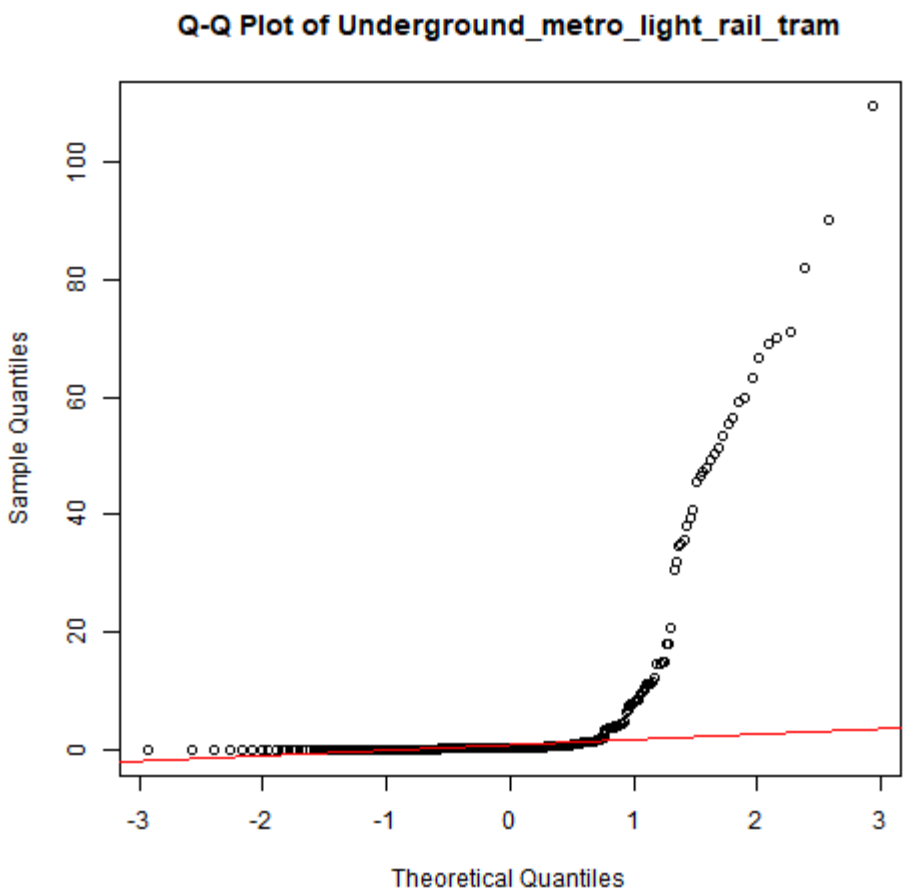

Figure 130 q-q plot of underground, metro, light rail

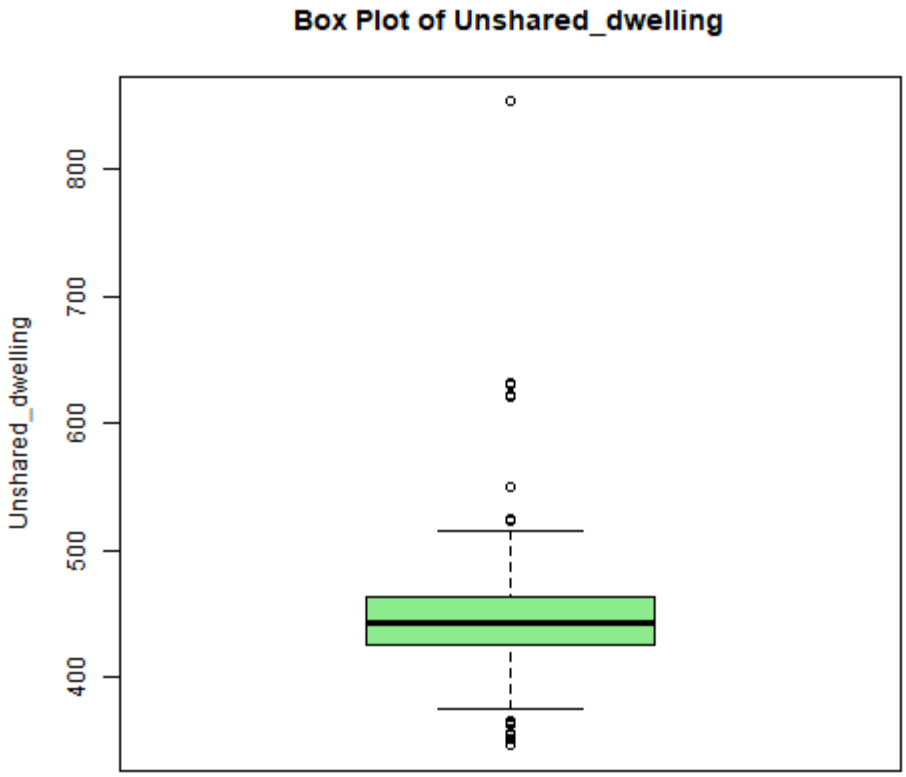

Figure 131 box plot of unshared dwelling

The box plot of covid deaths for people in unshared dwelling shows quite a bit many upward and downward outliers.

The upward outliers could be due to inability to take care of themselves and take their medications or go for regular health check-ups, while the downward outliers could be due to reduced social contact which reduces the risk of social contamination.

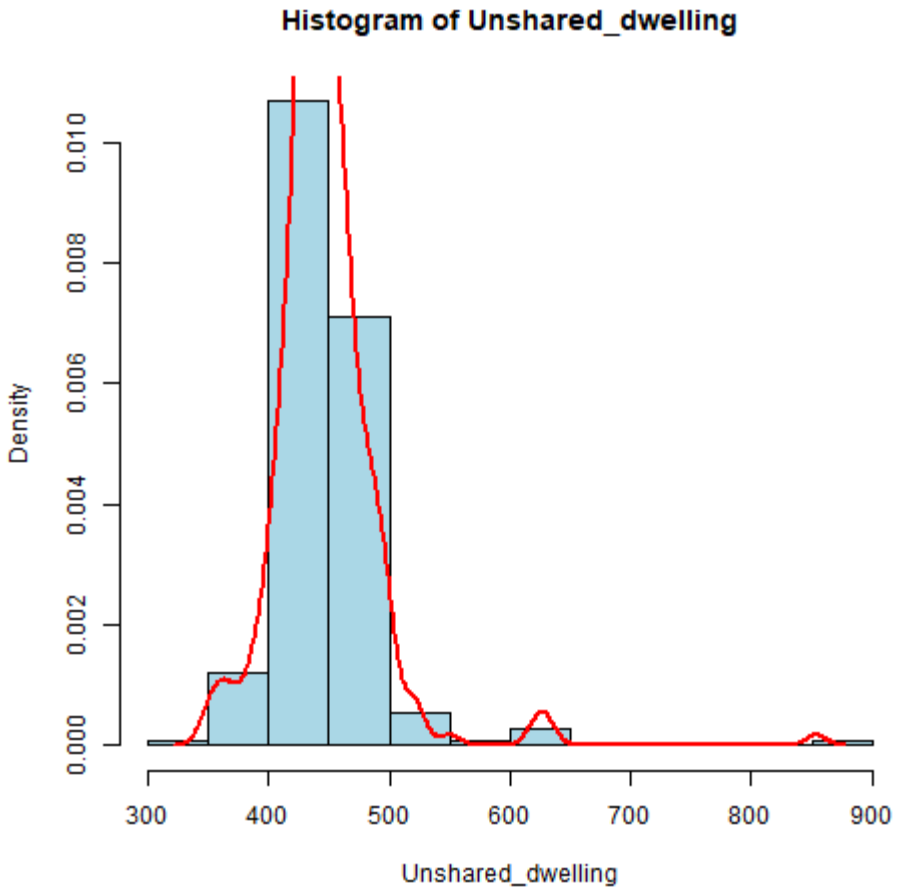

Figure 132 hist of unshared dwelling

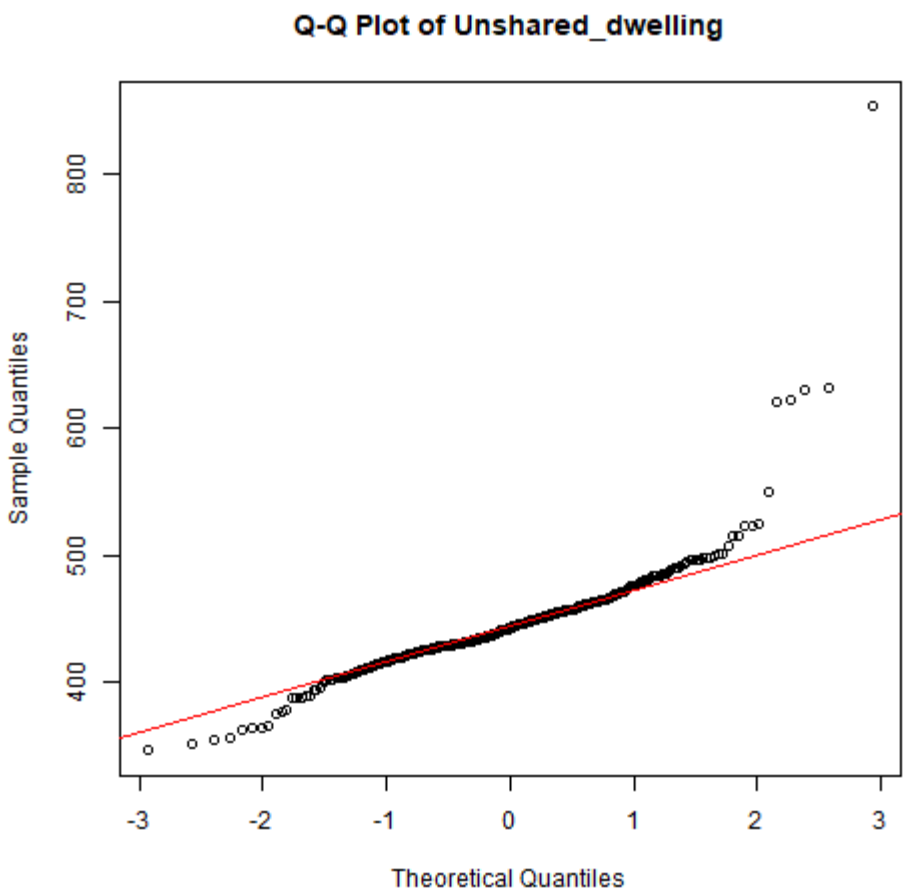

Figure 133 q-q plot of unshared dwelling

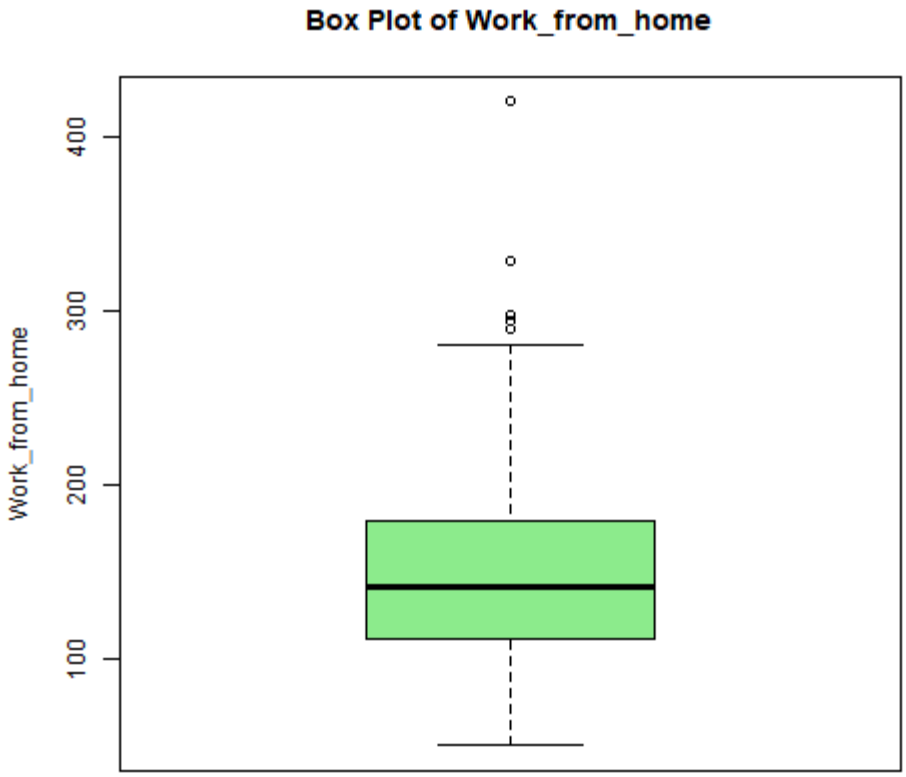

Figure 134 box plot of work from home

The box plot of covid deaths for people that work from home shows upward spikes which could be due to seclusion and absence of communal support.

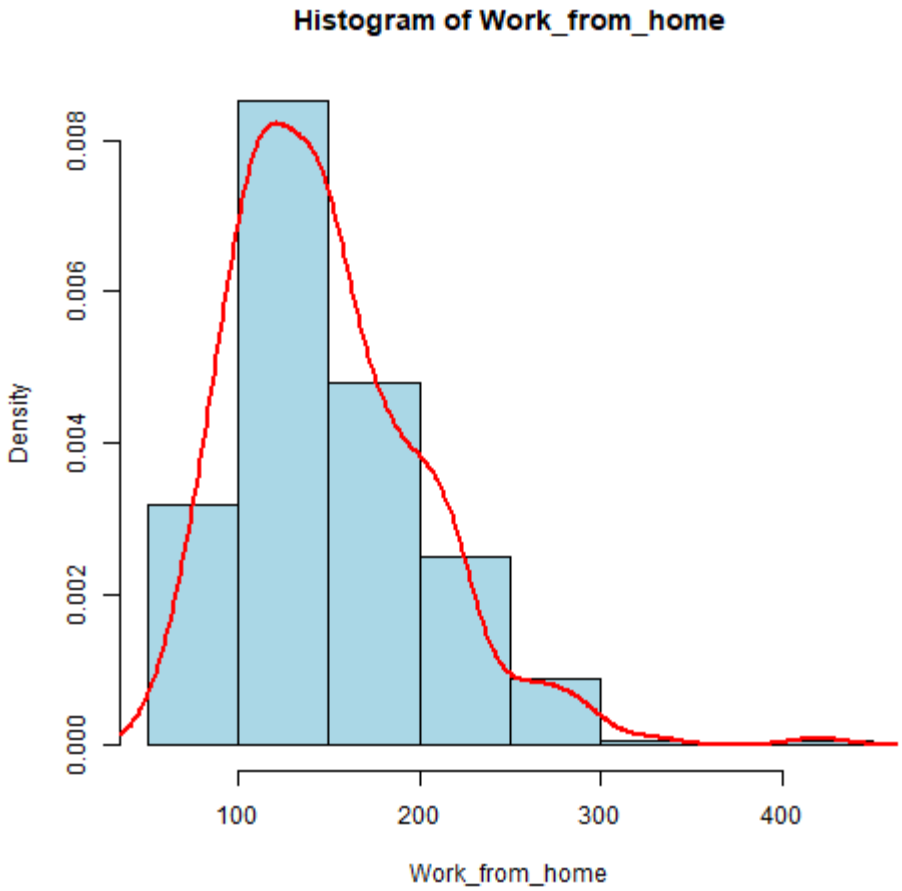

Figure 135 hist of work from home

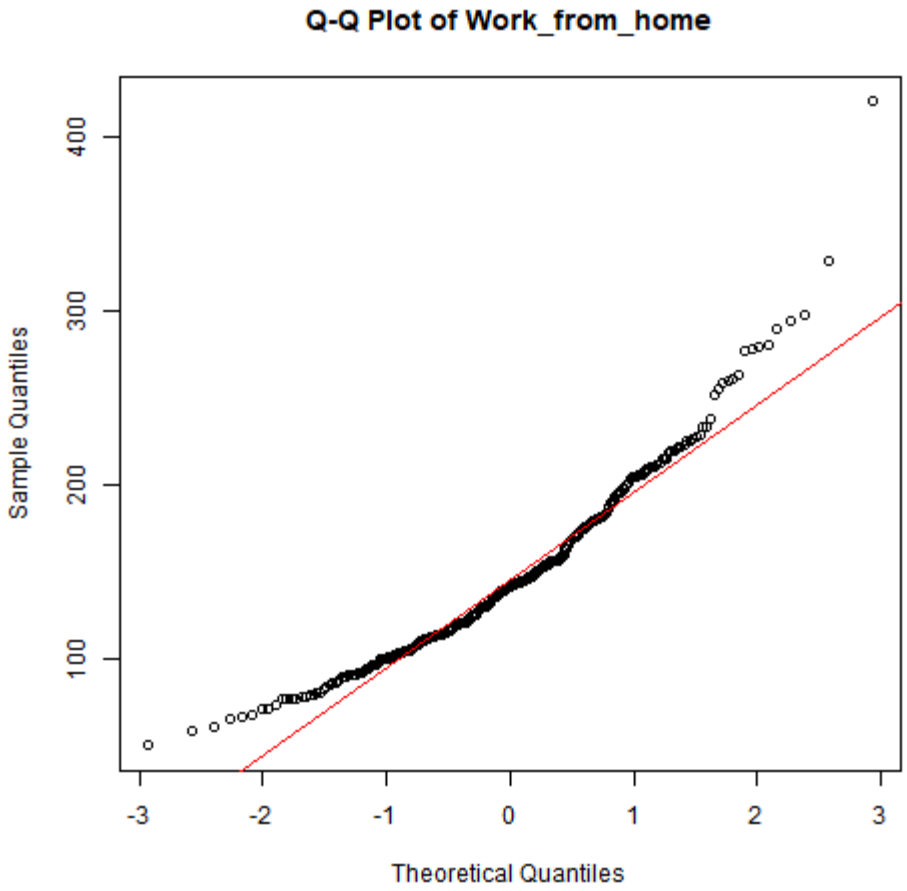

Figure 136 q-q plot of work from home

variable	n	shapiro_stat	shapiro_p	ks_stat	ks_p
total_deaths	296	0.957963982	1.57014771212353e-07	0.04443073	0.60296193
aged_40_to_44_years	296	0.973275213	2.53562432541553e-05	0.061132927	0.21857914
aged_45_to_49_years	296	0.984861436	0.003273211	0.064197695	0.17423484
aged_50_to_54_years	296	0.946189178	6.20038024054313e-09	0.100109037	0.00530148
aged_55_to_59_years	296	0.96656361	2.34823370006272e-06	0.080171307	0.04451596
aged_60_to_64_years	296	0.994159881	0.313947476	0.041875864	0.67696149
aged_65_to_69_years	296	0.997951855	0.975769574	0.02369405	0.99633063
aged_70_to_74_years	296	0.995961147	0.648185582	0.03842878	0.77453795
aged_75_to_79_years	296	0.996113285	0.681000383	0.039647842	0.74074546
aged_80_to_84_years	296	0.997332858	0.912105426	0.031464845	0.93124522
aged_85_years_and_over	296	0.995025729	0.456898159	0.039442727	0.74650377

Figure 137Shapiro wilk test and Kolmogorov Smirnov test

From the Shapiro Wilk test results, following the rule of thumb for if $p < 0.05$, we reject the null hypothesis, the researcher observes that for ages 40-44,45-49,50-54 and 55-59, we reject the null hypothesis. Meaning that we accept the alternative hypothesis which means the covid death distributions are not normal across those ages. There are cogent reasons why the deaths were not normally distributed across those ages which could be due to factors like increased family responsibility, work life, reduction in unnecessary social events, etc.

The Kolmogorov-Smirnov test p values are all > 0.05 so in that case, we fail to reject the null hypothesis.

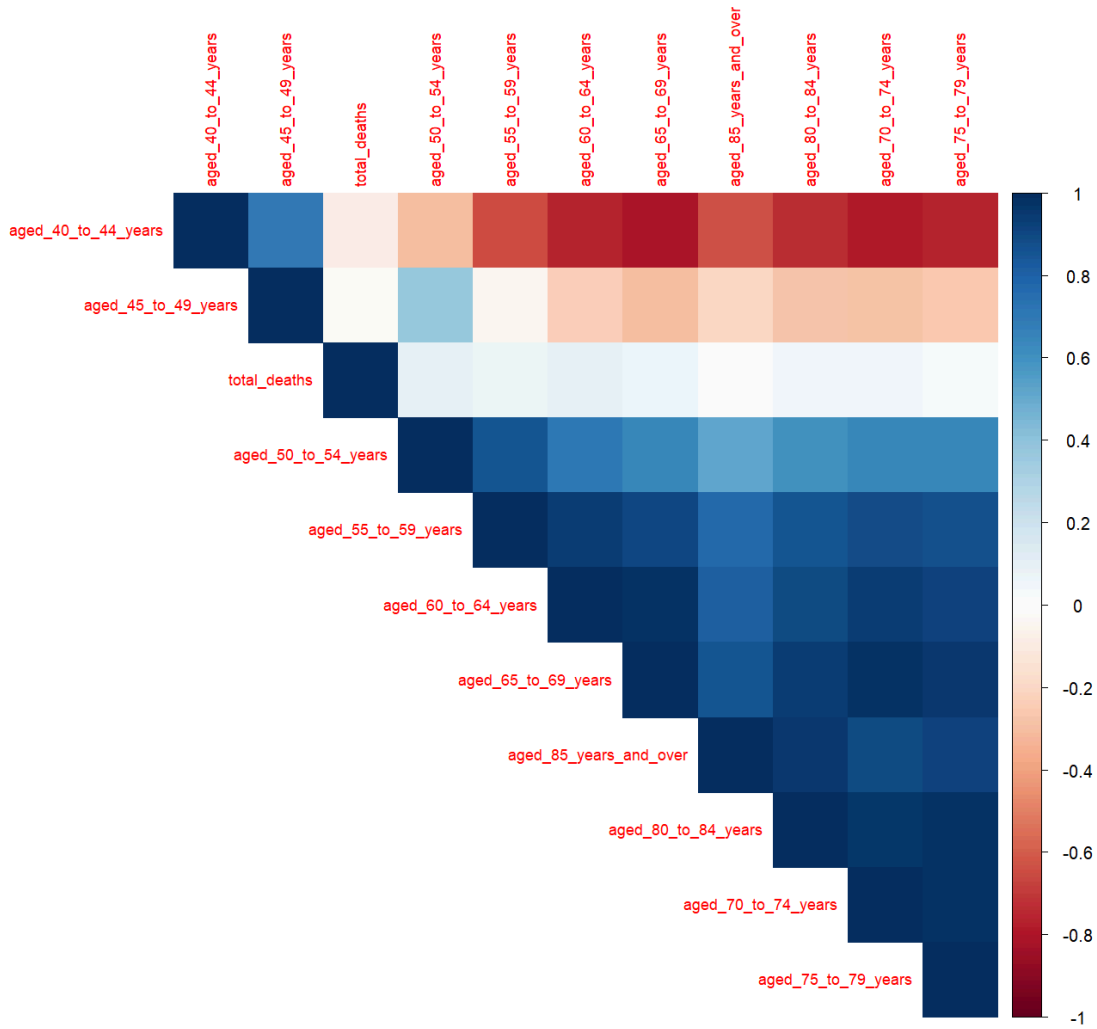

Figure 138 Correlation colour diagram

From the correlation colour diagram, we observe that the blue colour is positive correlation while the red colour is negative correlation.

The variables that show strong positive correlation with covid total deaths are: Shared dwelling: This is because living arrangements in shared spaces increase the risk of social contact which the covid virus is transmitted through.

Public transport also shows strong positive correlation with covid total deaths because there is increased possibility of transmission of the virus through social contact in public transport systems.

Not living in a couple is another variable that shows strong positive correlation with covid total deaths because individuals living alone lack support in health care and monitoring.

Poor health: is another variable with strong positive correlation with total deaths because people with preexisting health conditions are more likely to die from the covid virus.

Certain communal establishments (care homes without nursing: also experienced high covid mortality due to the absence of proper nursing care, detention centres: also, are likely to suffer medical neglect and poor hygiene during the pandemic which might have led to high covid mortality, guest houses: also, due to poor hygiene led to high covid mortality)

The variables that show negative correlation with the covid total deaths are:

Living in a couple: This is because people living as a couple help and assist each other in medical care and emotional support and they stood a better chance surviving the virus.

Private vehicle use: people in private cars also were less exposed to social contact which was a major cause of covid transmission which made them stand a better chance of surviving the virus.

Good dwelling conditions: good facilities with proper hygiene and good living standards reduced the risk of covid contamination and mortality.

	total_deaths	aged_40_to_44_years	aged_45_to_49_years	aged_50_to_54_year	aged_55_to_59_ye	aged_60_to_64_year	aged_65_t	aged_70_t	aged_75_t	aged_80_t	aged_85_years_and_over
total_deat	1	-0.085539644	-0.027960109	0.091102524	0.075433629	0.101237032	0.064065	0.055299	0.037969	0.052198	0.007083276
aged_40_t	-0.085539644	1	0.694883158	-0.306485411	-0.643353918	-0.76645943	-0.80074	-0.78677	-0.76546	-0.7391	-0.633500559
aged_45_t	-0.027960109	0.694883158	1	0.372971704	-0.04485309	-0.238778446	-0.30442	-0.28424	-0.26567	-0.27633	-0.203416914
aged_50_t	0.091102524	-0.306485411	0.372971704	1	0.852670674	0.707108298	0.648035	0.643947	0.642624	0.6036	0.516727887
aged_55_t	0.075433629	-0.643353918	-0.04485309	0.852670674	1	0.945667556	0.901602	0.886254	0.872089	0.854192	0.771022839
aged_60_t	0.101237032	-0.76645943	-0.238778446	0.707108298	0.945667556	1	0.975963	0.943518	0.915153	0.897924	0.819493898
aged_65_t	0.064065498	-0.800744606	-0.30441577	0.648035041	0.901602251	0.975963298	1	0.980762	0.9566	0.932743	0.85265285
aged_70_t	0.055299442	-0.786770301	-0.284240006	0.64394658	0.886254253	0.943518315	0.980762	1	0.986966	0.964565	0.892700715
aged_75_t	0.03796891	-0.765460703	-0.265667598	0.642623966	0.87208876	0.915152696	0.9566	0.986966	1	0.98132	0.918111618
aged_80_t	0.052197941	-0.739096955	-0.276329208	0.603599891	0.854191514	0.897923734	0.932743	0.964565	0.98132	1	0.953987379
aged_85_y	0.007083276	-0.633500559	-0.203416914	0.516727887	0.771022839	0.819493898	0.852653	0.892701	0.918112	0.953987	1

Figure 139 correlation matrix

The age groups 40-44, 45-49, 50-54 and 55-59 have p values < 0.05 meaning we reject the null hypothesis. We then accept the alternative hypothesis which means the distribution of covid deaths for these age groups is not normal. This could be due to factors such as increased family responsibility, work life and reduction in unnecessary social hangouts.

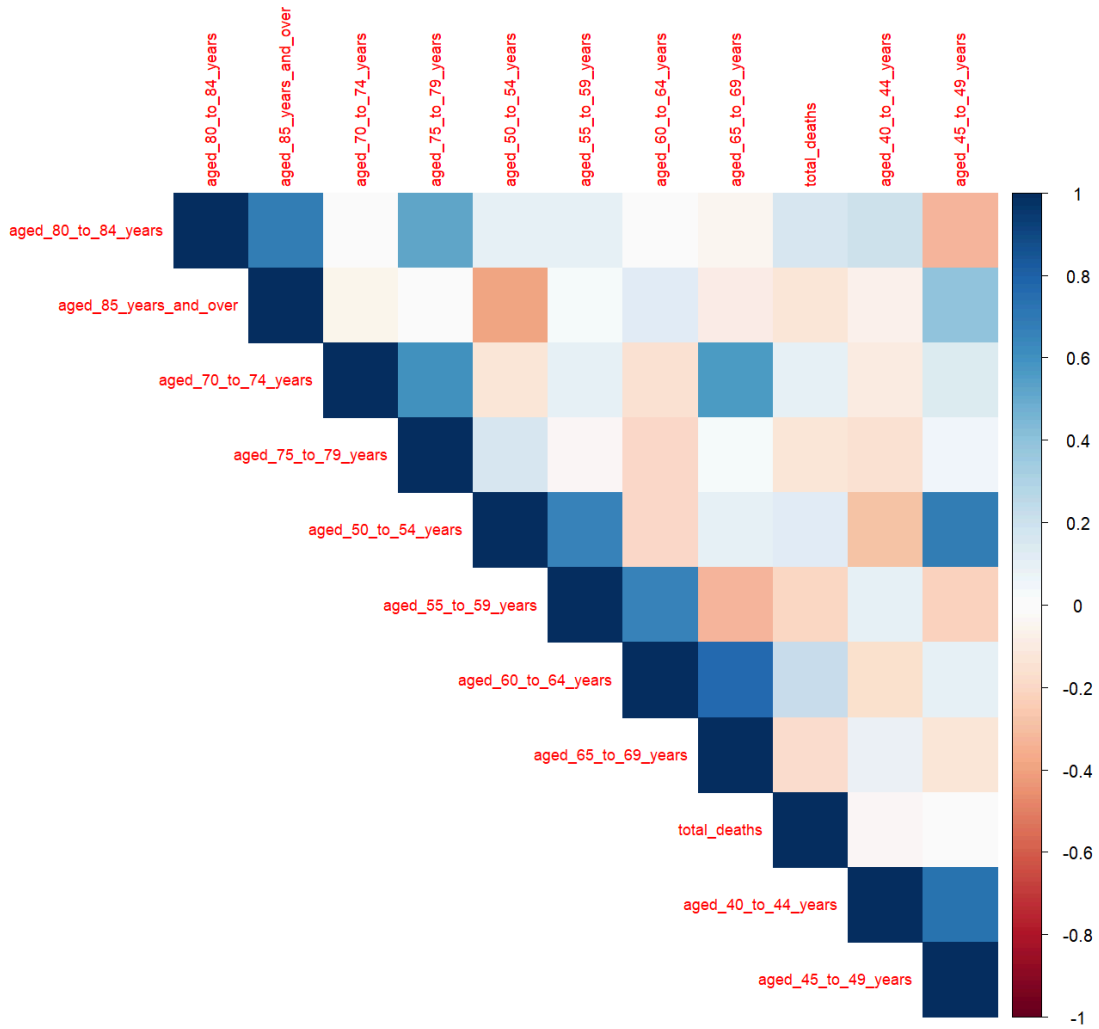

Figure 140 Partial correlation colour map

From the partial correlation colour diagram, we observe that the blue colour is positive correlation while the red colour is negative correlation.

The variables that show strong positive correlation with covid total deaths are: Shared dwelling: This is because living arrangements in shared spaces increase the risk of social contact which the covid virus is transmitted through.

Public transport also shows strong positive correlation with covid total deaths because there is increased possibility of transmission of the virus through social contact in public transport systems.

Not living in a couple is another variable that shows strong positive correlation with covid total deaths because individuals living alone lack support in health care and

monitoring.

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Certain communal establishments (care homes without nursing: also experienced high covid mortality due to the absence of proper nursing care, detention centres: also, are likely to suffer medical neglect and poor hygiene during the pandemic which might have led to high covid mortality, guest houses: also, due to poor hygiene led to high covid mortality)

The variables that show negative correlation with the covid total deaths are:

Living in a couple: This is because people living as a couple help and assist each other in medical care and emotional support and they stood a better chance surviving the virus.

Private vehicle use: people in private cars also were less exposed to social contact which was a major cause of covid transmission which made them stand a better chance of surviving the virus.

Good dwelling conditions: good facilities with proper hygiene and good living standards reduced the risk of covid contamination and mortality.

	total_deaths	aged_40_to_4	aged_45_to_49_y	aged_50_to_	aged_55_to_5	aged_60_t	aged_65_to_69_	aged_70_t	aged_75_t	aged_80_t	aged_85_y
total_deat	1	-0.03397753	-0.011155303	0.12072655	-0.20272343	0.228314	-0.175720809	0.100177	-0.12628	0.168234	-0.12488
aged_40_t	-0.033977533	1	0.738262505	-0.283136	0.10335051	-0.16892	0.08809426	-0.09828	-0.15949	0.206007	-0.06752
aged_45_t	-0.011155303	0.738262505	1	0.68927054	-0.21841434	0.108751	-0.134850678	0.138467	0.041081	-0.3234	0.390587
aged_50_t	0.120726549	-0.28313596	0.689270537	1	0.656278075	-0.19041	0.099622374	-0.13521	0.166386	0.106645	-0.39174
aged_55_t	-0.20272343	0.10335051	-0.21841434	0.65627808	1	0.656427	-0.322400592	0.104644	-0.03087	0.099438	0.023785
aged_60_t	0.22831406	-0.16891507	0.108751356	-0.1904143	0.656427088	1	0.760496649	-0.15346	-0.19148	0.006662	0.128343
aged_65_t	-0.175720809	0.08809426	-0.134850678	0.09962237	-0.322400592	0.760497	1	0.563496	0.023688	-0.04922	-0.08678
aged_70_t	0.100176765	-0.09828289	0.138466668	-0.1352054	0.104644431	-0.15346	0.563495624	1	0.604425	-0.00143	-0.05187
aged_75_t	-0.126283812	-0.15949295	0.041081044	0.16638632	-0.030869338	-0.19148	0.023687699	0.604425	1	0.521054	-0.00323
aged_80_t	0.168234489	0.206006769	-0.323403706	0.10664487	0.099438131	0.006662	-0.049222479	-0.00143	0.521054	1	0.683757
aged_85_y	-0.124881994	-0.06752284	0.390586849	-0.3917389	0.023784712	0.128343	-0.086775807	-0.05187	-0.00323	0.683757	1

Figure 141 Partial correlation matrix

The age groups 40-44, 45-49, 50-54 and 55-59 have p values < 0.05 meaning we reject the null hypothesis. We then accept the alternative hypothesis which means the distribution of covid deaths for these age groups is not normal. This could be due to factors such as increased family responsibility, work life and reduction in unnecessary social hangouts.

variable	msa
total_deaths	0.172415
aged_40_to_44_years	0.852877
aged_45_to_49_years	0.431305
aged_50_to_54_years	0.726914
aged_55_to_59_years	0.840404
aged_60_to_64_years	0.83633
aged_65_to_69_years	0.855897
aged_70_to_74_years	0.892559
aged_75_to_79_years	0.895107
aged_80_to_84_years	0.868563
aged_85_years_and_over	0.864833

Figure 142 KMO per item

Kaiser-Meyer-Olkin factor adequacy

Call: psych::KMO(r = cor_matrix)

Overall MSA = 0.83

MSA for each item =

total_deaths	aged_40_to_44_years	aged_45_to_49_years
0.17	0.85	0.43
aged_50_to_54_years	aged_55_to_59_years	aged_60_to_64_years
0.73	0.84	0.84
aged_65_to_69_years	aged_70_to_74_years	aged_75_to_79_years
0.86	0.89	0.90
aged_80_to_84_years	aged_85_years_and_over	
0.87	0.86	

If MSA > 0.6, we proceed to factor analysis.

From the table, we see that the age variables 45-49, 50-54, 55-59, 60-64, 65-69, 70-74, 75-79, 80-84 and 85 and above have MSA > 0.6, so we select them for factor analysis.

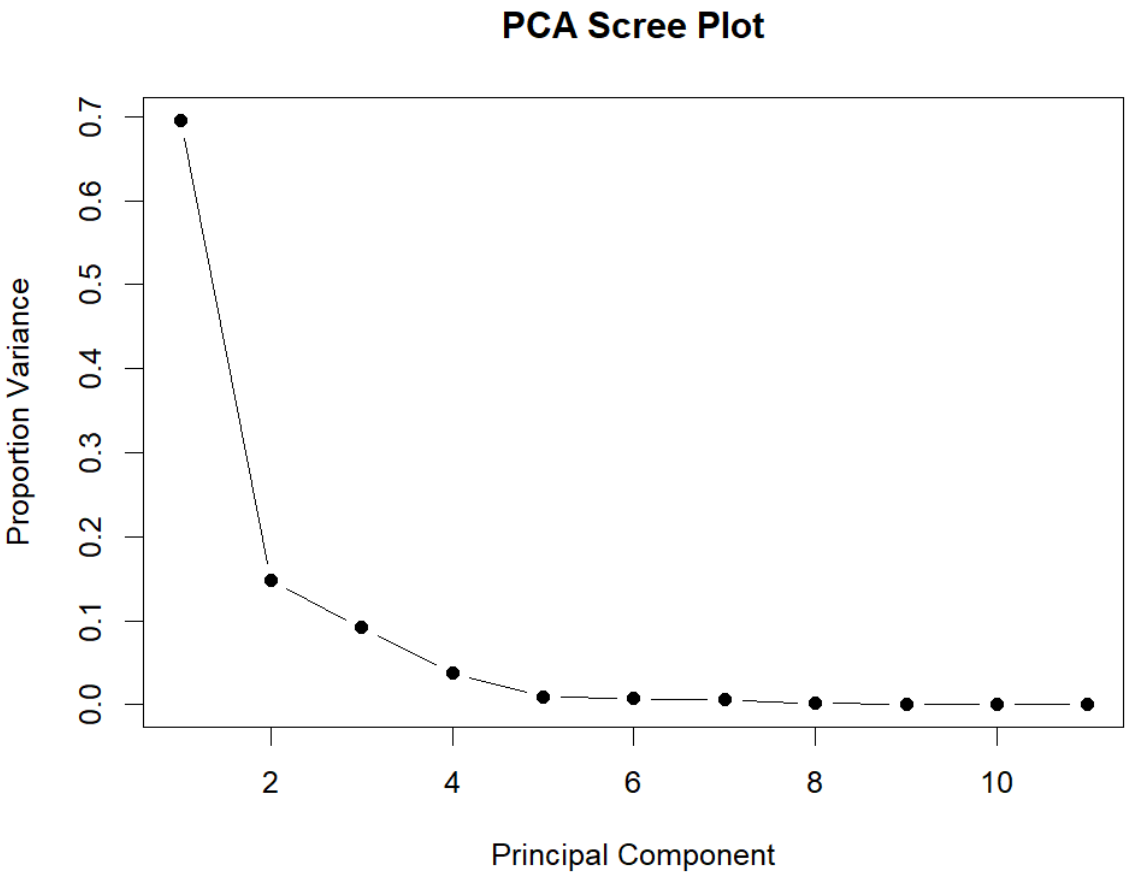

Figure 143 PCA scree (elbow) plot

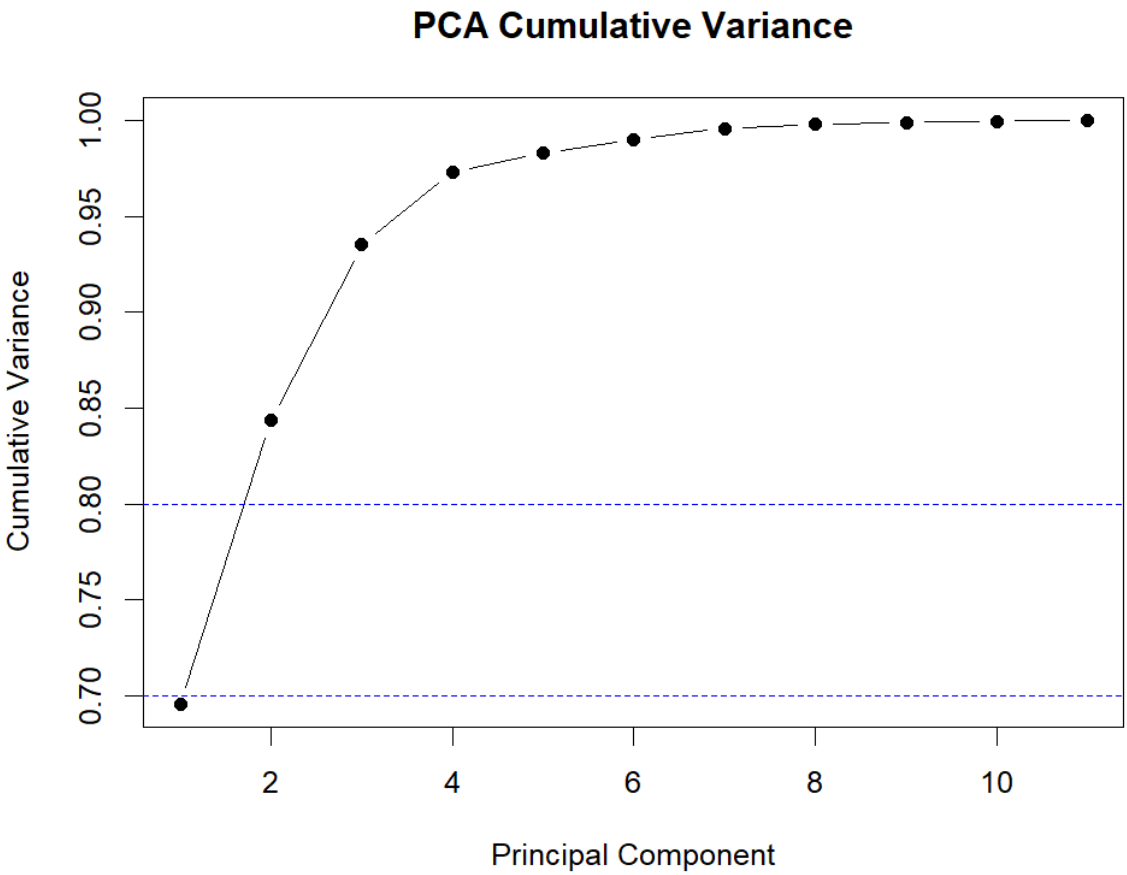

Figure 144 Cumulative scree plot for greater accuracy

At 95% accuracy, we take 4 points.

PC1 (69.6%): aged_70_to_74_years(0.357), aged_75_to_79_years(0.357), aged_65_to_69_years(0.354), aged_80_to_84_years(0.352), aged_60_to_64_years(0.348), aged_55_to_59_years(0.338)

PC2 (14.8%): aged_45_to_49_years(-0.739), aged_50_to_54_years(-0.504), aged_40_to_44_years(-0.395), aged_55_to_59_years(-0.201), aged_65_to_69_years(0.034), aged_60_to_64_years(-0.029)

PC3 (9.2%): total_deaths(0.988), aged_50_to_54_years(0.080), aged_85_years_and_over(-0.077), aged_40_to_44_years(-0.068), aged_60_to_64_years(0.063), aged_55_to_59_years(0.045)

	PC1	PC2	PC3
total_deaths	-0.04479	0.005839	0.987532
aged_40_to_44_years	-0.2766	-0.39529	-0.06775
aged_45_to_49_years	-0.08106	-0.73881	-0.02605
aged_50_to_54_years	0.256412	-0.50378	0.079958
aged_55_to_59_years	0.338388	-0.20084	0.044691
aged_60_to_64_years	0.347566	-0.02929	0.063469
aged_65_to_69_years	0.354197	0.034335	0.01766
aged_70_to_74_years	0.357227	0.026066	-0.00092
aged_75_to_79_years	0.356865	0.014873	-0.02426
aged_80_to_84_years	0.351735	0.027837	-0.01755
aged_85_years_and_over	0.331243	0.005377	-0.0767

Figure 145 PCA Rotated loadings

Section 5: Implementation

Research methodology.

The researcher employed exploratory data analytical techniques to derive insights from trends across different living and working conditions.

Recommendations.

Teenagers and Young adults should quit drinking and smoking as this exposes them to high mortality during outbreaks of diseases and flu.

Social contact should be done with observance of hygiene protocols, or better still be minimised during disease or viral outbreaks.

People of all age groups should have regular health check-ups.

Early age children should be properly vaccinated and protected, especially during disease or flu outbreaks.

Section 6: Discussion:

Following a comprehensive overview and analysis of the dataset, the researcher observed that the most important variables were specific Age distributions, living arrangements and Mode of transport to work.

In the Age groups, ages 15-25 experienced high mortality due to some lifestyle patterns that are synonymous with youths like excessive smoking, drinking, clubbing (leading to exposed social risk)

The researcher then noticed a sharp decline in mortality rates from ages 35-50 due to increased sense of responsibility, family life, restriction of social hangouts, reduction of drinking and smoking and improved medical checkup consciousness.

The researcher noticed a decrease as well in mortality from 60-75+ due to delicate medical care, restriction of movement and nursing care.

In the mode of transport to work, people with private vehicles with doors and windows were less exposed to the virus mortality, while people that used public transport services like trains, buses were more exposed to the virus due to increased social contact.

In terms of living arrangement, people that lived as a couple interestingly stood better chance of surviving the virus due to companionship and mutual support in medication, health monitoring and general care than people who lived alone.

Also, people who lived in good dwelling conditions with good and hygienic facilities stood better chance of surviving the virus than people who lived in bad living conditions.

Section 7: Conclusion

COVID-19 is a deadly virus, but it is beatable and can be dealt with if the people affected by it take some certain precautions like:

Prioritizing timely vaccination,

Taking medications as at when due,

Practicing safe social distancing,

Reducing unnecessary social and clubbing events,

Reducing alcohol and smoking,

Taking regular medical checkups,

And applying hand sanitizers, wearing hand gloves and nose masks especially when using public transport.

Appendix:

My final .csv dataset: u3059227_DS7006_CW2_data.csv

My SQL script.txt

My R Script.txt

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